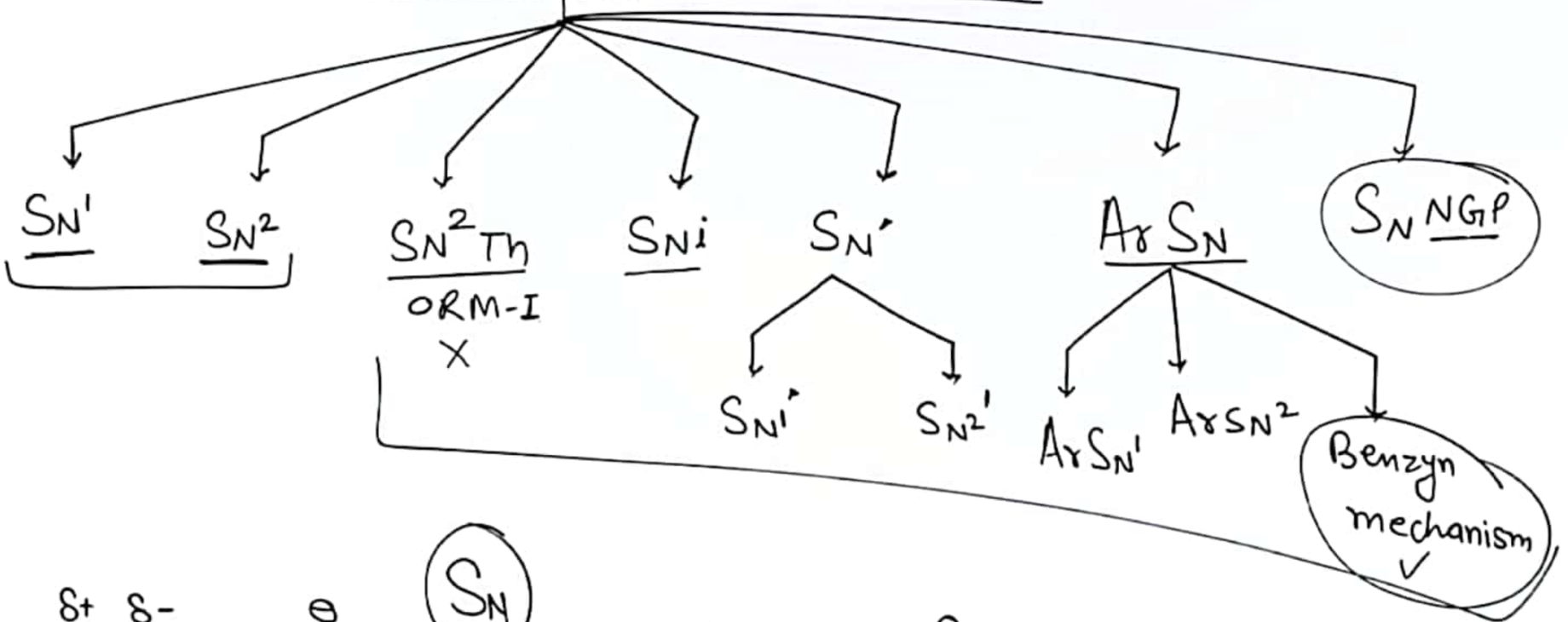
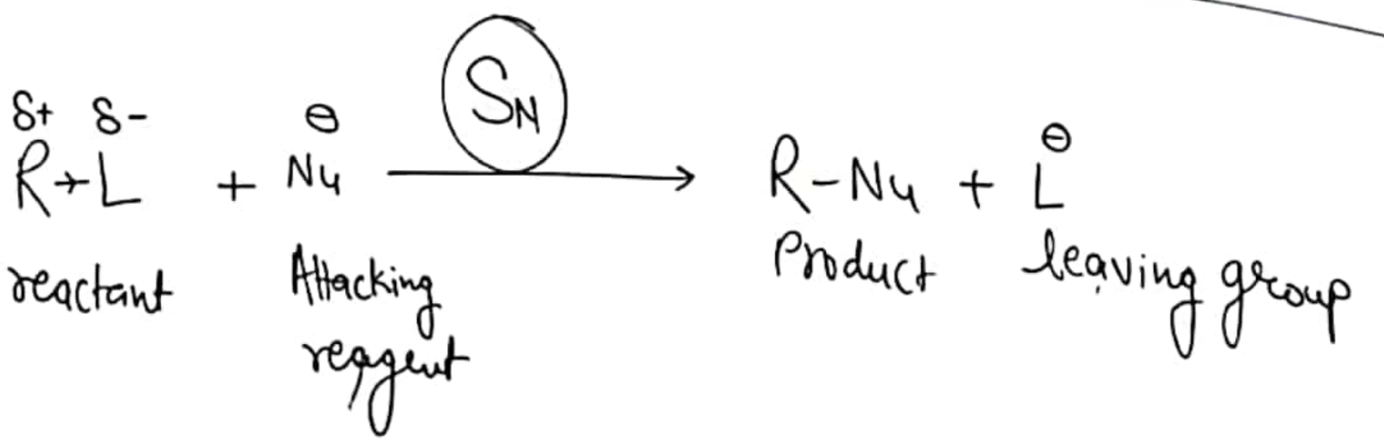
ORGANIC REACTION MECHANISM - III

ORM-III

Nucleophilic Substitution Reactions (S_N)

Alkyl halide
Alcohol
ether





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ORGANIC REACTION MECHANISM – III

ORM-III

S_N1 (Unimolecular Nu. Sub. Rx^n)

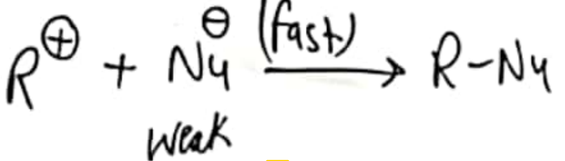
① Two step mechanism

② Mechanism

Step.1



Step.2

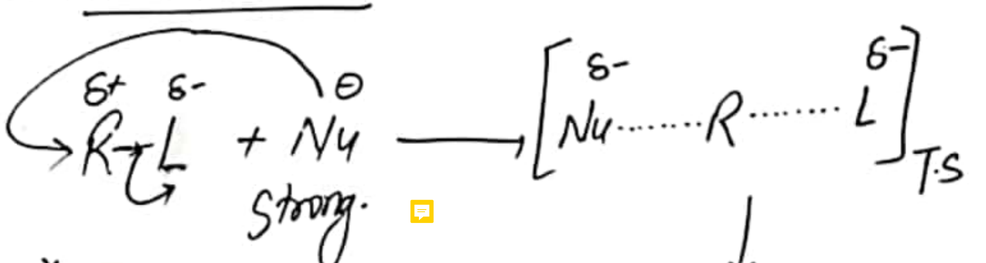


S_N2 (Bimolecular Nu. Sub. Rx^n

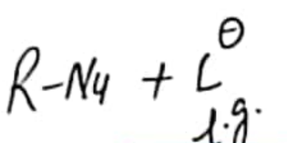
① Single step concerted mechanism

Where bond formation and bond breaking take place simultaneously (T.S.)

(2) Mechanism







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ORGANIC REACTION MECHANISM - III

ORM-III

S_N1 (Unimolecular Nu. Sub. Rx^n)

(3) Kinetics of S_N1

⇒ Rate of S_N1 reaction $\propto [R-L]^1 [Nu^{\ominus}]^0$

⇒ 1st order reaction.

(4) Nucleophile :-

Weak nucleophiles are favourable for S_N1 reaction.

⇒ Generally neutral nucleophiles, like $H_2\ddot{O}$, $R-\ddot{O}H$, H_2S, CH_3COOH etc

S_N2 (Bimolecular Nu. Sub. Rx^n

(3) Kinetics for S_N2

⇒ Rate of $S_N2 \propto [R-X]^1 [Nu^-]^1$

⇒ 2nd order reaction

(4) Nucleophiles

Strong nucleophiles are favourable for S_N2 Rx^n

⇒ Generally anionic nucleophile

$H^{\ominus}$, $X^{\ominus}$, $R^{\ominus}$, $NH_2^{\ominus}$, $OH^{\ominus}$, $OR^{\ominus}$, $SH^{\ominus}$, $SR^{\ominus}$, $R-COO^{\ominus}$, $CN^{\ominus}$, NH_3, $R-NH_2$, $R-NH-R$, $\ddot{P}R_3$ etc

some neutral nucleophiles are also strong Nu-

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ORGANIC REACTION MECHANISM - III

ORM-III

S_N1 (Unimolecular Nu. Sub. Rx^n)

(5) Leaving group

⇒ reactivity for S_N1 ∝ leaving ability

⇒ R—I > R-Br > R-Cl > R-F (Reactivity)

↑

No S_N1 Rx^n

(6) Solvent:

Polar protic solvents are favourable for S_{N1}

S_N2 (Bimolecular Nu. Sub. Rx^n

(5) Leaving group.

⇒ same

⇒ R-I > R-Br > R-Cl > R-F (Reactivity)

↑

No S_N2 Rx^n

(6) Solvents:

⇒ Polar aprotic solvent - -

nucleophilicity increases in presence of polar aprotic solvents

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ORGANIC REACTION MECHANISM - III

ORM-III

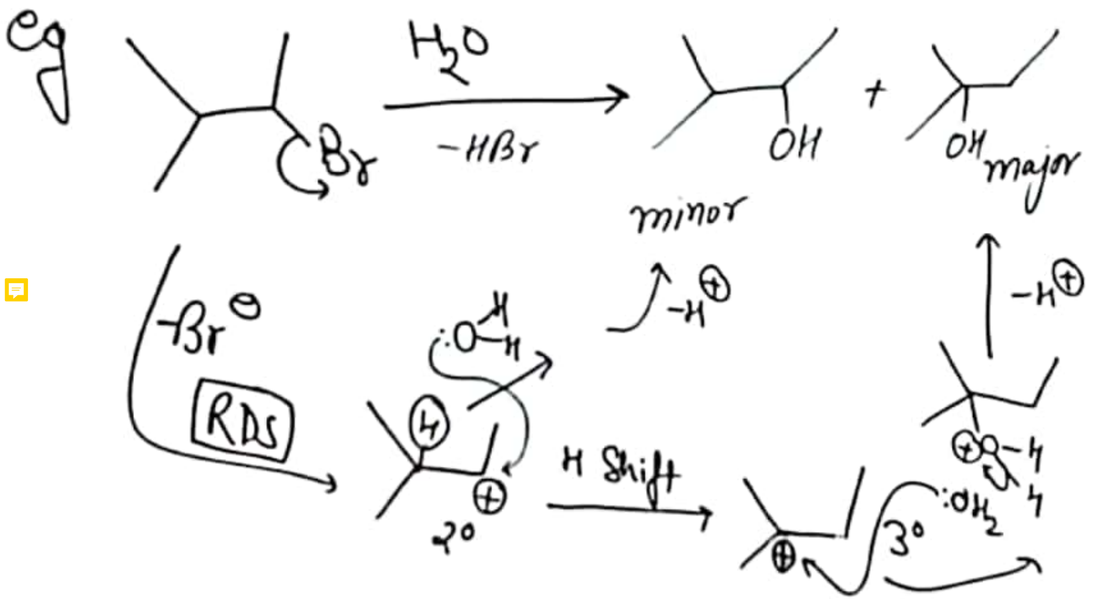
S_N1 (Unimolecular Nu. Sub. Rx^n)

(7) Rearrangement:-

Intermediate = free carbocation.

↓

Rearrangement in carbocation is possible

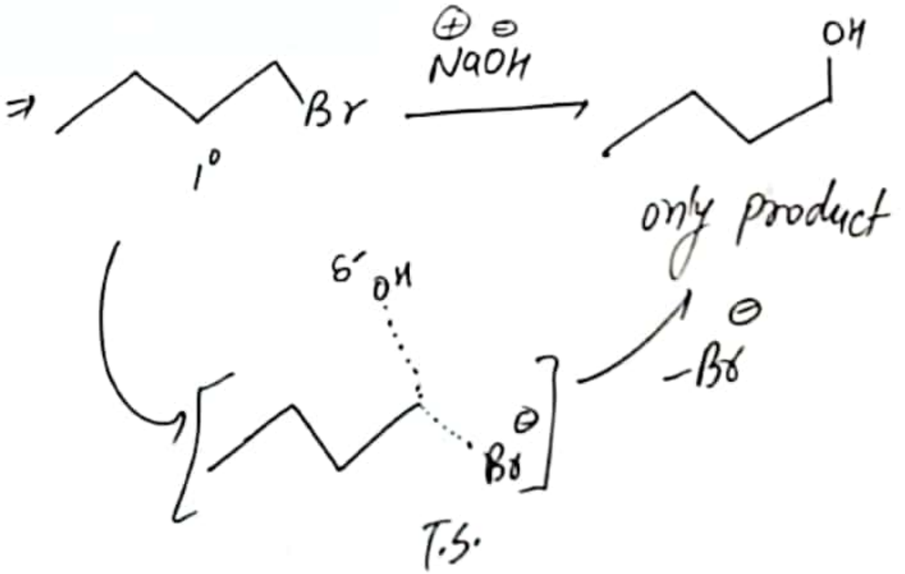


S_N2 (Bimolecular Nu. Sub. Rx^n

(7) Rearrangement

↳ No Carbocation intermediate

↳ No Rearra.



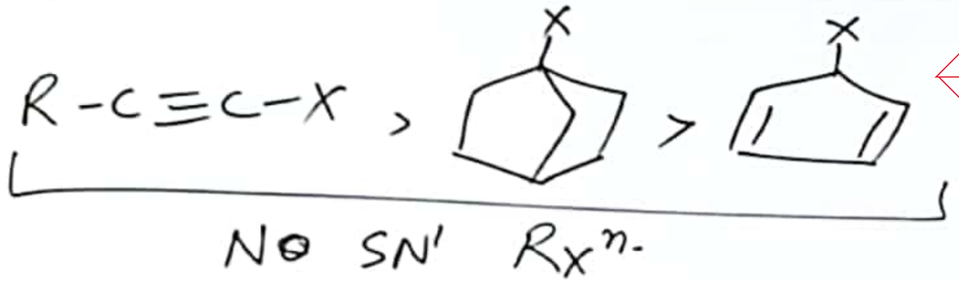
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⊗ Reactivity on the basis of alkyl group
Reactivity of SN1 ∝ Stability of carbocation intermediate of 1st step (RDS)

ORGANIC REACTION MECHANISM – III

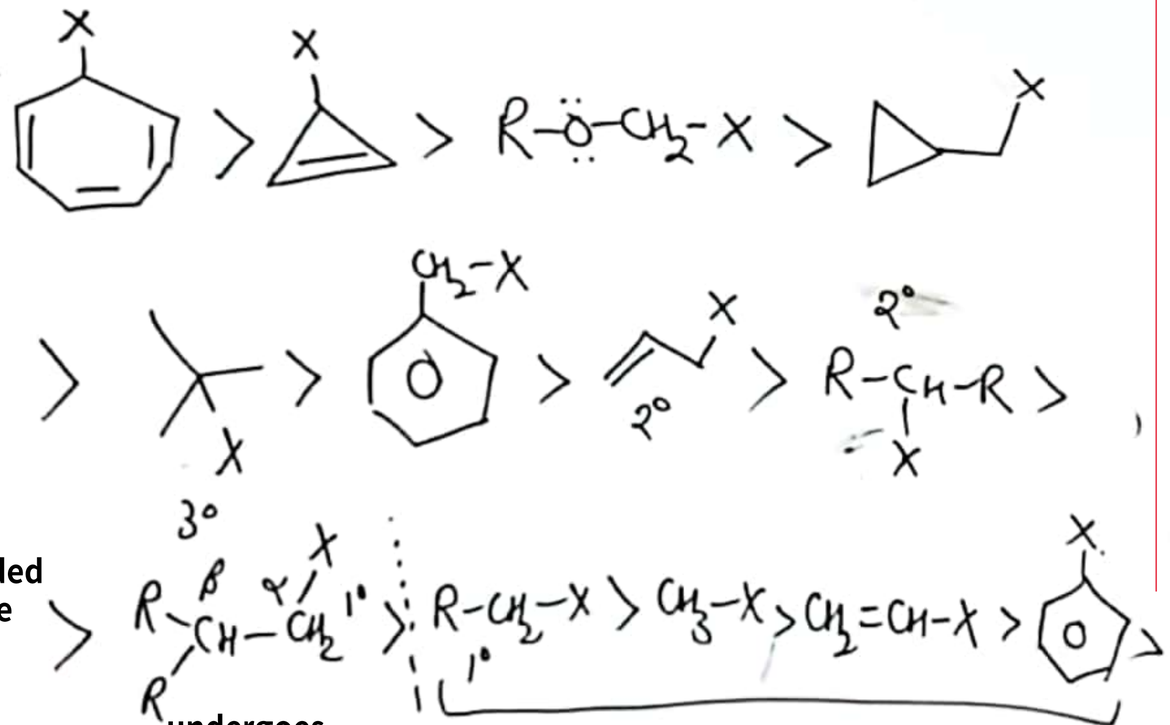
ORM-III

SN1 (Unimolecular Nu. Sub. Rxn)



No SN1 Rxn.

General order:



SN2 (Bimolecular Nu. Sub. Rxn

⊗ Reactivity for SN2 ∝ 1 / crowding on carbon of leaving group.

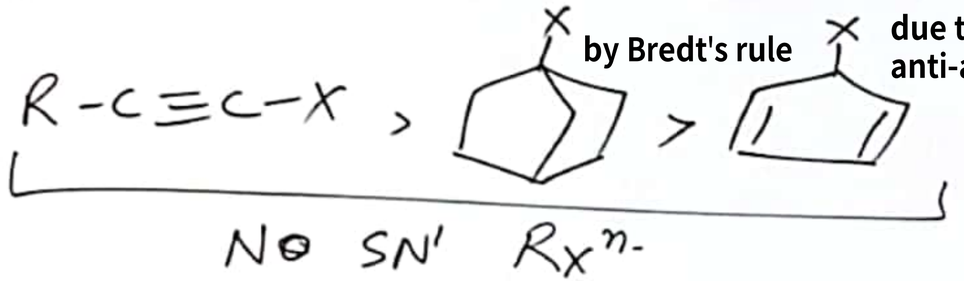
⇒ Reactivity ∝ e⁻ dificiency on carbon of l.g.

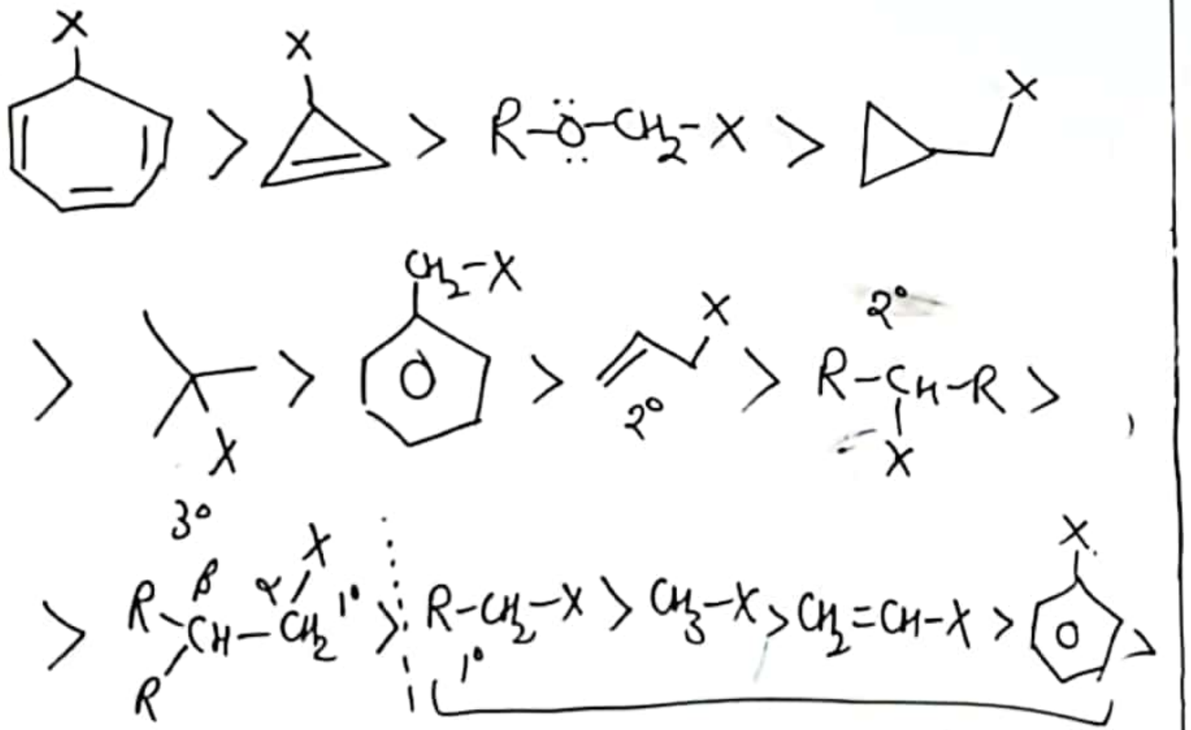
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ORGANIC REACTION MECHANISM - III

ORM-III

S_N1 (Unimolecular Nu. Sub. R_xn)

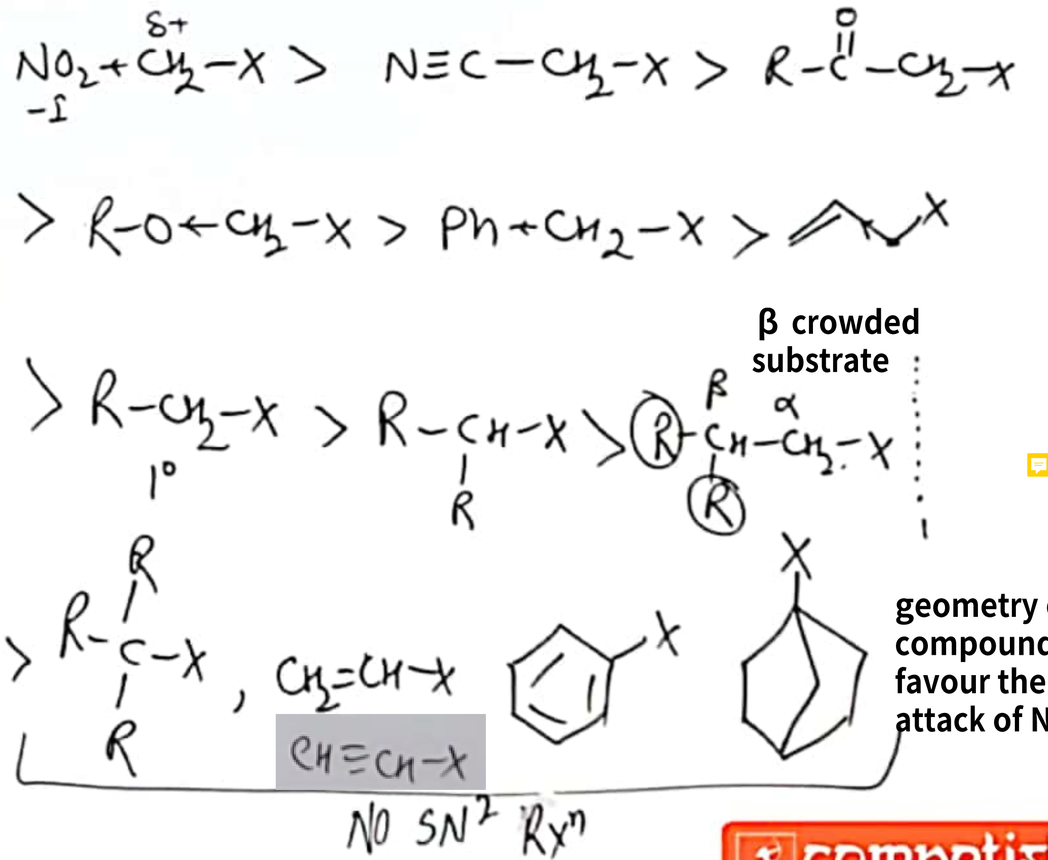




S_N2 (Bimolecular Nu. Sub. R_xn

⑧ Reactivity

General order



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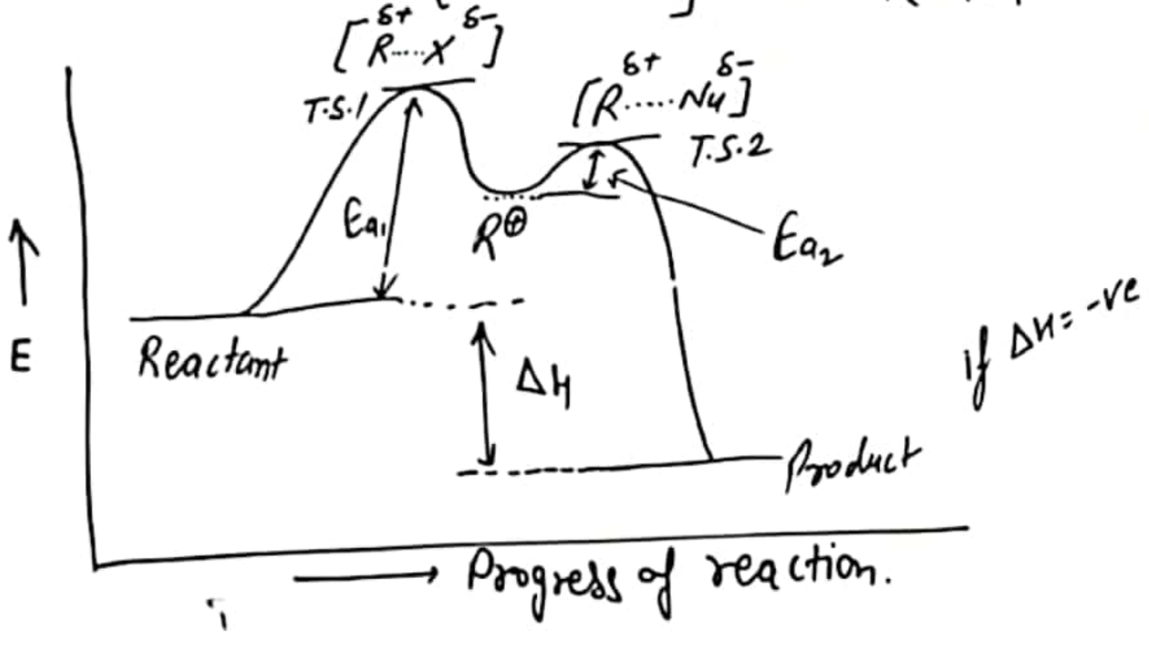
ORGANIC REACTION MECHANISM - III

S_N1 (Unimolecular Nu. Sub. Rx^n)

⑨ Energetics of S_N1

$$R-X \longrightarrow [R^{\delta+} \cdots X^{\delta-}] \longrightarrow R^{\oplus} + X^{\ominus}$$

$$R^{\oplus} + N_4^{\ominus} \longrightarrow [R^{\delta+} \cdots N_4^{\delta-}] \longrightarrow R-N_4$$

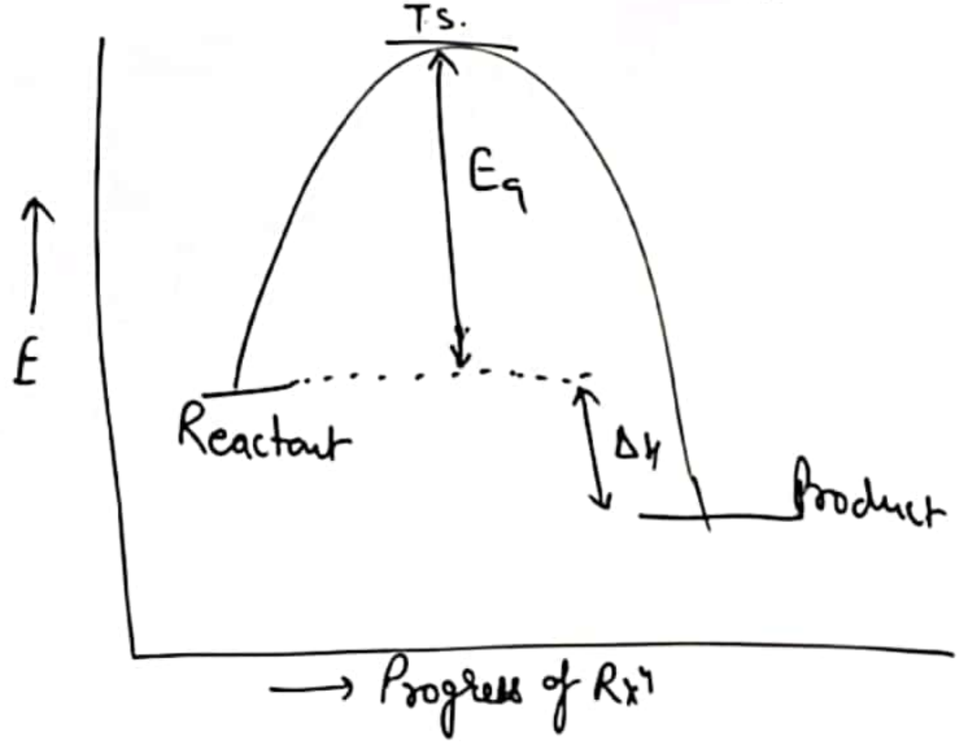


if ΔH = -ve

S_N2 (Bimolecular Nu. Sub. Rx^n)

⑨ Energetics - - - -

$$R-X + N_4^{\ominus} \longrightarrow [N_4^{\delta-} \cdots R^{\delta+} \cdots X^{\delta-}]_{TS} \rightarrow R-N_4 + X^{\ominus}$$



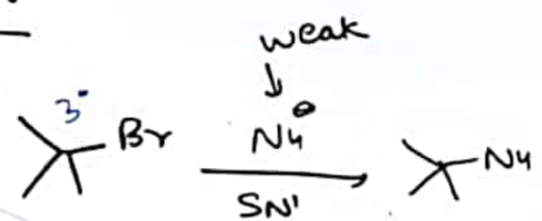
ORGANIC REACTION MECHANISM - III

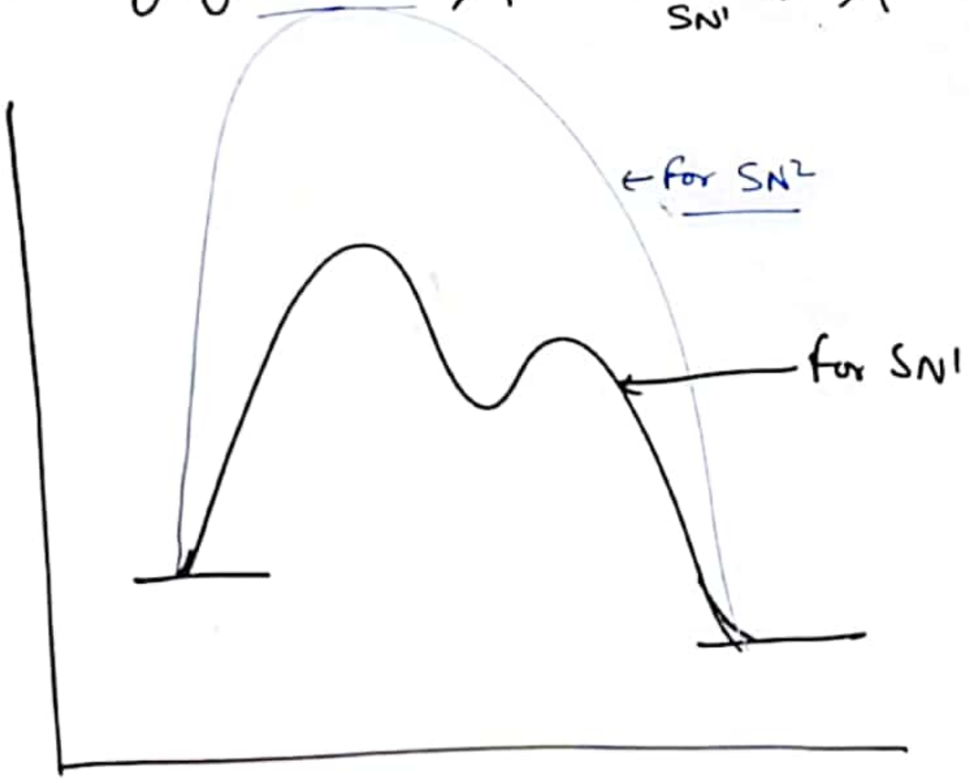
ORM-III

S_N1 (Unimolecular Nu. Sub. Rxn)

⑨ Energetics of S_N1

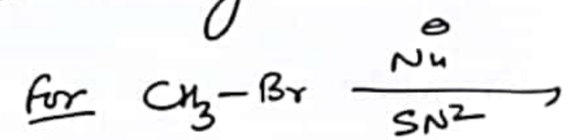
for 3° alkyl group

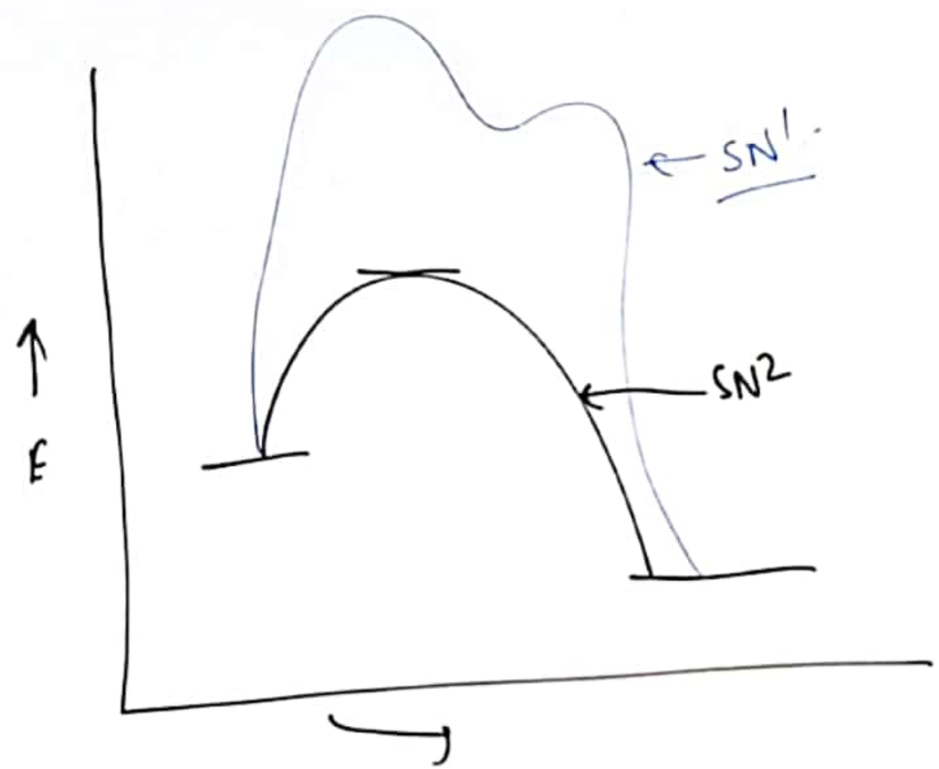




S_N2 (Bimolecular Nu. Sub. Rxn)

⑨ Energetics ----

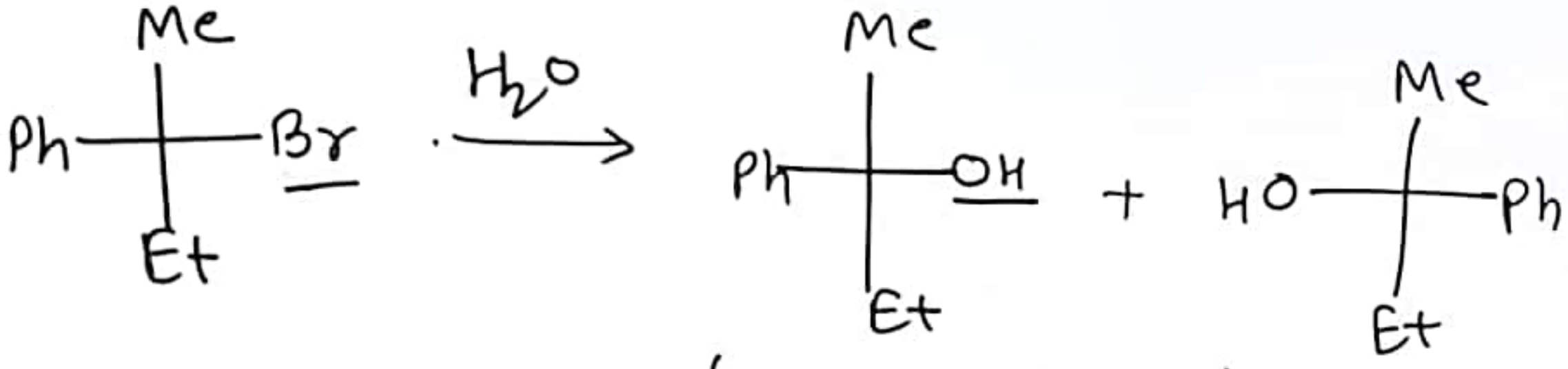




(10) Stereochemistry of SN1

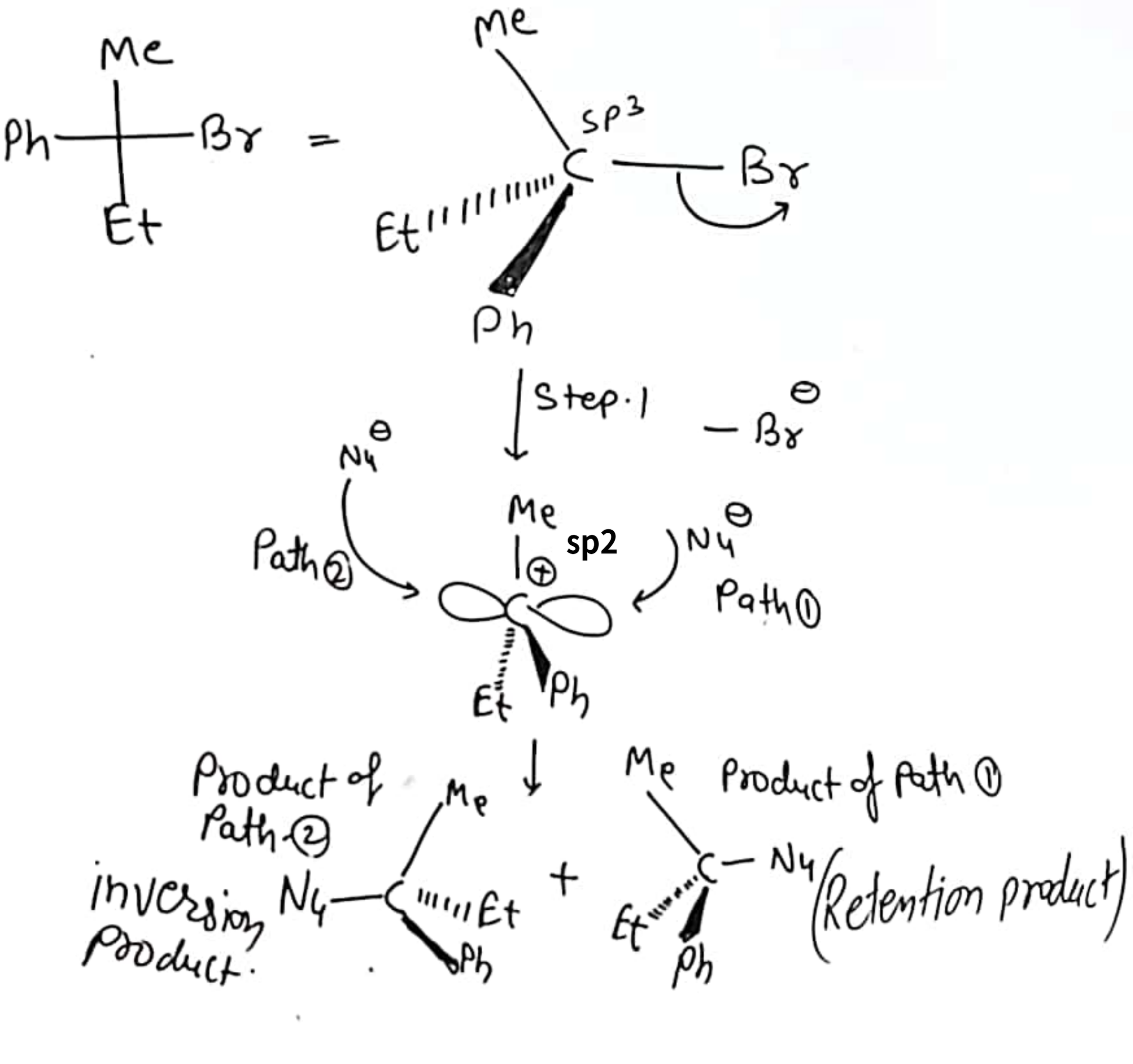
When l.g. is present on chiral C then in SN1 reaction reaction generally more than one stereoisomeric products are form (generally mix. of enantiomers) due formation of planar carbocation

(10) Stereochemistry of SN1



S_N1 (Unimolecular Nu. Sub. Rx^n)

(10) Stereochemistry of SN1



(10) Stereochemistry of S_N1

Experimently 100% racemic mix. does not form due to repulsion b/w l.g$^{\ominus}$ & Nu$^{\ominus}$ in initial moments of formation of carbocation. (% inversion product > % Retention product)

⇒ % Racemisation $\propto$ rate of formation of $C^{\oplus}$

⇒ " " " $\propto$ stability of intermediate

⇒ " " " $\propto$ stability of l.g.

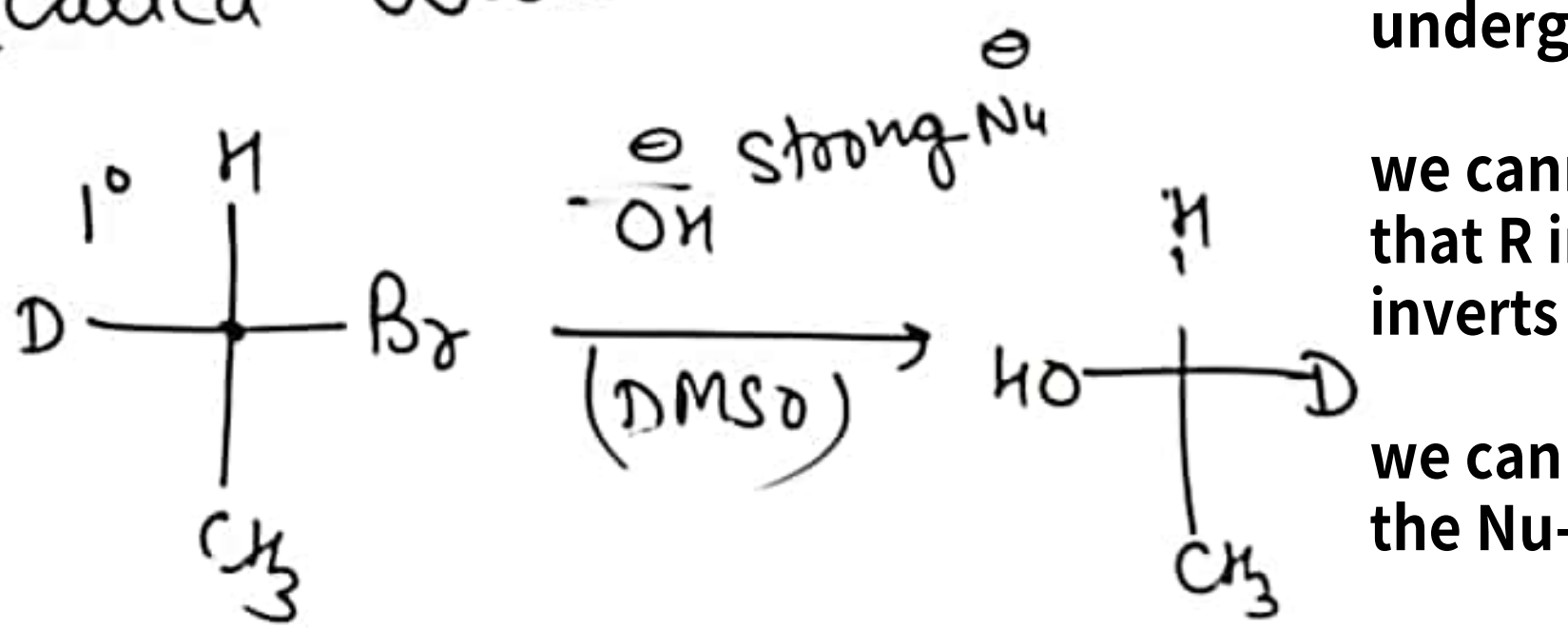
⇒ " " $\propto$ Polarity of solvent

SN^2 (Bimolecular Nu. sub. Rxn

(10) Stereochemistry for SN^2

When l.g. is present on chiral carbon then only inversion product form.

(called "walden inversion")



only inversion Product

only the side of the Nu- undergoes inversion

we cannot say with surety that R inverts to S or + inverts to -

we can only say the side of the Nu- gets inverted

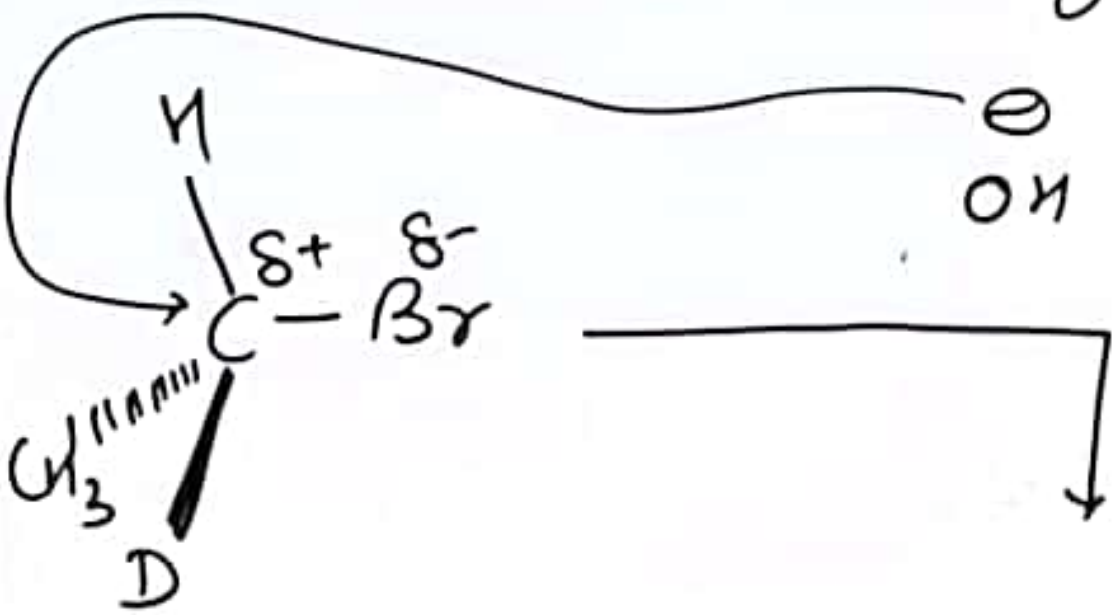
if $[\alpha] = +x^\circ$

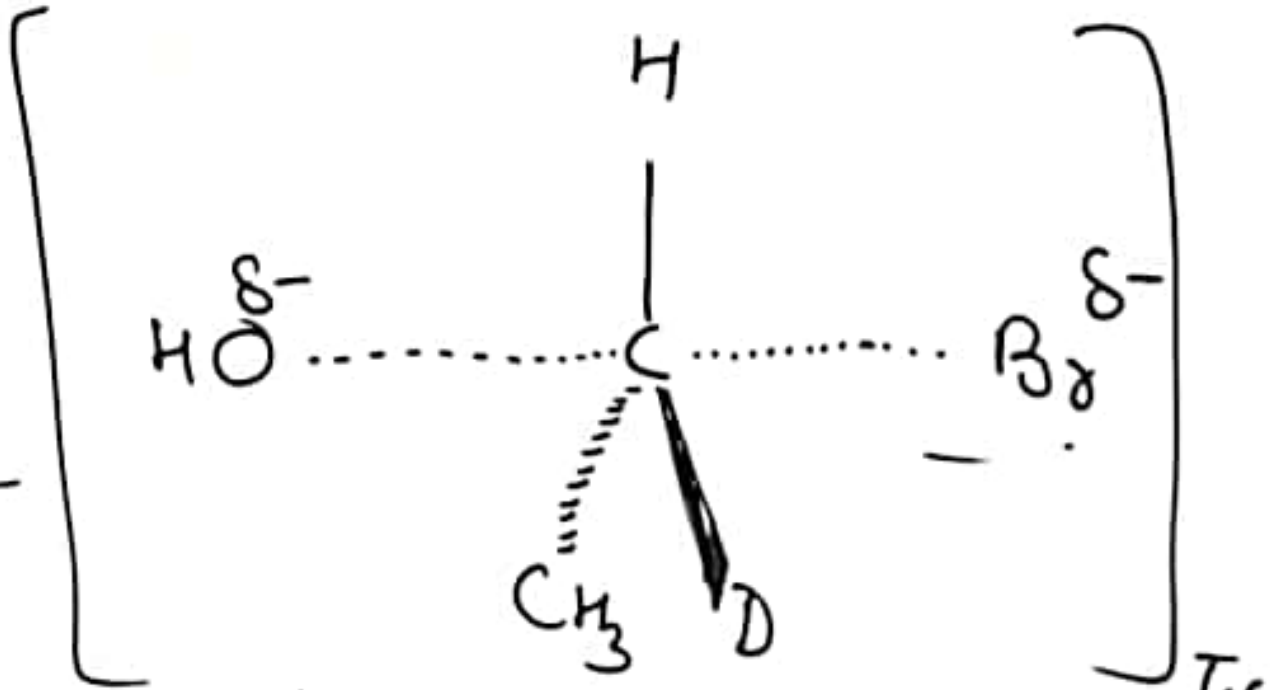
$[\alpha] \neq -x^\circ$

R S

SN^2 (Bimolecular Nu. sub. Rxn

(10) Stereochemistry for SN^2

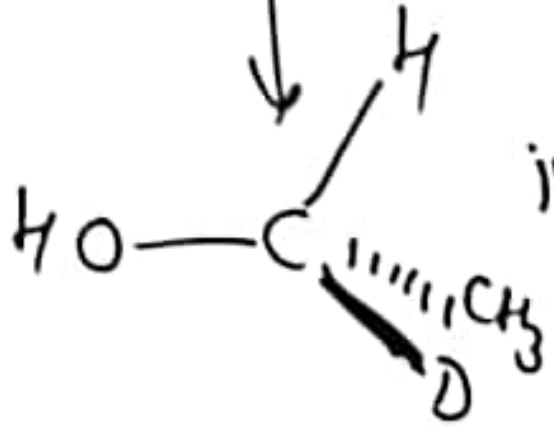




TBP

Trigonal bipyramidal

T.S

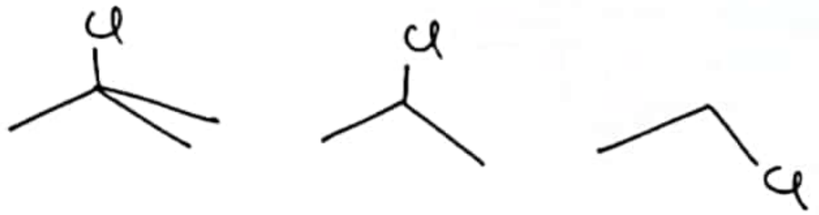


inversion product

ORGANIC REACTION MECHANISM – III

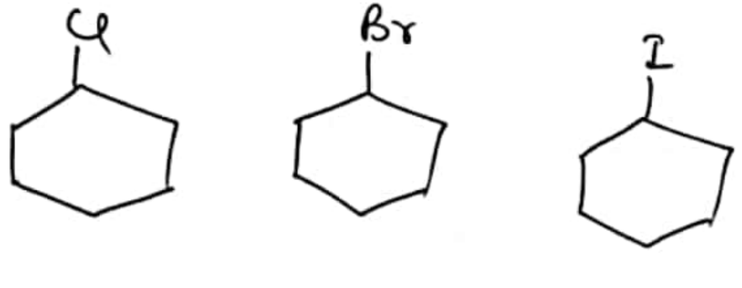
ORM-III

Q Give order of reactivity for S_N1 and S_N2 R_x^n.

① 

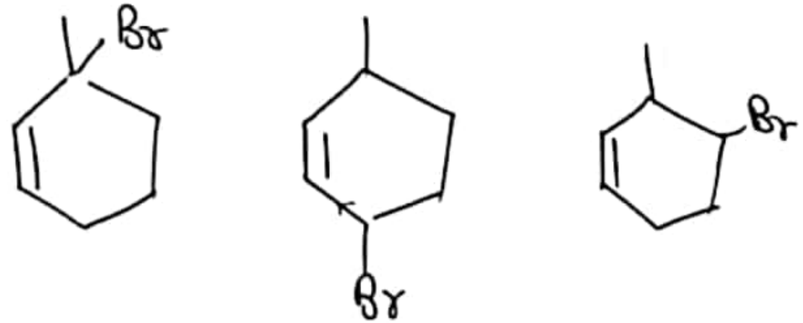
→ Reactivity of S_N1 = a > b > c (No S_N1 R_x^n)

→ " " " S_N2 = c > b > a (No S_N2 R_x^n)

② 

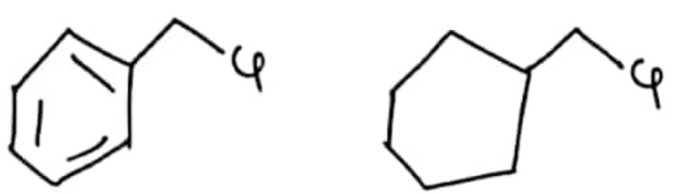
→ S_N1 → c > b > a

→ S_N2 = c > b > a

③ 

→ S_N1 → a > b > c

→ S_N2 → b > c > a a least due to steric crowding of tertiary group, despite it being an allylic position

(4) 

→ S_N1 → a > b

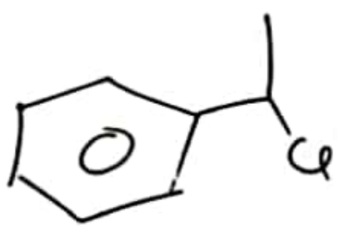
→ S_N2 → a > b

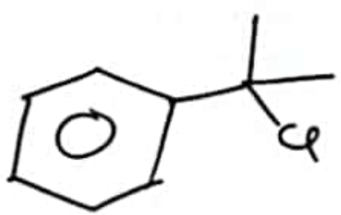
ORGANIC REACTION MECHANISM – III

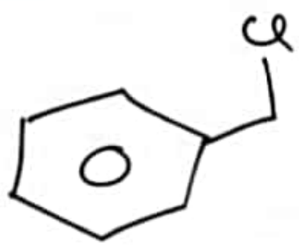
ORM-III

Give order of reactivity for S_N1 and S_N2 Rxn.

(5)





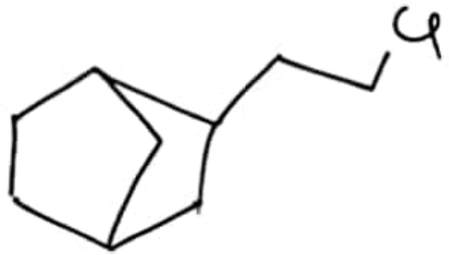


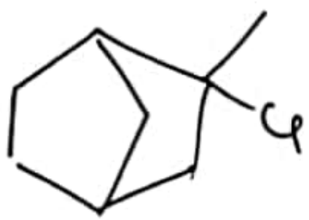
will give SN1 due to benzylic pos.

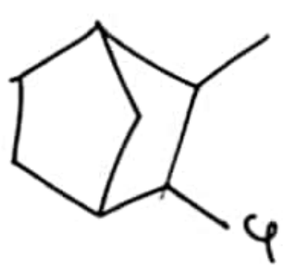
$\Rightarrow S_N1 \Rightarrow b > a > c$

$\Rightarrow S_N2 \Rightarrow c > a > b$

(6)





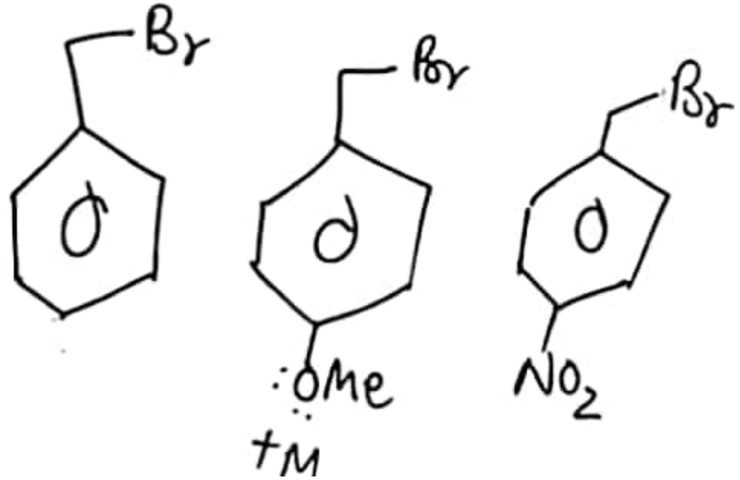




$= S_N1 \Rightarrow b > c > \underline{a > d}$ ← NO S_N1

$= S_N2 = a > c > \underline{b > d}$ ← NO S_N2

(7)



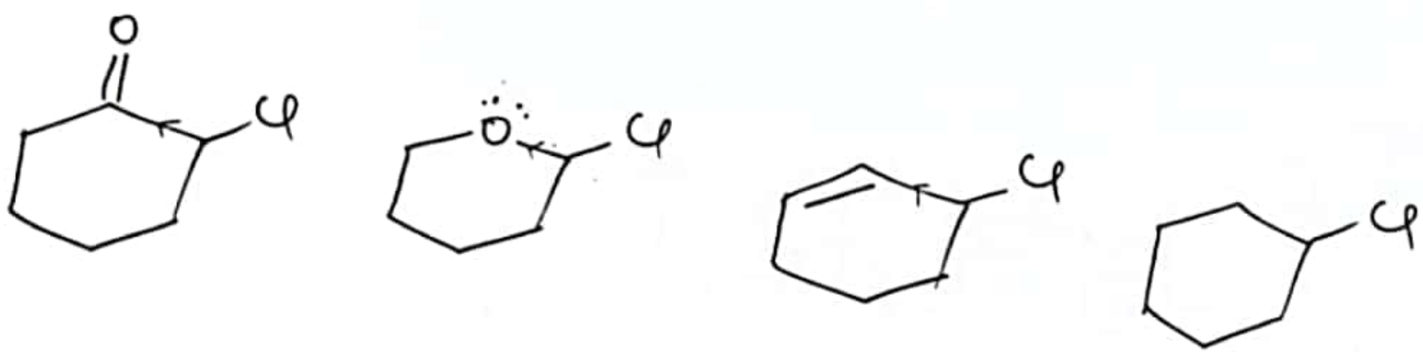
$\Rightarrow SN1 = b > a > c$

$\Rightarrow SN2 = c > a > b$

ORGANIC REACTION MECHANISM – III

Q Give order of reactivity for S_N1 and S_N2 Rxn.

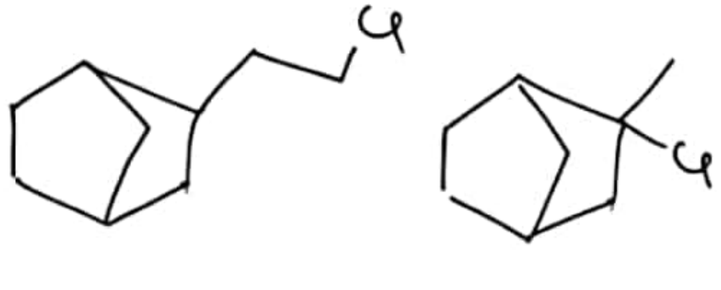
(8)

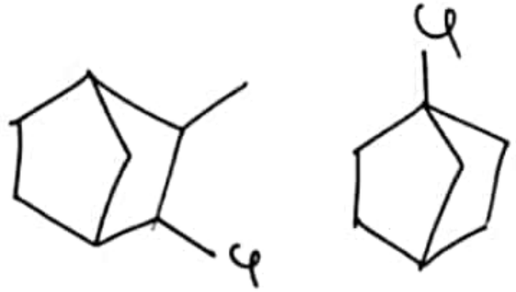


S_N1 = b > c > d > a ← No S_N1

S_N2 = a > b > c > d

(6)

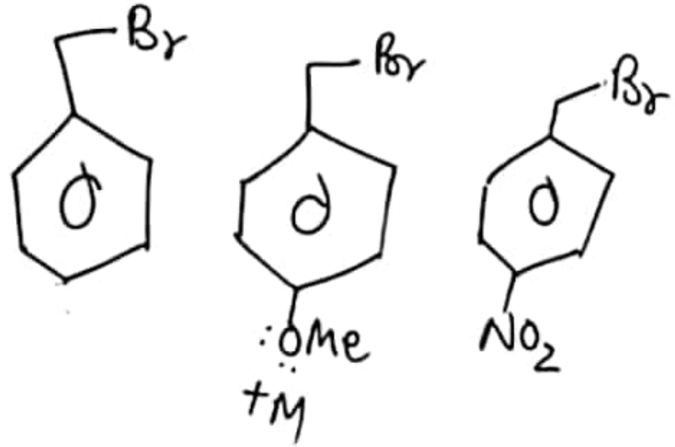




$\Rightarrow S_N1 = b > c > \underline{a > d}$ ← No S_N1

$\Rightarrow S_N2 = a > c > \underline{b > d}$ ← No S_N2

(7)



$\Rightarrow SN1 = b > a > c$

$\Rightarrow SN2 = c > a > b$

ORGANIC REACTION MECHANISM – III

ORM-III

Q How many of the following can give white ppt with $AgNO_3$ aq. solution ?

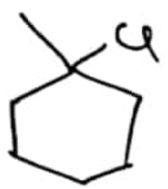
if the cation formed by the removal of Cl- is stable, then ppt. with AgNO3 will be favourable; else the rex. won't go in forward direction

AgI = yellow ppt

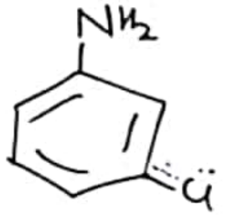
AgBr = Pale yellow ppt

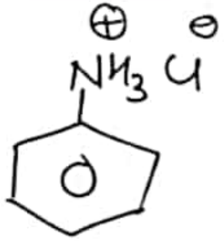
AgCl = White ppt

AgF = Soluble in water

✓ ① 

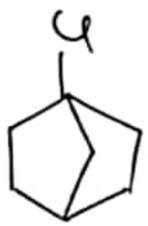
✓ ② HCl

✗ ③ 

✓ ④ 

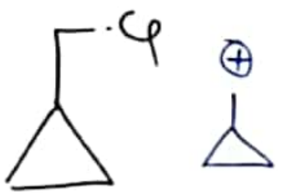
the acyl cation will be resonance stabilised

✓ ⑤ $CH_3-\overset{\overset{:O:}{\|}}{C}-Cl$

✗ ⑥ 

✓ ⑦ 

✓ ⑧ 

✓ ⑨ 

✗ ⑩ 

✗ ⑪ CH_3-Cl

✗ ⑫ 

✗ ⑬ $CH_2=CH-Cl$

Ans = 7

ORGANIC REACTION MECHANISM – III

(I) General SN reaction of alkyl halide

(II) Ammonolysis of alkyl halide.

(III) Reactions of halogen exchange in R–X

(IV) SN Rx^n of ambident nucleophile in R–X

(V) Wurtz Rx^n

(VI) S_N Rx^n of organometallic comp.

(VII) Williamson's ether synthesis (WES)

(VIII) SN Reaction of allylic position [SN' ⟨ SN'' / SN²']

(IX) Intramolecular SN^2 Rx^n

(X) SN Rx^n of R–OH

(XI) " " " ethers

(XII) " " " epoxides

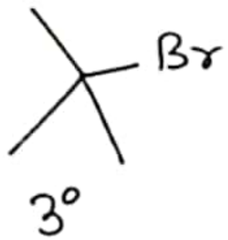
(XIII) " " Ar–X
(ArSN)

ORGANIC REACTION MECHANISM – III

ORM-III

(I) General SN Reaction of Alkyl halides:

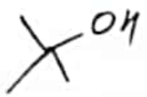
eg 1



$H_2\ddot{O}$, Δ SN'



H_2O + Acetone SN'



$H_2O + AgNO_3$ SN'

or $AgNO_{3aq.}$

" +

Ag^+, Hg^{+2}

in presence of these 2 metal ions, alkyl halide gives SN1 rex. easily (due to precipitation)

AgOH SN'

"

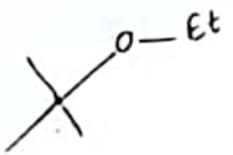
Ag_2O(moist) SN'

"

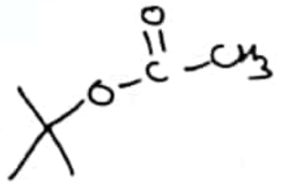
despite being a strong Nu- , it cannot make the tertiary undergo SN2

☆ KOH + H_2O SN'

EtOH SN'

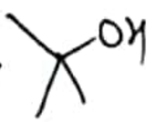


CH_3–COOH SN'

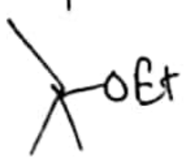


EtOH + H_2O

S_N1

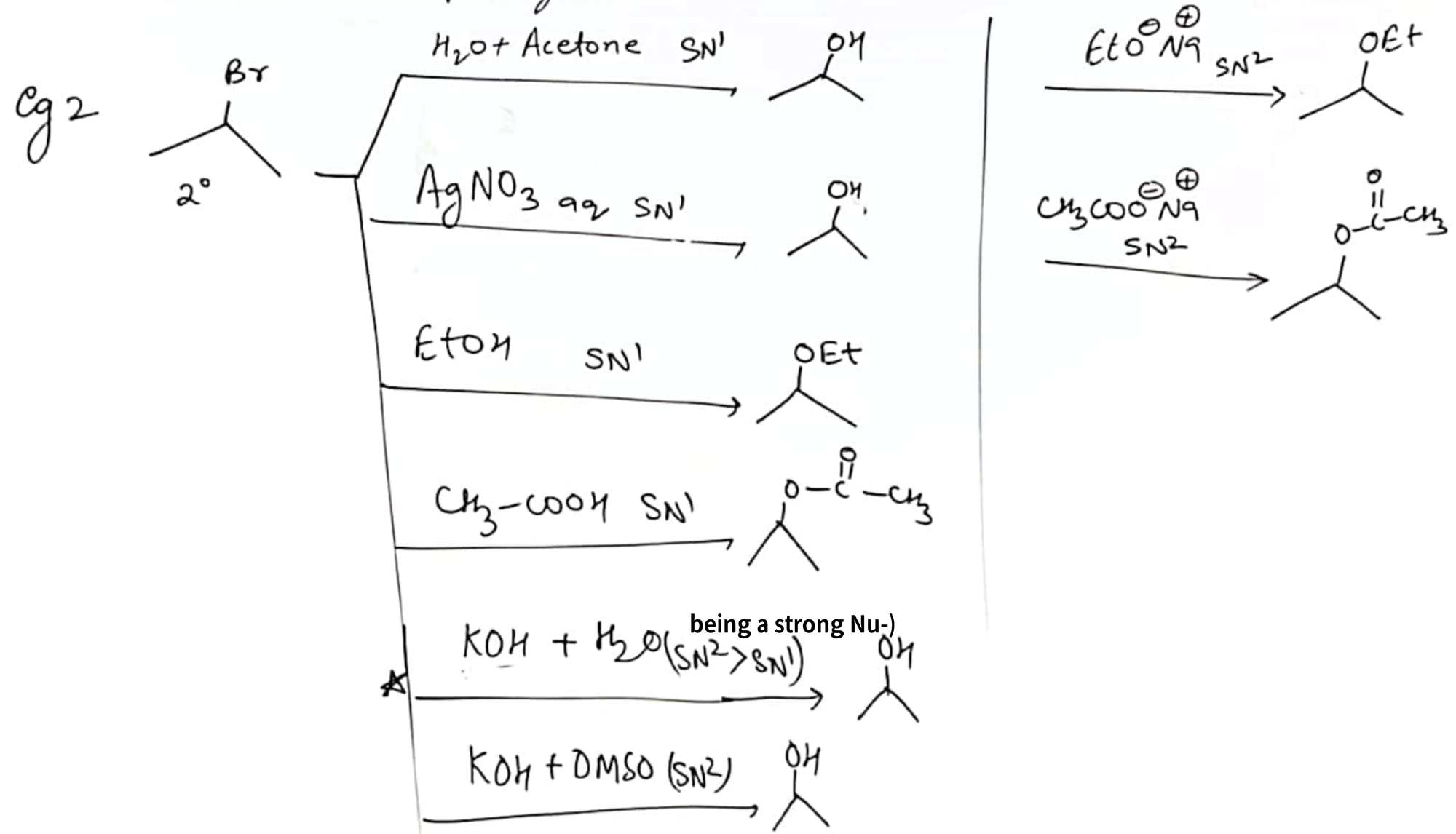


+



ORGANIC REACTION MECHANISM – III

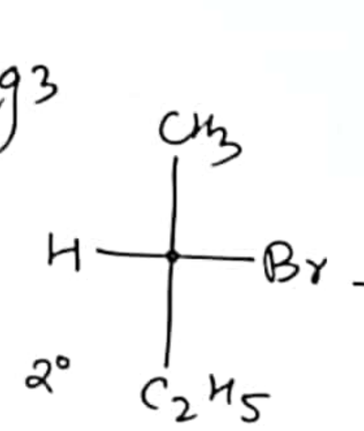
(I) General SN Reaction of Alkyl halides:



ORGANIC REACTION MECHANISM – III

(I) General SN Reaction of Alkyl halides:

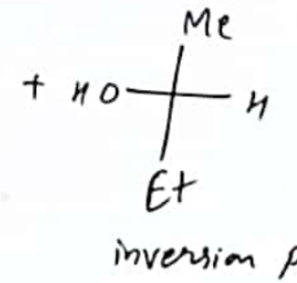
Eg 3



H_2O + Acetone
(SN1)



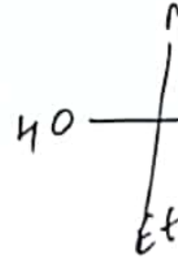
Retention



} Enantiomers.

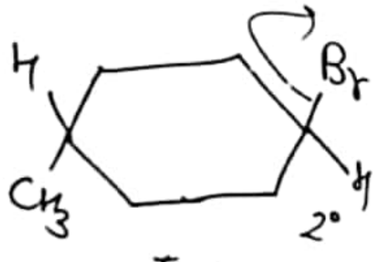
inversion product

KOH + DMSO
SN^2



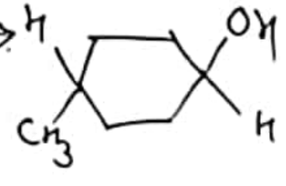
only inversion product

Eg 4

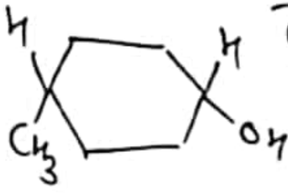


Trans

$AgNO_3 + H_2O$
(SN1)

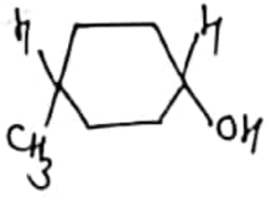


+



} mix. of geo. iso. (diastereomers)

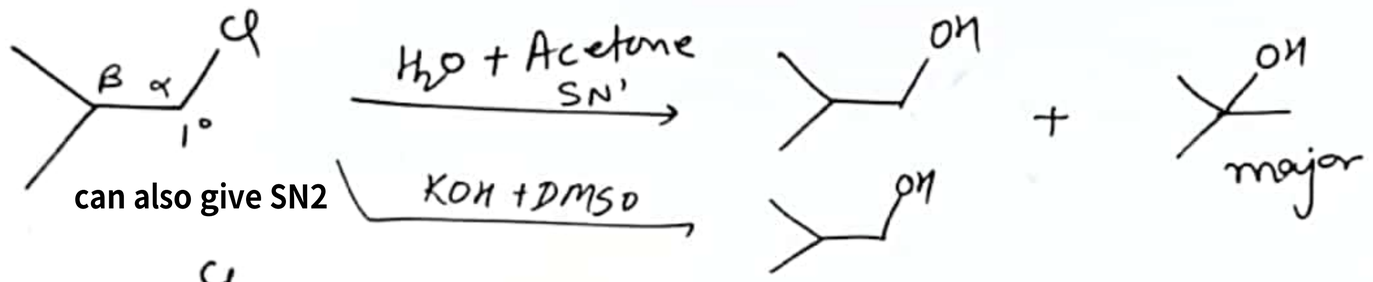
KOH + DMSO
SN2



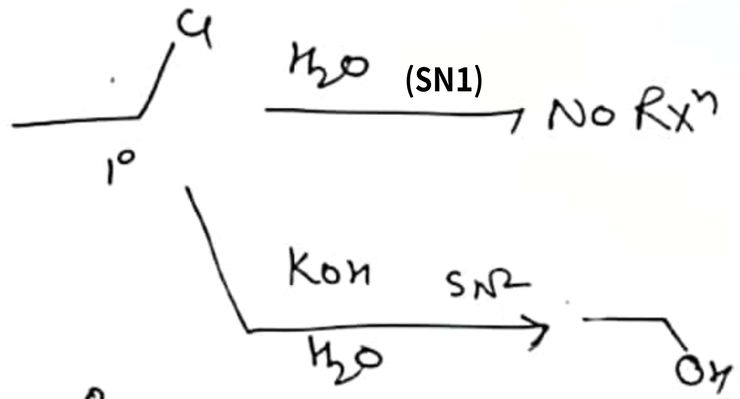
ORGANIC REACTION MECHANISM – III

(I) General SN Reaction of Alkyl halides:

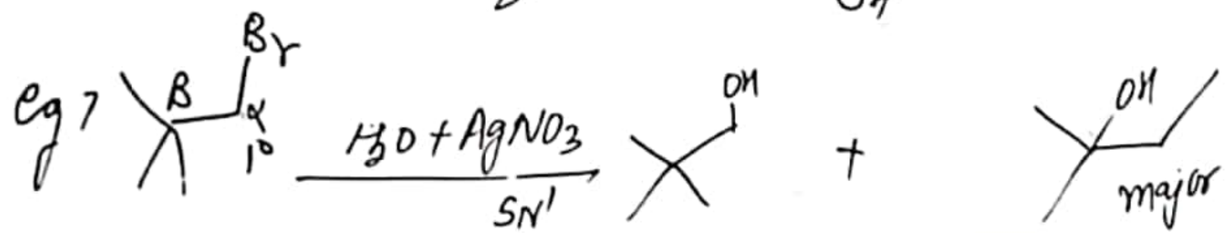
eg 5



eg 6

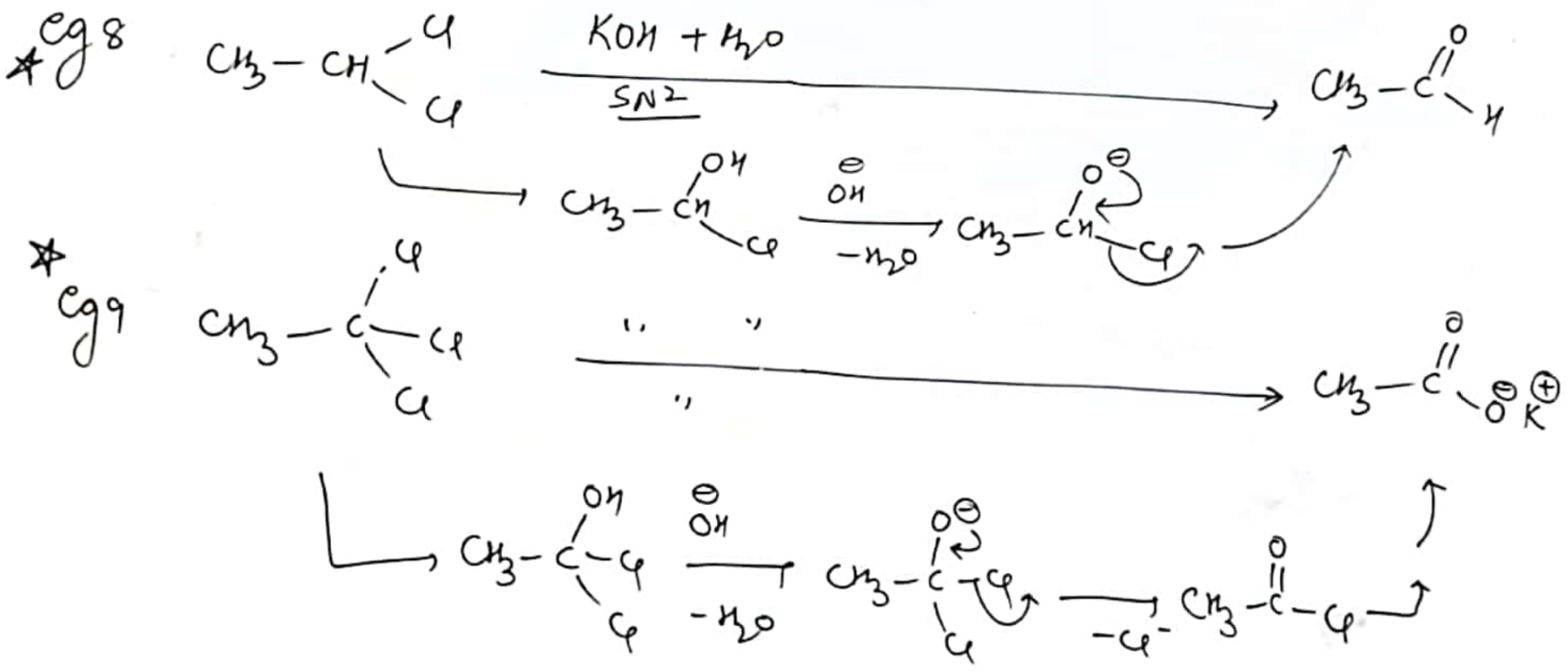


eg 7



ORGANIC REACTION MECHANISM – III

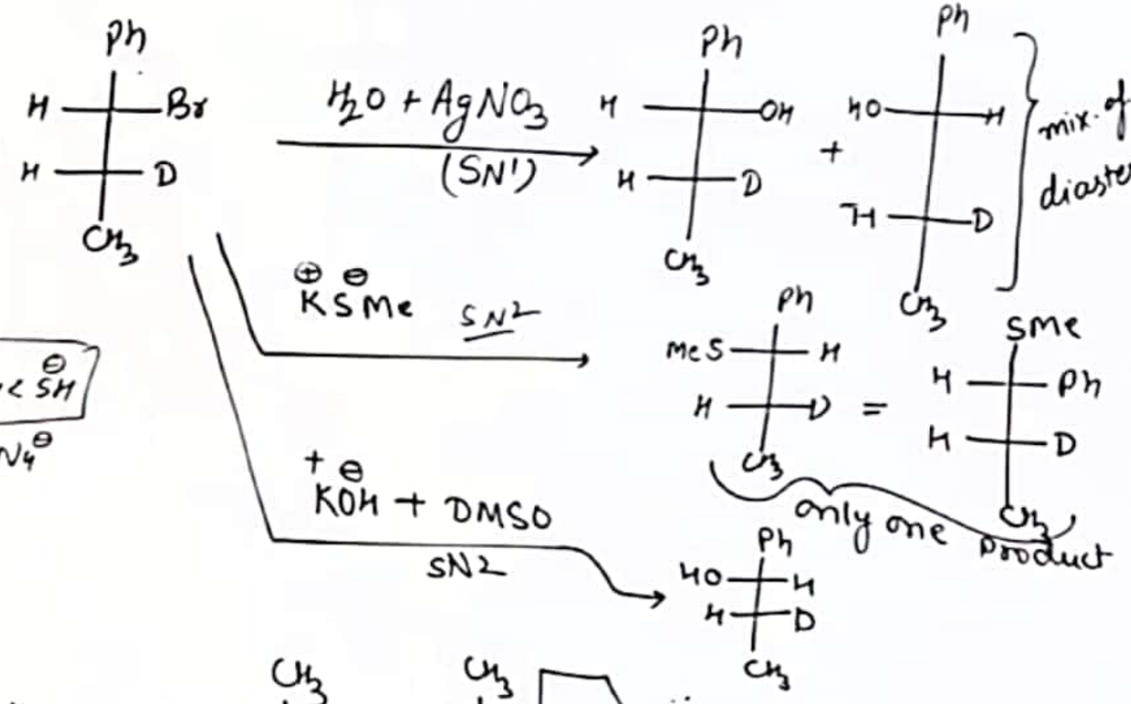
(I) General SN Reaction of Alkyl halides:



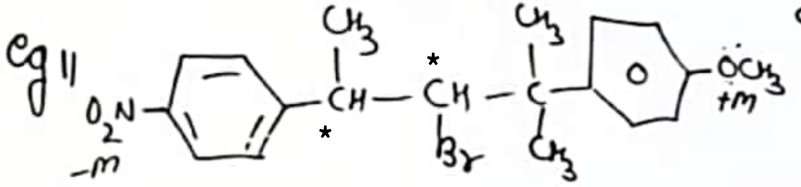
ORGANIC REACTION MECHANISM - III (ORM - III)

(I) General S_N Reactions :-

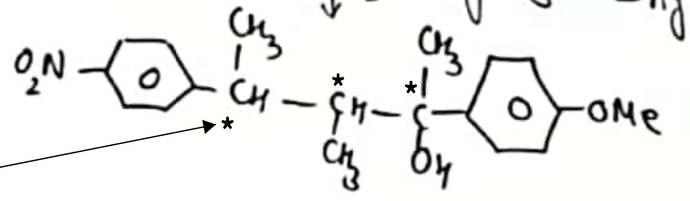
Eg 10



Eg 11



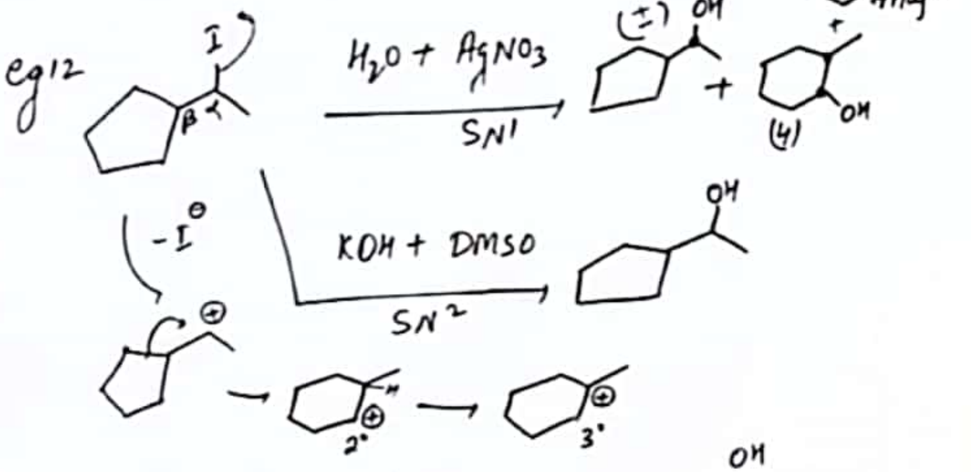
SN1 | $H_2O + AgNO_3$ $-AgBr\downarrow$



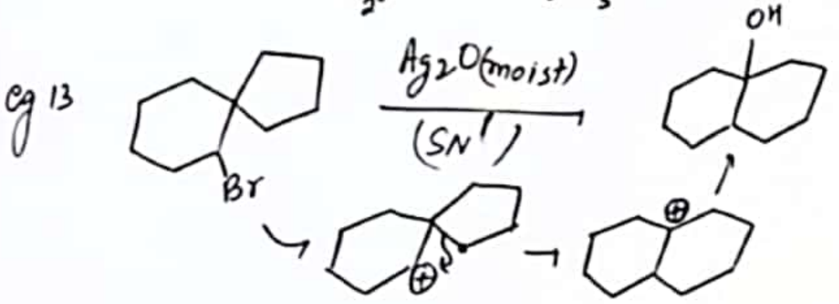
configuration is fixed here

total 4 products

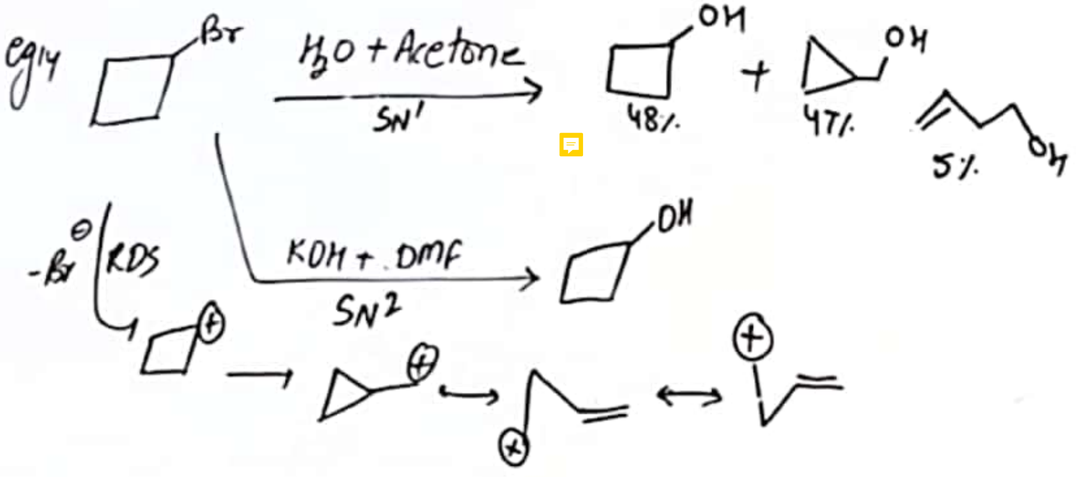
Eg 12



Eg 13

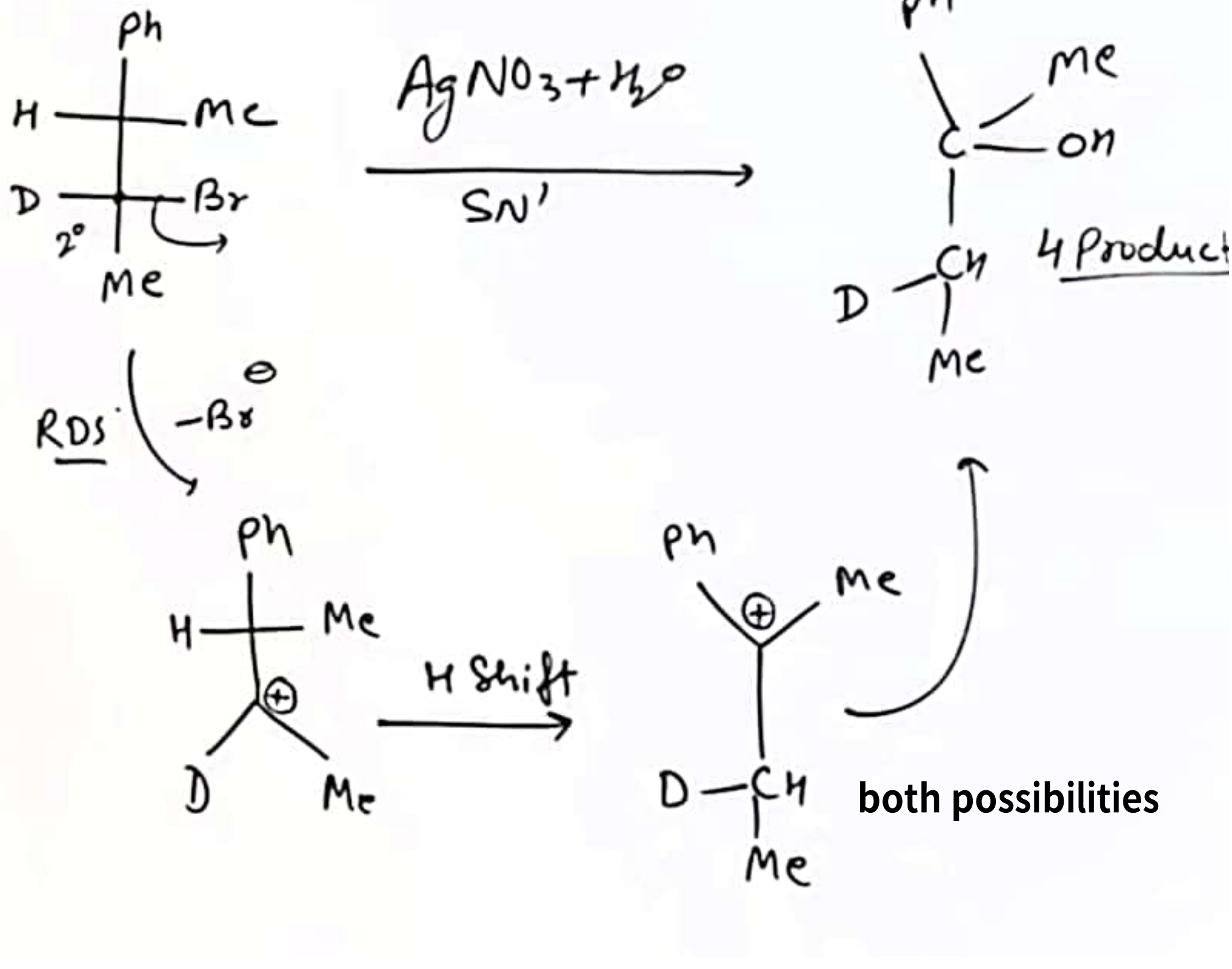


Eg 14



(I) General S_N Reactions :—

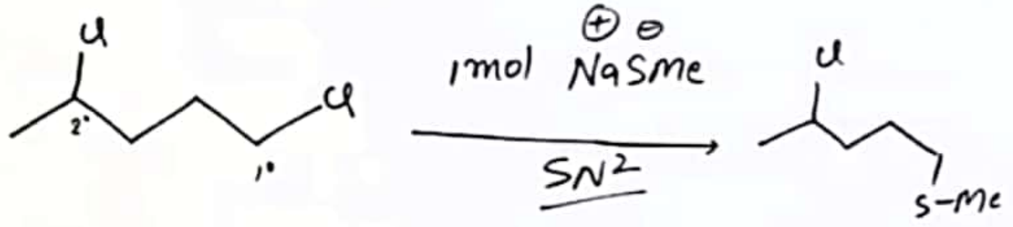
eg 15



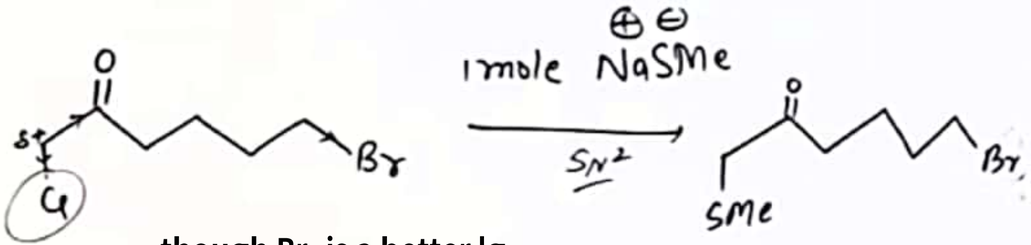
ORGANIC REACTION MECHANISM - III (ORM - III)

(I) General S_N Reactions :-

eg 18

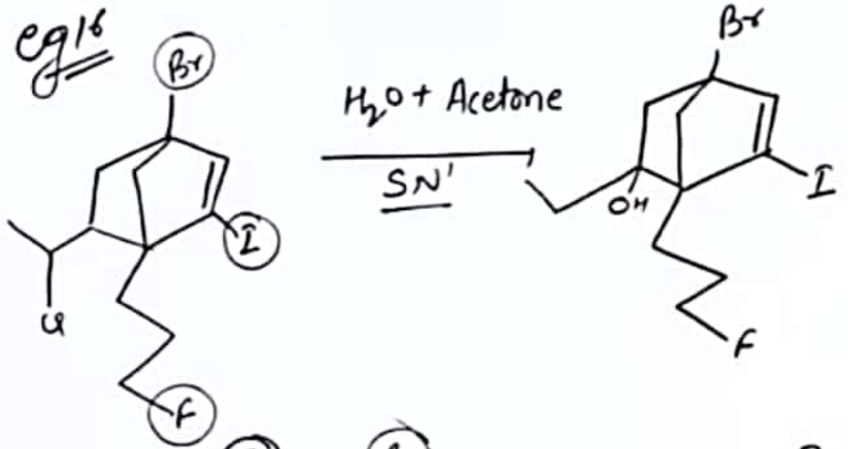


eg 19

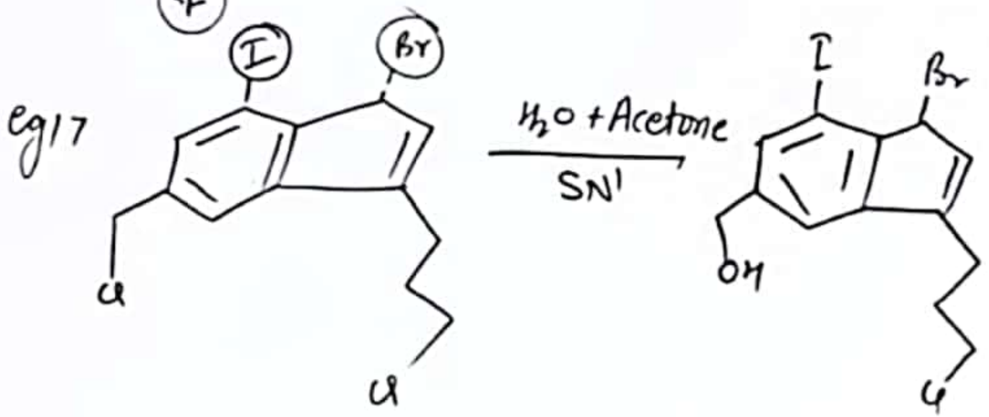


though Br- is a better lg
yet, Cl- is not that a bad leaving group and we have an EWG in adjacent position; hence it is more favoured

eg 16



eg 17



simple primary; no SN1

ORGANIC REACTION MECHANISM - III (ORM - III)

(I) General S_N Reactions :-

eg 20

CH_3-Br

1°

① H_2O + Acetone → No Rx^n

② EtOH → No Rx^n

③ KOH + H_2O S_N2 → CH_3-OH

④ KOH + DMSO (S_N2) → CH_3-OH

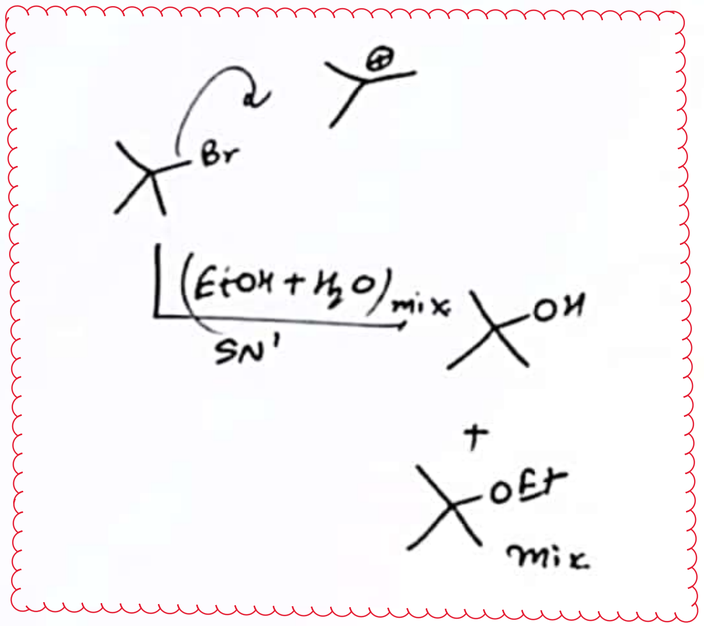
⑤ $EtO^{\ominus}Na^{\oplus}$ S_N2 → $CH_3-O-C_2H_5$

$EtO^{\ominus} > Ph-O^{\ominus}$

Nucleoph.

⑥ $Ph-O^{\ominus}Na^{\oplus}$ S_N2 → CH_3-O-Ph

⑦ $(EtO^{\ominus} + Ph-O^{\ominus})_{mix}$ → CH_3-O-Et (S_N2)



ORGANIC REACTION MECHANISM - III (ORM - III)

generally, acid-base rxn. > intramolecular > intermolecular

(I) General S_N Reactions :-

eg 20

CH_3-Br (1°)

① H_2O + Acetone → No Rx^n

② EtOH → No Rx^n

③ KOH + H_2O S_N2 → CH_3-OH

④ KOH + DMSO (S_N2) → CH_3-OH

⑤ $EtO^{\ominus}Na^{\oplus}$ S_N2 → $CH_3-O-C_2H_5$

$EtO^{\ominus} > Ph\text{-}O^{\ominus}$

Nucleoph.

⑥ $Ph\text{-}O^{\ominus}Na^{\oplus}$ S_N2 → CH_3-O-Ph

7 $(EtO^{\ominus} + Ph\text{-}O^{\ominus})_{mix.}$ S_N2 → CH_3-O-Et

WES

⑧ EtOH + Base → CH_3-OEt (S_N2; $EtO^{\ominus}$)

⑨ $(EtOH + Na)_{mix}$ → $EtO^{\ominus}Na^{\oplus} + H_2$,,

⑩ $Ph-\ddot{O}H$ → No Rx^n

⑪ $Ph-OH + Na$ → CH_3-OPh ($Ph\text{-}O^{\ominus}$)

⑫ $Ph-OH$ (1 mol) + $EtO^{\ominus}Na$ (1 mol) → CH_3-O-Ph ($Ph\text{-}O^{\ominus} + EtOH$)

⑬ $Ph-OH + EtO^{\ominus}_{excess}$ → CH_3-O-Et ($Ph\text{-}O^{\ominus} + EtOH + EtO^{\ominus}$)

Williamson's ether Synthesis.

ORGANIC REACTION MECHANISM - III (ORM - III)

(I) General S_N Reactions :-

eg 20 CH_3-Br (1°)

(14) $Na^{+}\,{}^{\ominus}SH$ S_N2 → CH_3-SH thioalcohol

(15) Na_2S^{-2} S_N2 → CH_3-S-CH_3 thioether

(16) $MeSH + Na$ S_N2 → CH_3-S-CH_3 ; $MeSH + Na \rightarrow Me-S^{\ominus}$

(17) NaN_3 S_N2 → CH_3-N_3

(18) $NaCN$ S_N2 → CH_3-CN

(19) $NaNH_2$ → CH_3-NH_2

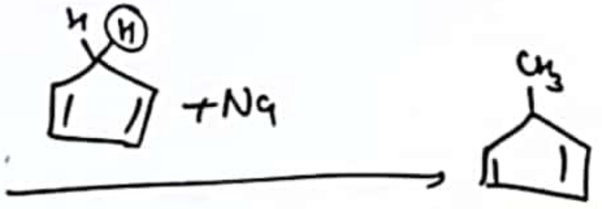
(20) CH_3Na or CH_3Li → CH_3-CH_3

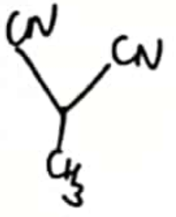
(21) $R\,MgX$ S_N2 → $R-CH_3 + Mg(X)Br$

22 R_2CuLi S_N2 → CH_3-R Corey-House Synthesis of alkane

(23) $R-C\equiv C^{\ominus}\,Na^{\oplus}$ S_N2 → $R-C\equiv C-CH_3$

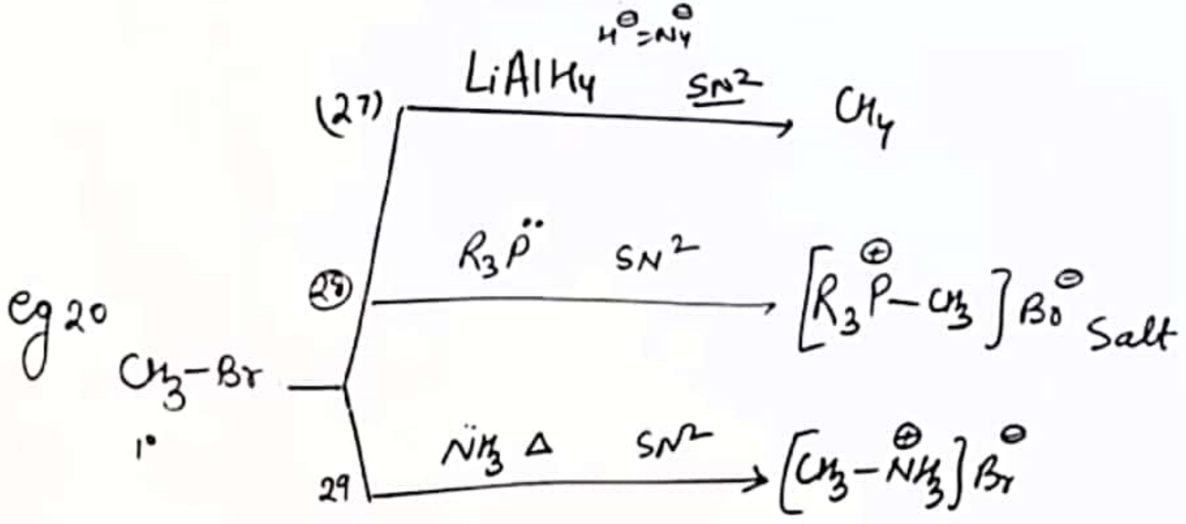
(24) $R-C\equiv C-H$ (C is sp; H is acid) $+ Na$ (S_N2) → $R-C\equiv C-CH_3$; $R-C\equiv C^{\ominus}\,Na^{\oplus}$

(25)  $+ Na$ →

(26)  $+ NaOH$ → 

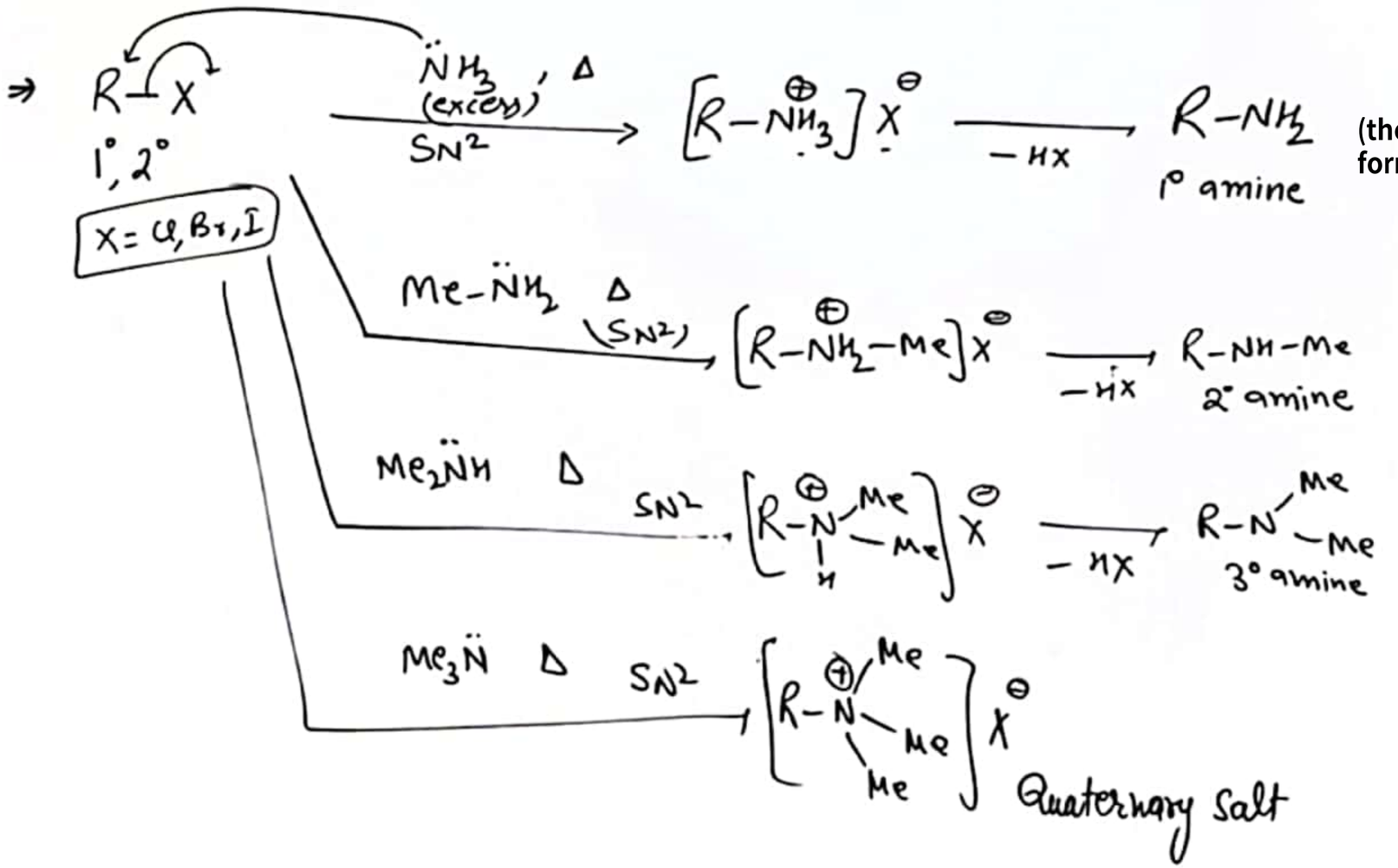
ORGANIC REACTION MECHANISM - III (ORM - III)

(I) General S_N Reactions :-



ORGANIC REACTION MECHANISM - III (ORM - III)

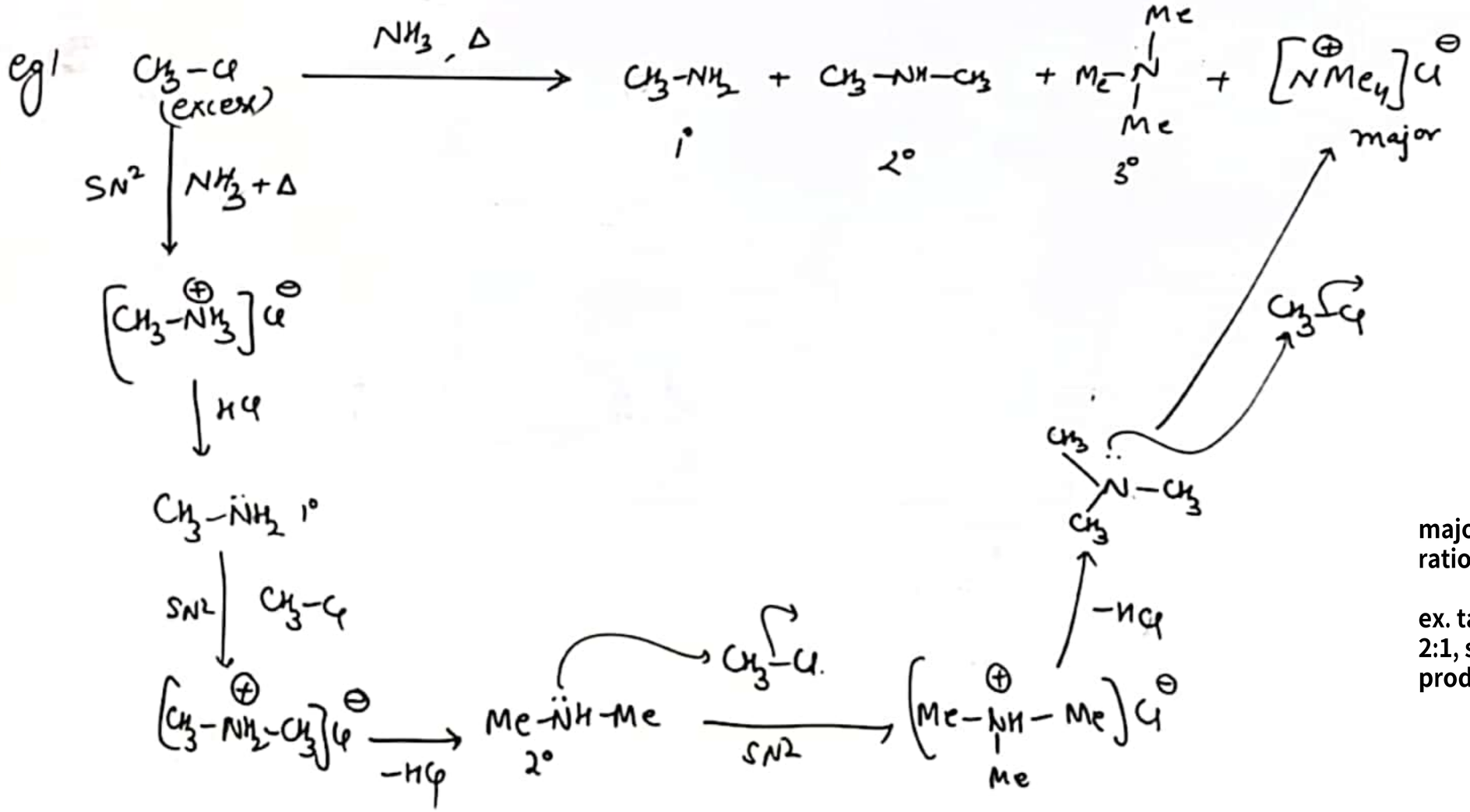
(II) Ammonolysis of alkyl halide :- (S_N2)



(the formed HX reacts with NH3 forming a salt)

ORGANIC REACTION MECHANISM - III (ORM - III)

(II) Ammonolysis of alkyl halide :- (SN^2)

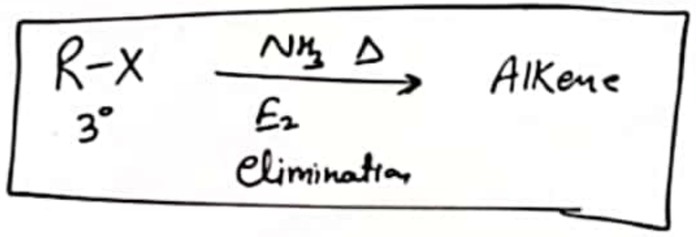


major product depends on the ratio of RX and NH3 taken

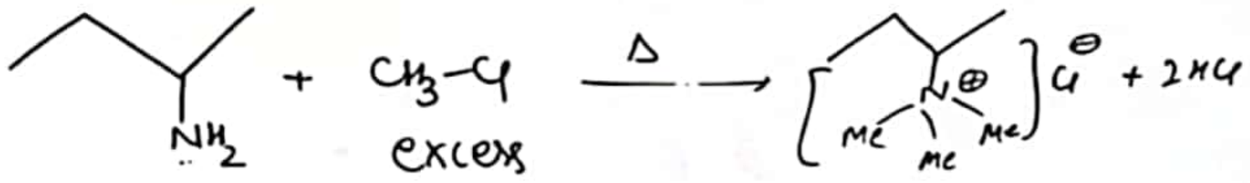
ex. taking RX and NH3 in the ratio 2:1, secondary amine will be major product

ORGANIC REACTION MECHANISM - III (ORM - III)

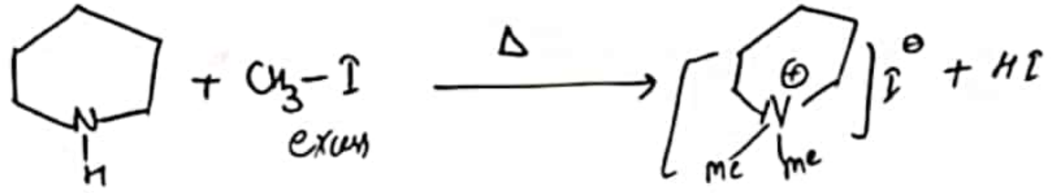
(II) Ammonolysis of alkyl halide :- (SN^2)



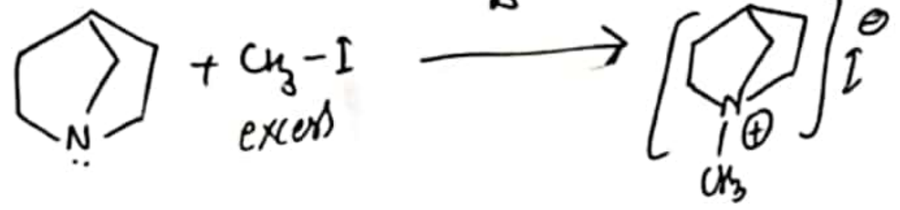
eg 2



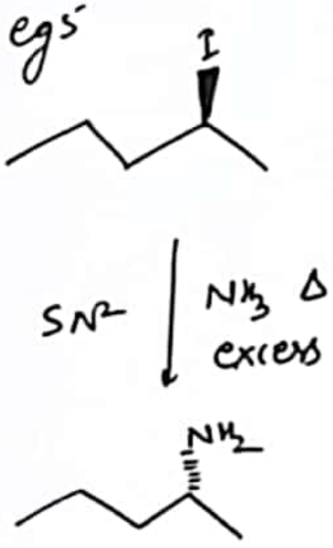
eg 3



eg 4



eg 5



ORGANIC REACTION MECHANISM - III (ORM - III)

(III) S_N Reactions of halogen exchange:—

the driving force behind these reactions is mainly the precipitation of the salts

$\Rightarrow$ $R\overset{\delta+}{-}X$ ($X = Cl, Br$) $\xrightarrow[\text{(Finkelstein Reaction)}]{NaI + \text{Acetone}\ \ S_N2}$ $R-I + NaX\downarrow$ (Ppt)

- NaI soluble in acetone
- NaCl insoluble in acetone
- NaBr insoluble " "

$F^{\ominus}\cdots\overset{\delta+}{R}-\overset{\delta-}{X}$ ($X = Cl, Br, I$) $\xrightarrow[\text{(Swartz Reaction)}]{AgF_{aq.} \,/\, Hg_2F_2 \,/\, CoF_2}$ $R-F + AgX\downarrow$ (Ppt)

- AgF soluble in H_2O
- AgCl insoluble in H_2O
- AgBr " "
- AgI " "

Nucleophilicity $F^{\ominus} < Cl^{\ominus} < Br^{\ominus} < I^{\ominus}$ in polar protic solvent

$R-X \xrightarrow{NaF_{aq.}}$ ND $R_X{}^n$ (weak $N^{\ominus}$ in H_2O)

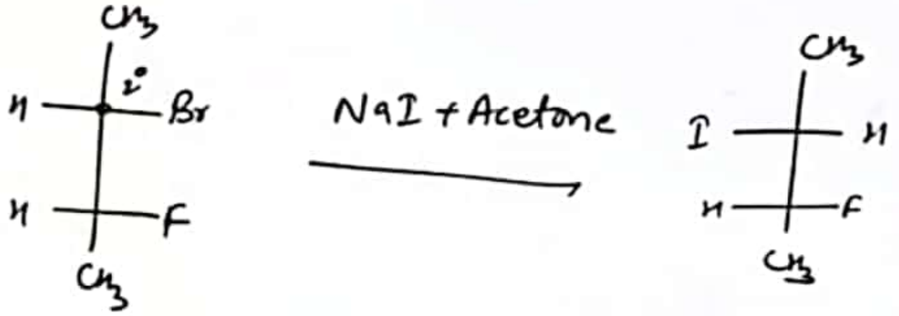
$R-X \xrightarrow[S_N2]{NaF + DMF}$ $R-F$ (strong $N^{\ominus}$)

Nucleophilicity in polar aprotic solvent $\Rightarrow F^{\ominus} > Cl^{\ominus} > Br^{\ominus} > I^{\ominus}$

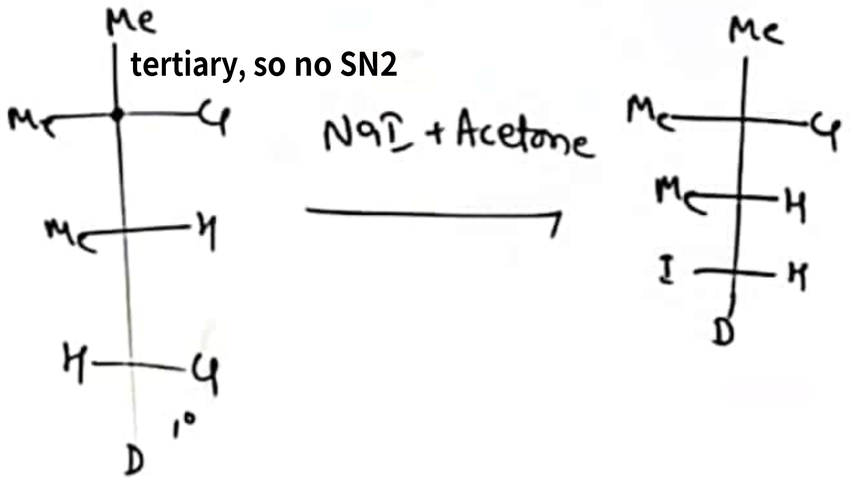
ORGANIC REACTION MECHANISM - III (ORM - III)

(III) S_N Reactions of halogen exchange :—

eg 1

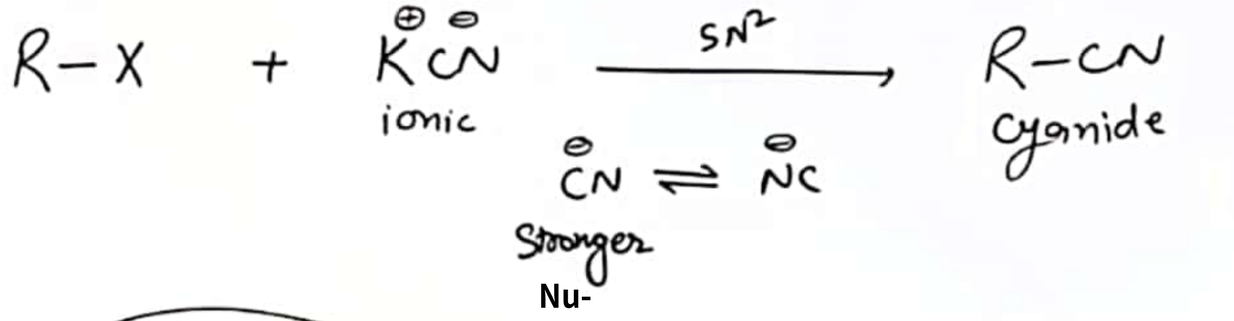


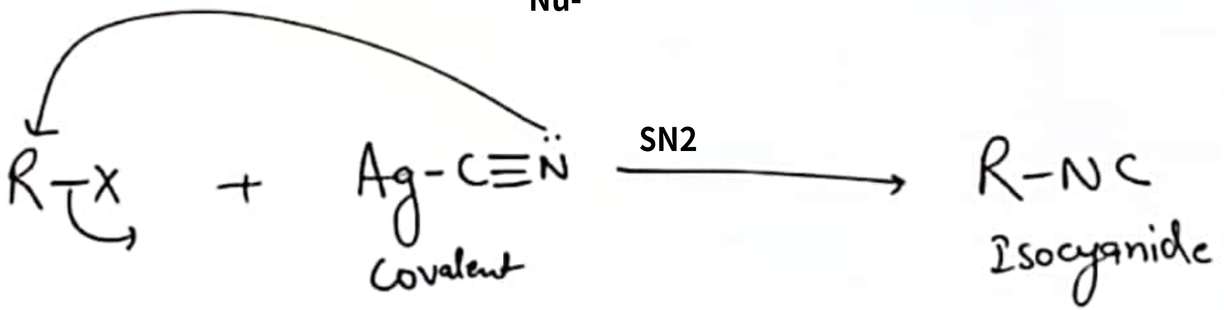
eg 2



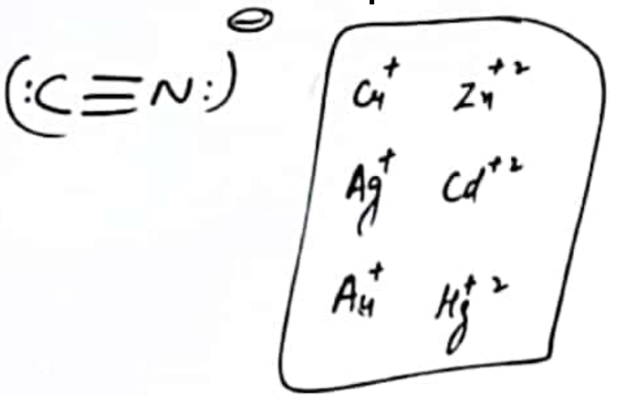
ORGANIC REACTION MECHANISM - III (ORM - III)

(IV) SN Reaction of ambident nucleophiles



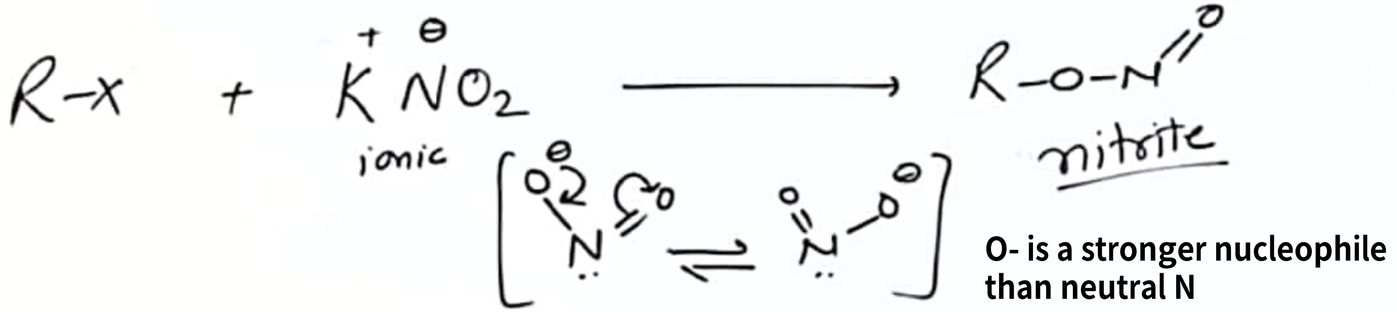


pseudo-inert gas configuration (high covalent character in compounds of these ions)

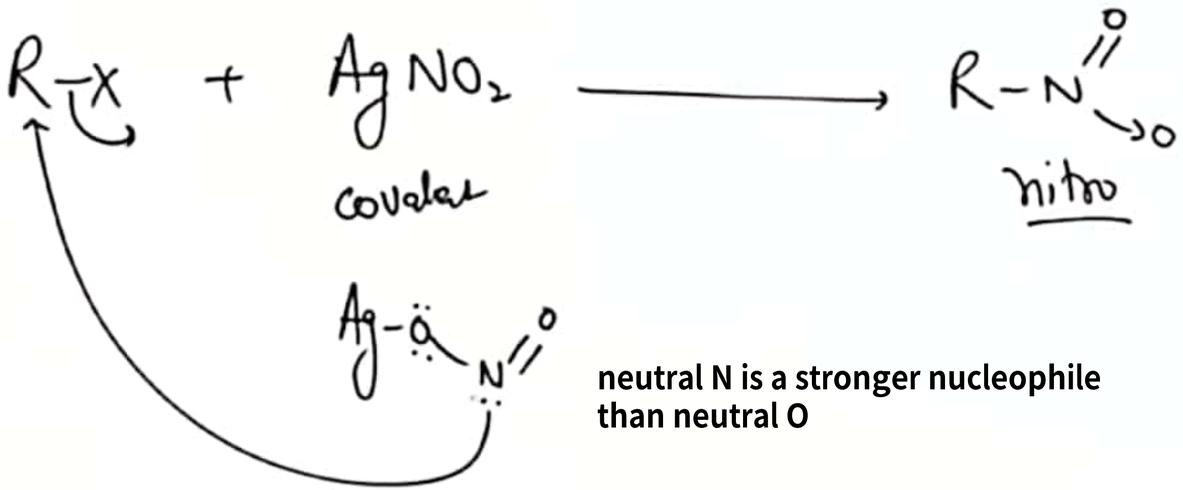


ORGANIC REACTION MECHANISM - III (ORM - III)

(IV) SN Reaction of ambident nucleophiles



O- is a stronger nucleophile than neutral N



neutral N is a stronger nucleophile than neutral O

ORGANIC REACTION MECHANISM - III (ORM-III)

(V) **Wurtz Reaction:-**

$$R-X + 2Na + X-R \xrightarrow[\text{solvent}]{\text{dry ether}} R-R + 2NaX$$

R–R : alkane

mechanism

(a) Ionic mechanism (S_N2)

$$2Na \longrightarrow 2Na^{\oplus} + 2e^{\ominus}$$

$$R-X + 2e^- \longrightarrow R^{\ominus} + X^{\ominus}$$

$$R^{\ominus} + R-X \xrightarrow{S_N2} R-R + X^{\ominus}$$

Strong Nucleophile

$1^\circ > 2^\circ > 3^\circ$ probability of SN2

(b) Free-Radical mechanism

$$\cdot Na \longrightarrow Na^+ + e^-$$

$$R-X + e^- \longrightarrow R^\cdot + X^{\ominus}$$

$$R^\cdot + R^\cdot \longrightarrow R-R$$

$3^\circ > 2^\circ > 1^\circ$ probability of free radical mechanism

ORGANIC REACTION MECHANISM - III (ORM-III)

(V) Wurtz Reaction:-

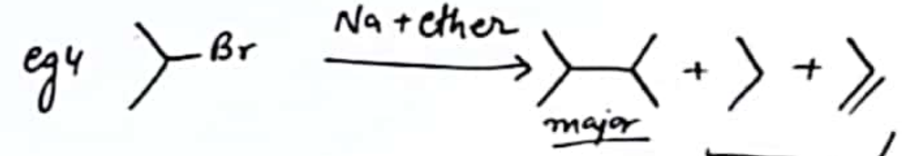
eg 1 $CH_3-Cl \xrightarrow{Na + ether} CH_3-CH_3$

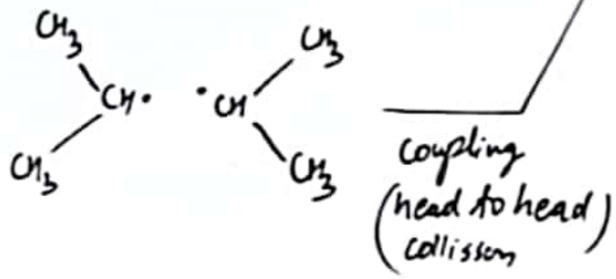
eg 2 $C_2H_5-Cl \xrightarrow{Na + ether} C_2H_5-C_2H_5$

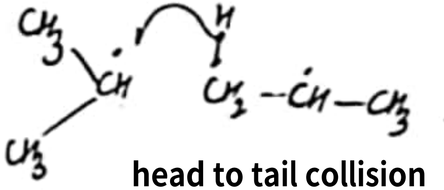
eg 3 $[CH_3-Cl + C_2H_5-Cl]_{mix} \xrightarrow{Na + Et_2O}$ CH_3-CH_3 + $C_2H_5-C_2H_5$ + $CH_3-CH_2-CH_3$

$R-X + R'-X \xrightarrow{Na+ether}$ $R-R$, $R-R'$, $R'-R'$

thus, alkanes of the type R-R' cannot be formed exclusively using Wurtz reaction

eg 4 

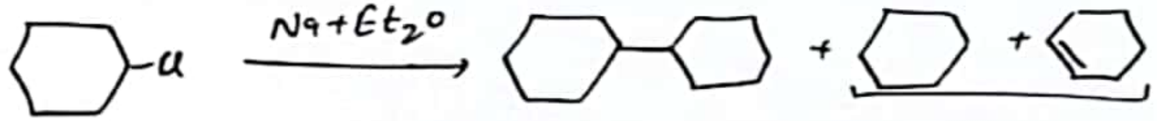


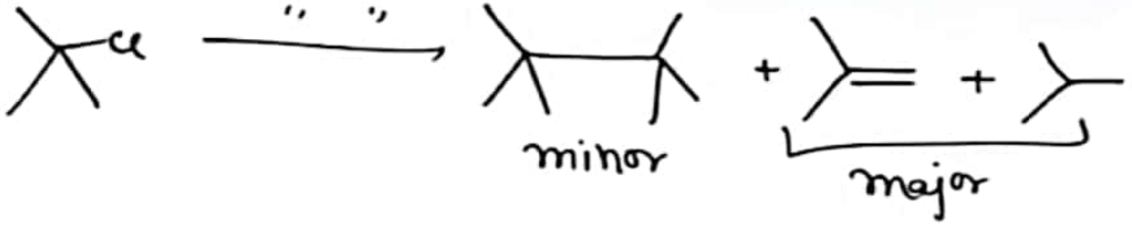


Note for 3° alkyl halide in wurtz Rx^n ⇓ Major product (disproportionation)

ORGANIC REACTION MECHANISM - III (ORM-III)

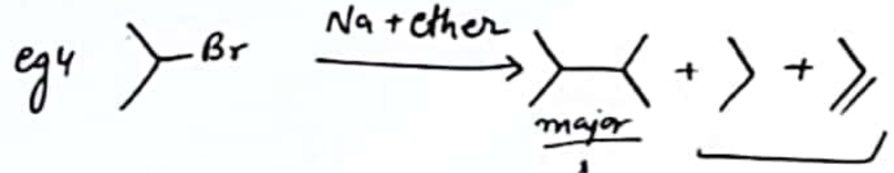
(V) Wurtz Reaction:-

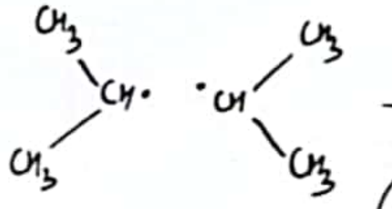
eg 5 

eg 6 

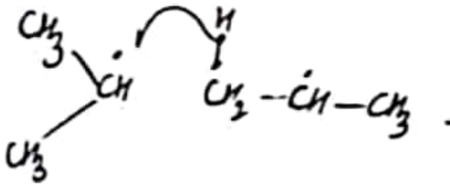
eg 7 $Ph-CH_2-Cl \xrightarrow{Na,\ Et_2O} Ph-CH_2-CH_2-Ph$

1,2 - Diphenyl ethane

eg 4 



Coupling (head to head collision)



Disproportionation.

Note for 3° alkyl halide in wurtz Rxn

⇓

Major product (disproportionation)

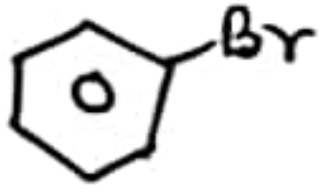
(V) Wurtz Reaction:-

Wurtz- Fittig Rxn

$$Ar-X + R-X \xrightarrow{Na + ether} Ar-R$$

Fittig Rxn

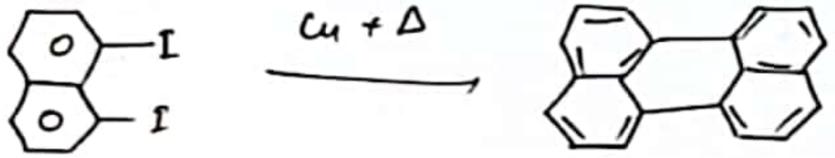
$$Ar-X \xrightarrow{Na + ether} Ar-Ar$$

Eg  $\xrightarrow{Na + ether}$ 

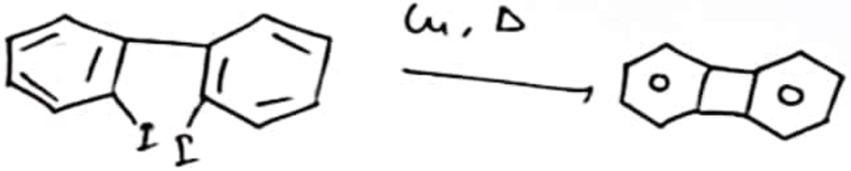
Biphenyl

ORGANIC REACTION MECHANISM - III (ORM-III)

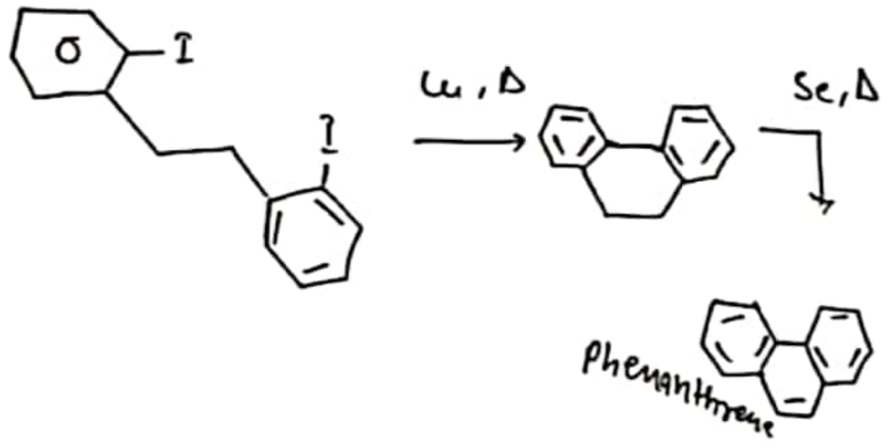
eg 4



eg 5



eg 6



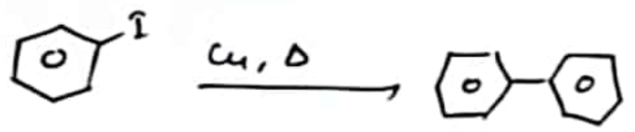
Ullamann's Reaction:-

$$2Ar\text{-}I \xrightarrow[-CuI]{Cu\ \Delta} Ar\text{-}Ar$$

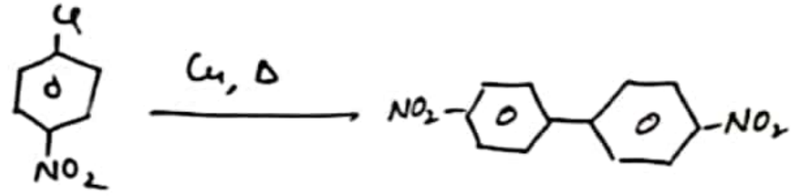
Radical mech.

rex. with Iodides takes place easily
for Chlorides and Bromides, EWG must be present at o,p on the ring

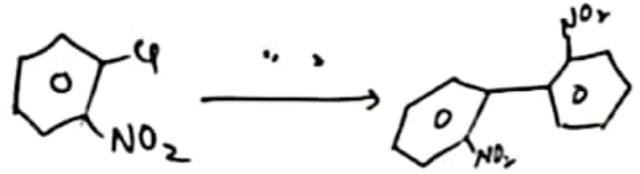
eg 1



eg 2

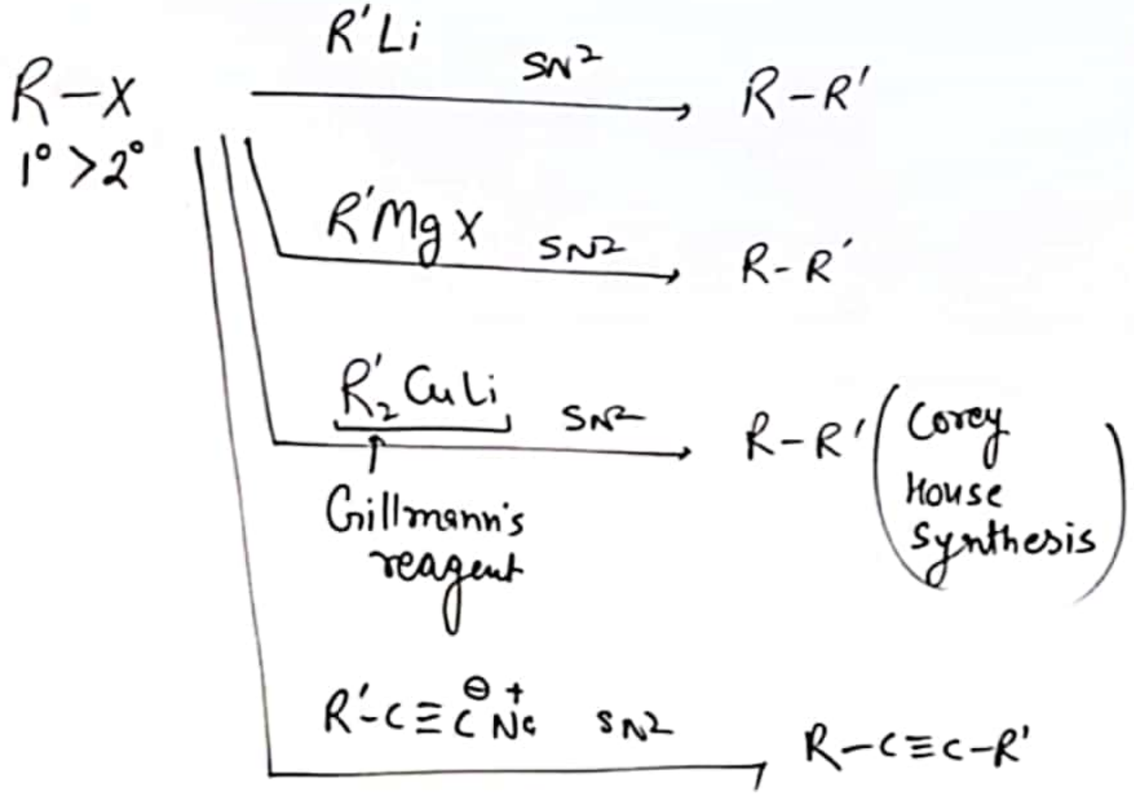


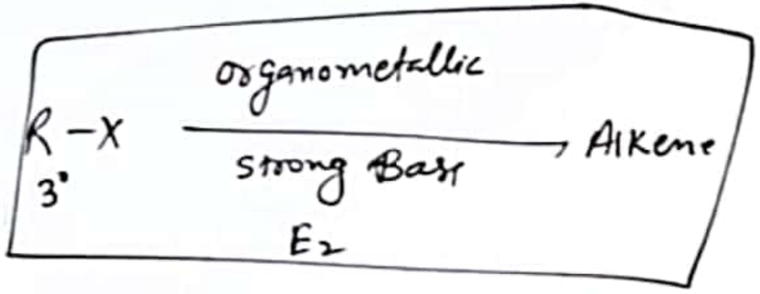
eg 3

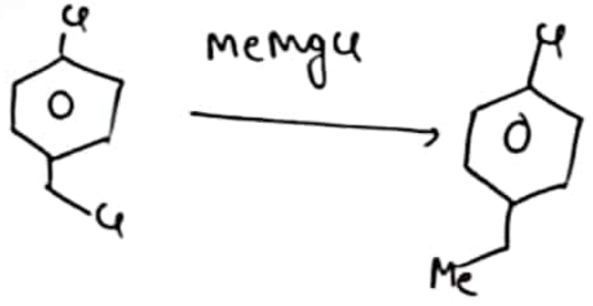


ORGANIC REACTION MECHANISM - III (ORM-III)

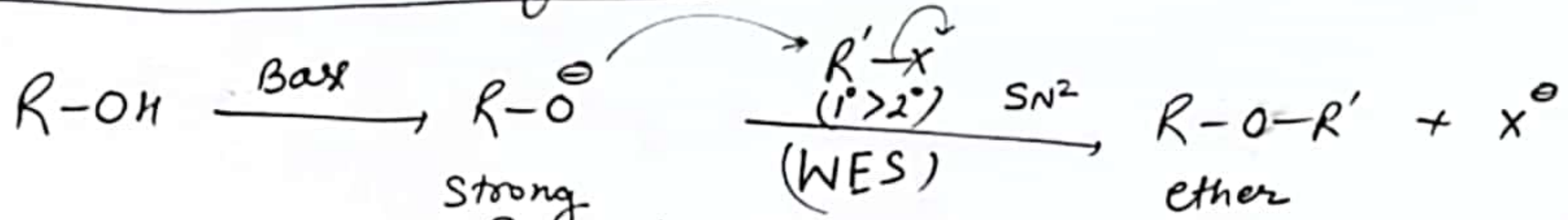
(VI) S_N2 Reaction of organometallic compound.

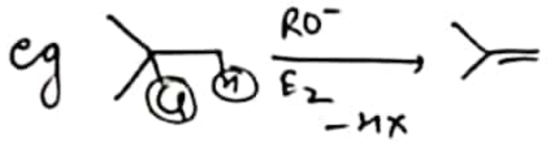




eg 

(VII) Williumson's Ether Synthesis:- (WES)



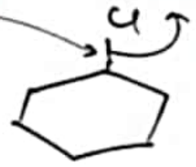
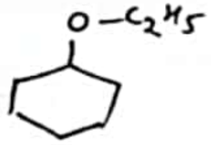
eg 

R-X 1°, 2°	$R'O^{\ominus}$ / NH_3, Δ / $R'MgX$ / R'_2CuLi / $LiAlH_4$ (SN2 Reaction)	R-O-R' / R-NH₂ / R-R' / R-R' / R-H
R-X 3°	" / " / " / ; / ,, E2 = Elimination (-HX)	Alkene

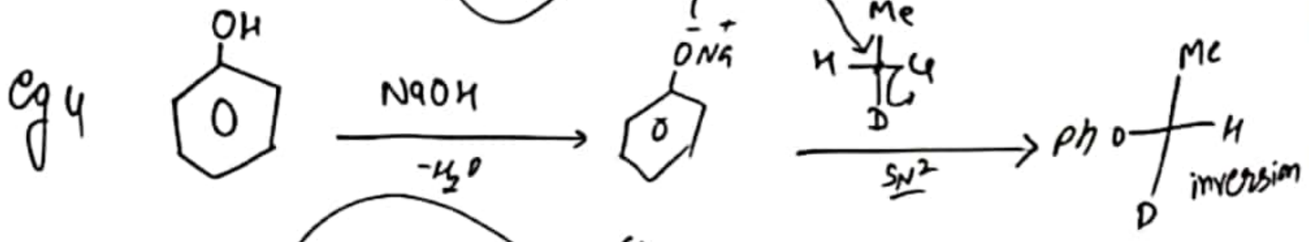
ORGANIC REACTION MECHANISM - III (ORM-III)

(VII) Williumson's Ether Synthesis:- (WES) (S_N2)

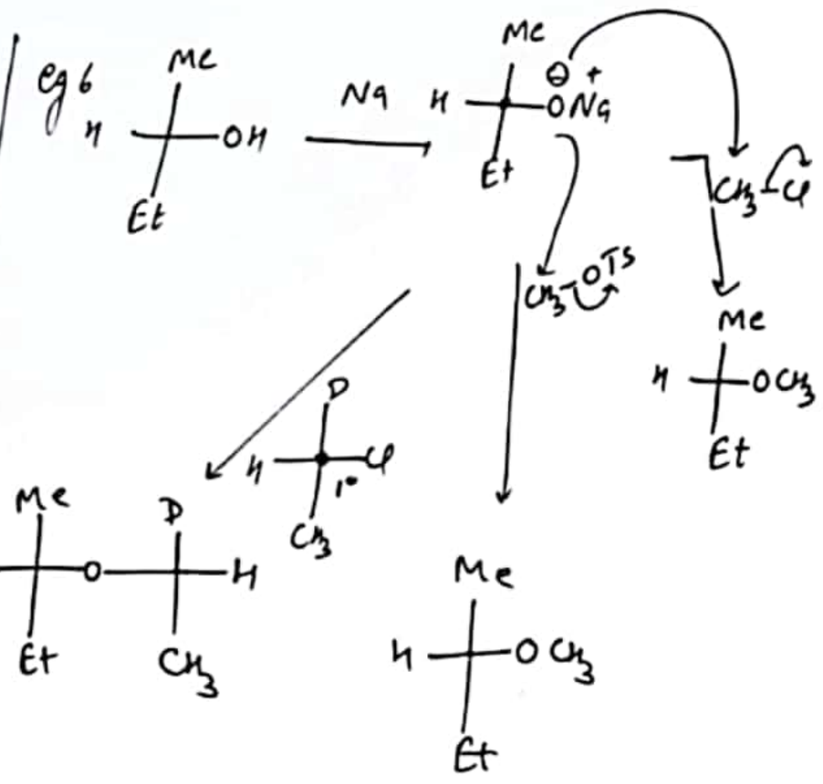
eg 1 $CH_3-\overset{\ominus}{O}Na + \text{C}_6\text{H}_5(CH_2-Cl) \longrightarrow CH_3-O-CH_2-Ph$

eg 2 $C_2H_5-\overset{\ominus}{O}Na +$  $\longrightarrow$ 

eg 3 

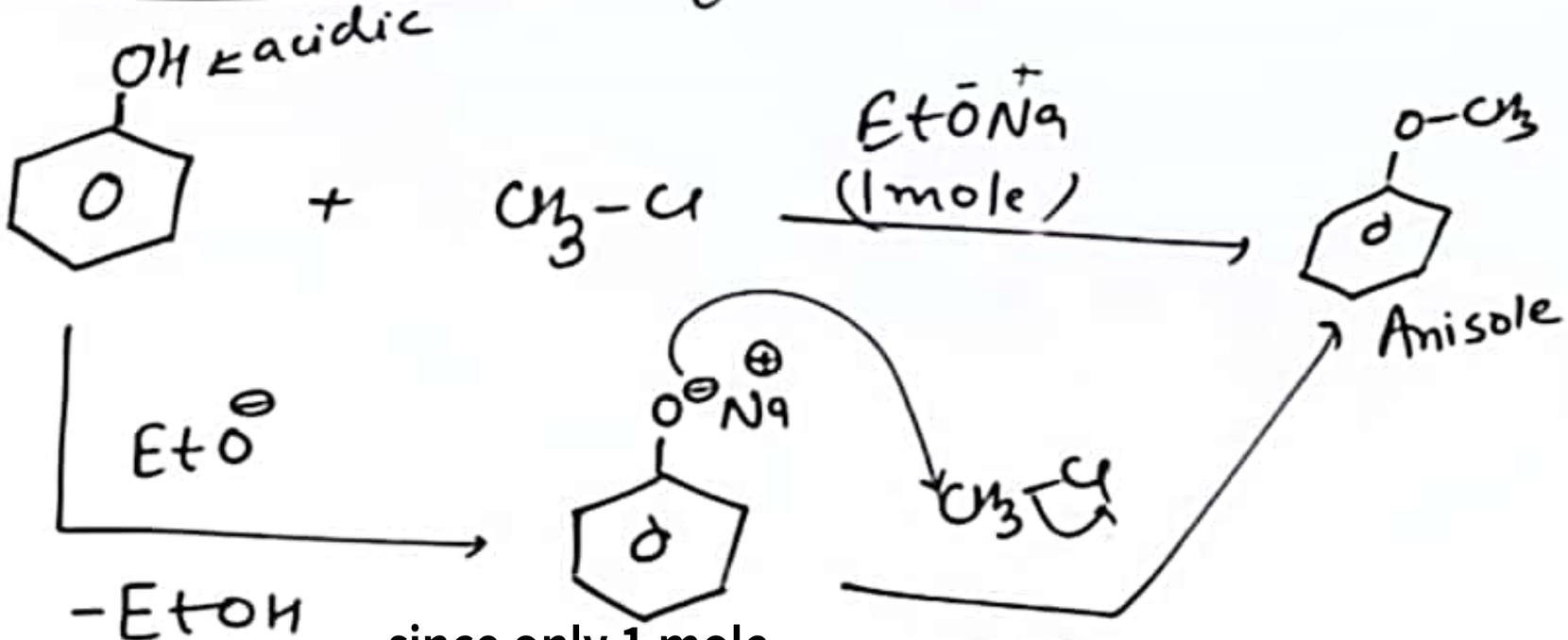
eg 4 

eg 5 $C_2H_5O^-Na^+ + Cl-CH_2-\underset{D}{\overset{CH_3}{C}}(H) \longrightarrow C_2H_5O-CH_2-\underset{D}{\overset{CH_3}{C}}(H)$

eg 6 

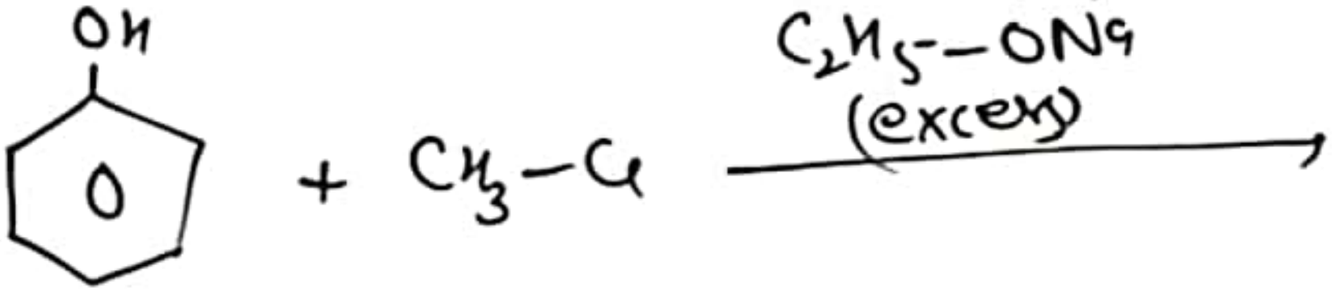
(VII) Williumson's Ether Synthesis:- (WES) (S_N2)

eg 7



since only 1 mole EtO- was taken, hence only PhO- is the final Nu-

eg 8

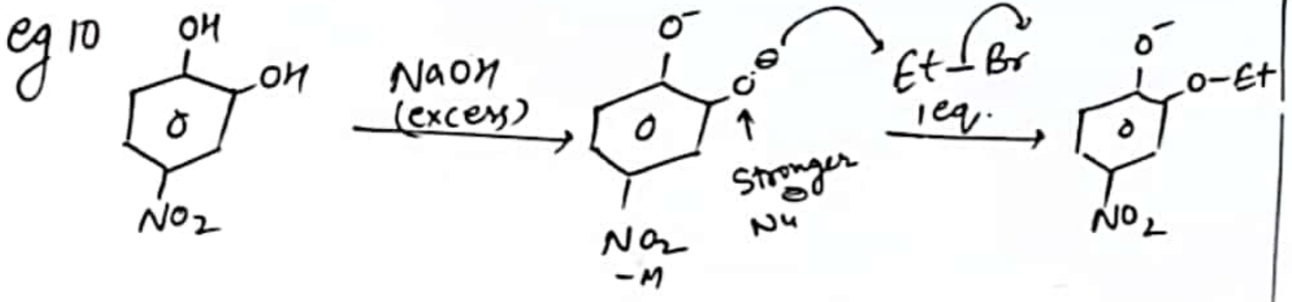


eg 9

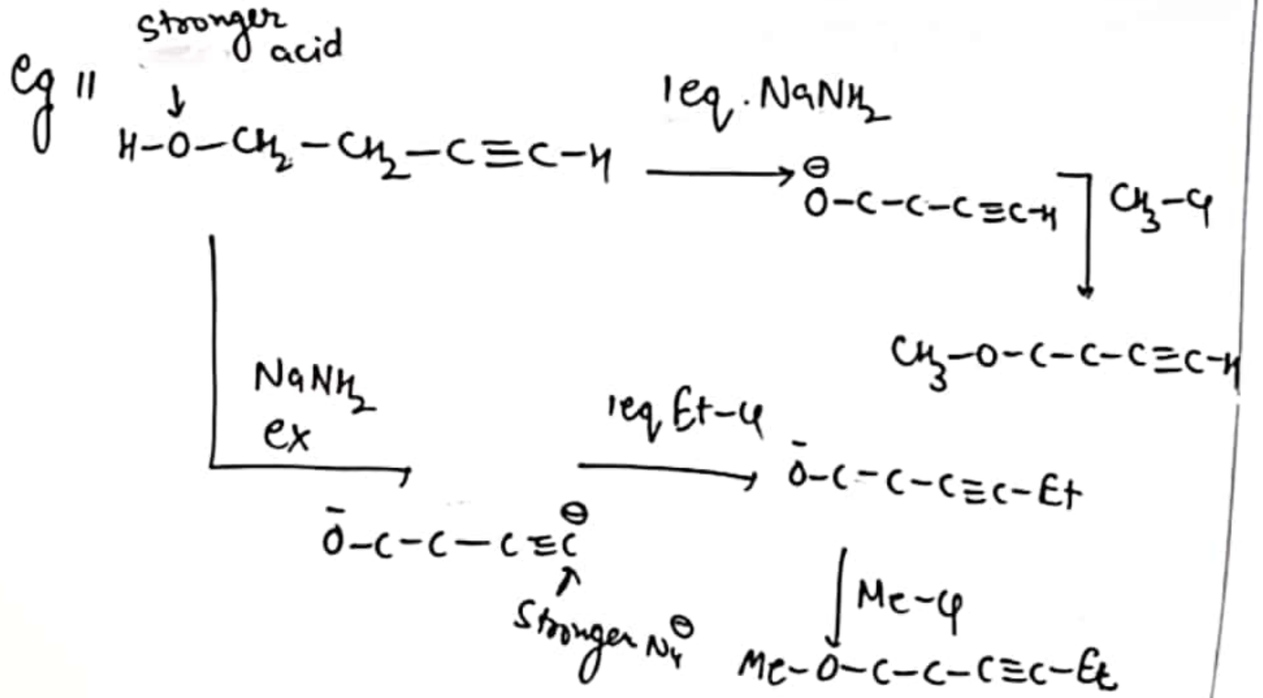
ORGANIC REACTION MECHANISM - III (ORM-III)

(VII) Williumson's Ether Synthesis:- (WES) (S_N2)

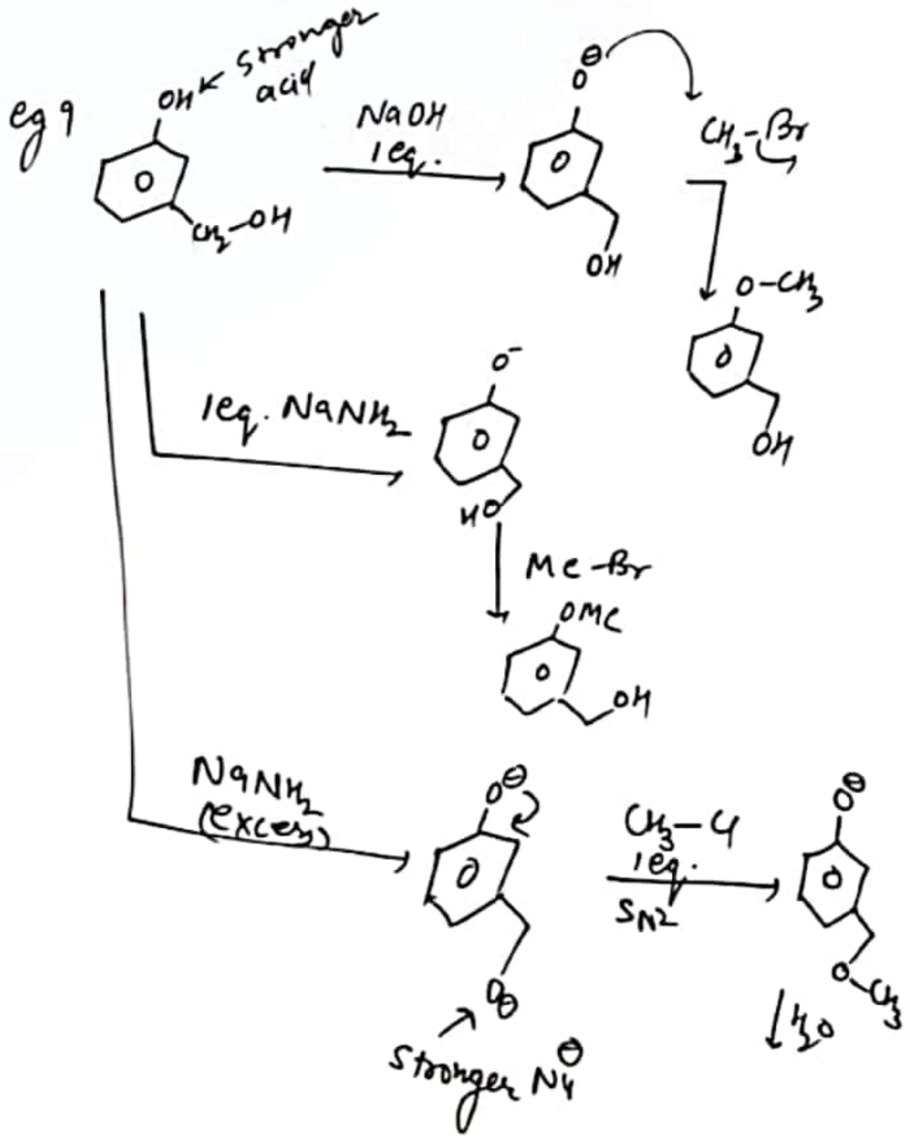
eg 10



eg 11



eg 9



ORGANIC REACTION MECHANISM - III (ORM-III)

(VII) Williumson's Ether Synthesis:- (WES) (S_N2)

Q Give most suitable reactants for formation of following ether (by EWS)

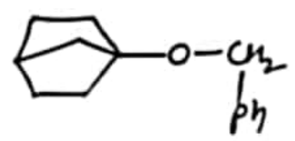
① 

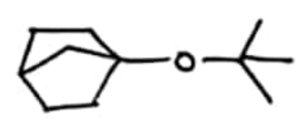
② 

③ 

④ 

⑤ 

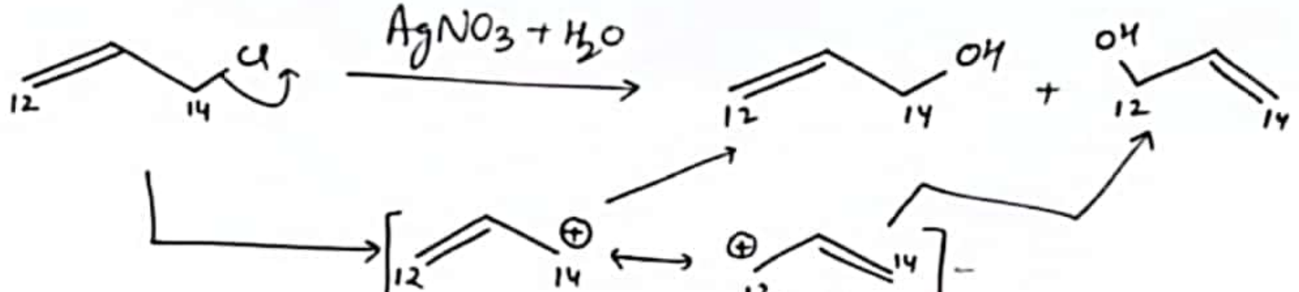
⑥ 

⑦ 

(VIII) Allylic S_N Reaction = S_N' → S_N1' / S_N2'

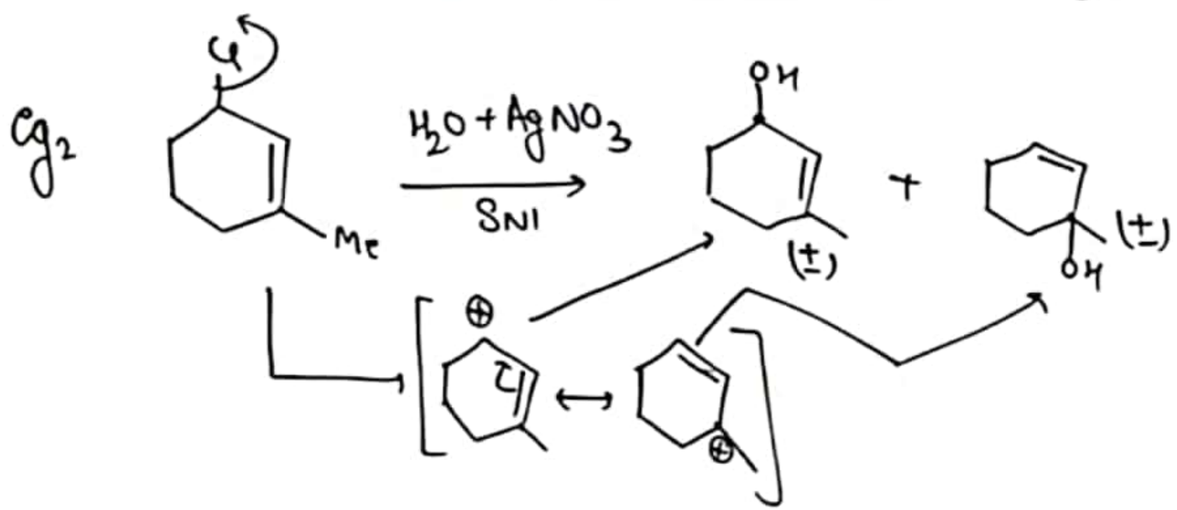
→ S_N1'

Eg 1



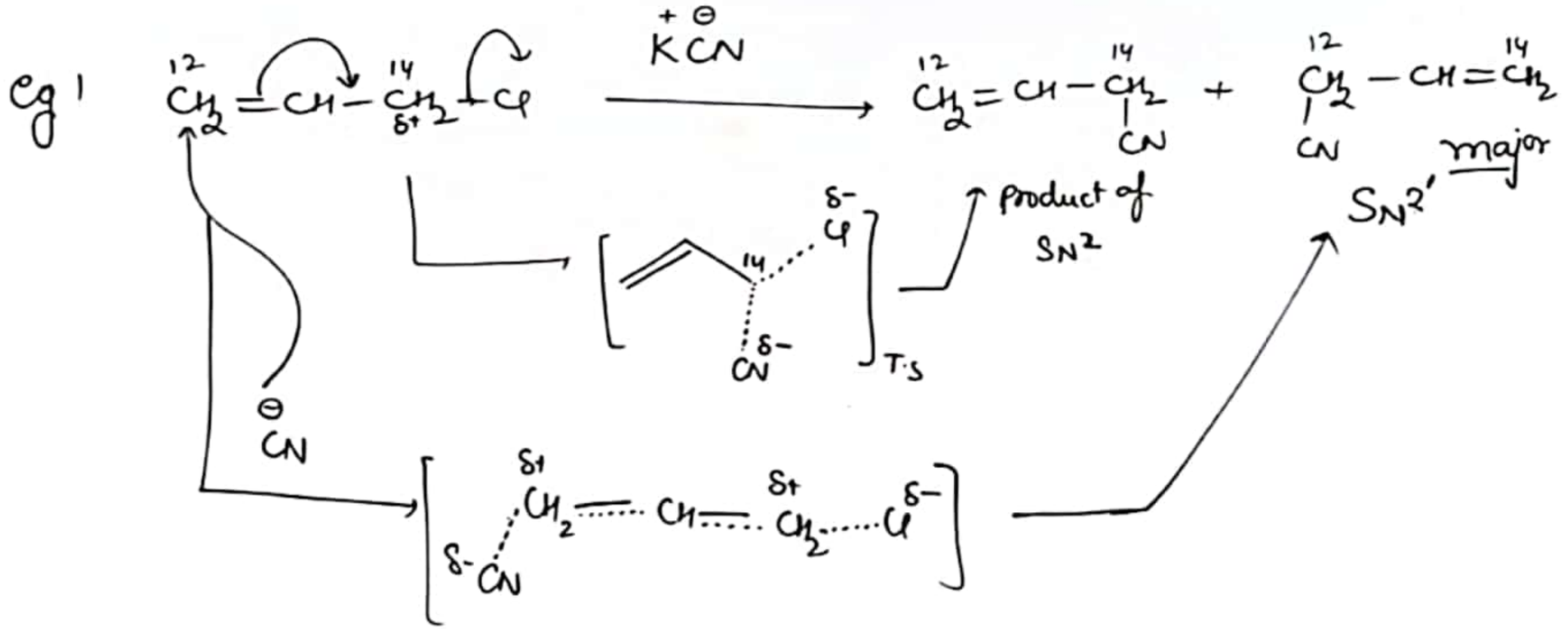
both products obtained in equal amounts

Eg 2



(VIII) Allylic S_N Reaction = S_N' → S_N1' / S_N2'

⇒ S_N2' Reaction.

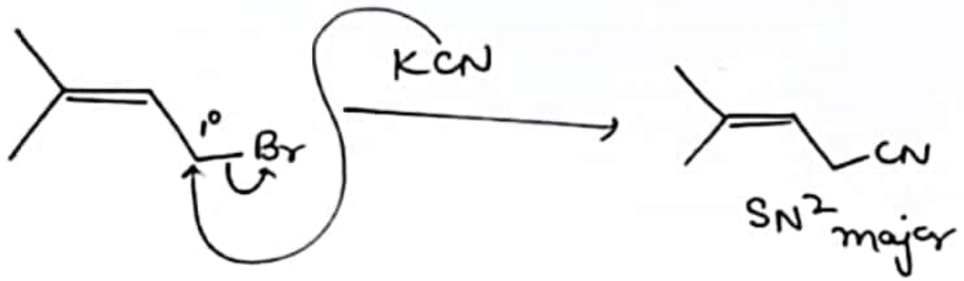


due to the comparatively less crowding on C12 than on C14

(VIII) Allylic S_N Reaction = S_N' → S_N1' / S_N2'

⇒ S_N2' Reaction.

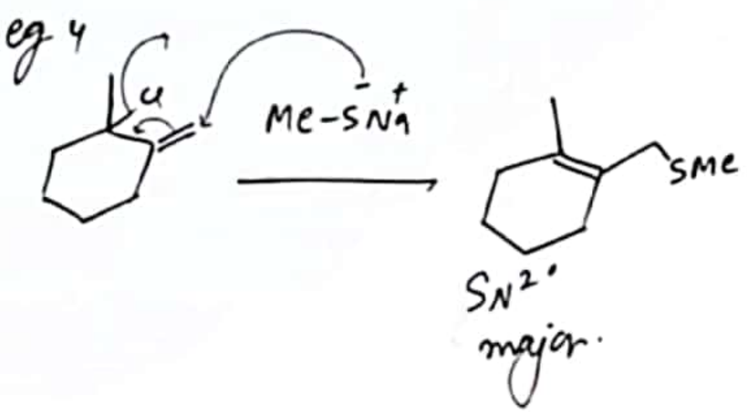
eg 2



eg 3



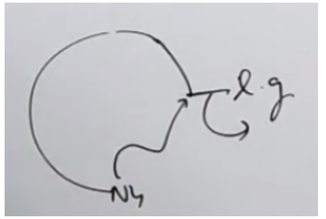
eg 4



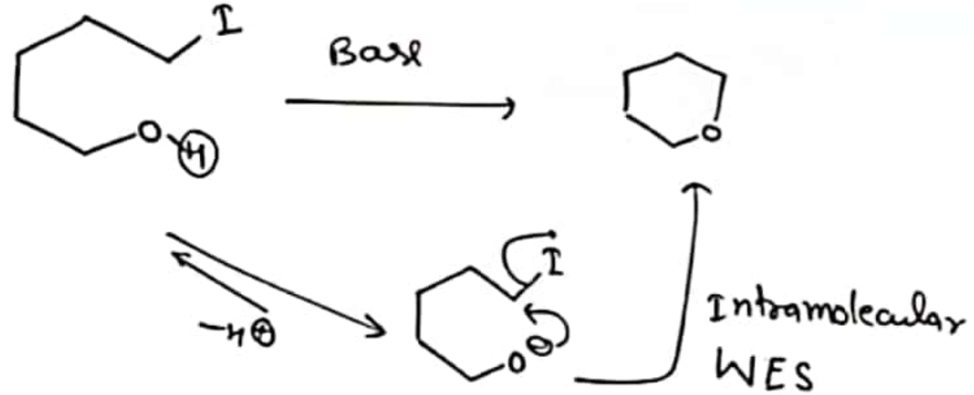
ORGANIC REACTION MECHANISM - III (ORM-III)

(IX) Intramolecular S_N2 Reaction:-

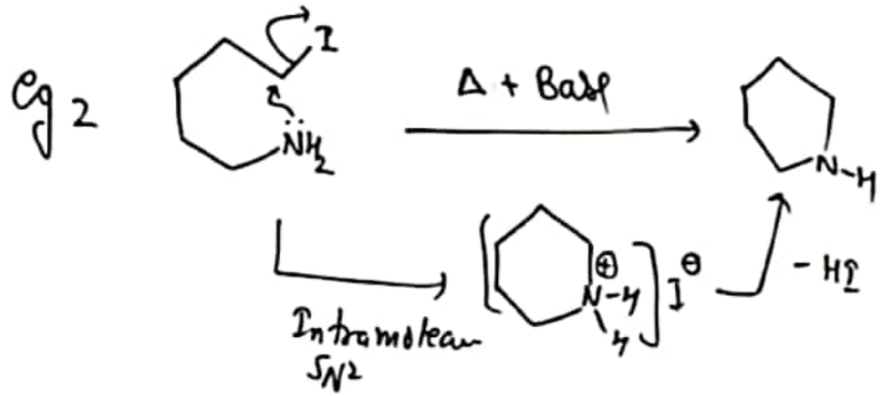
When l.g and strong nucleophile both are present in same molecule then intramolecular S_N2 reaction possible.



eg 1

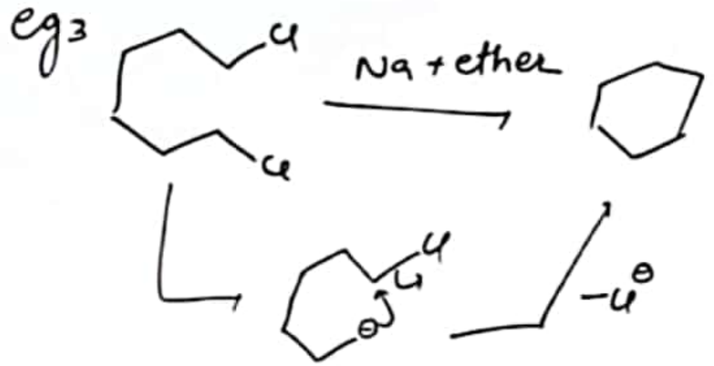


eg 2



Base given to absorb HI

eg 3

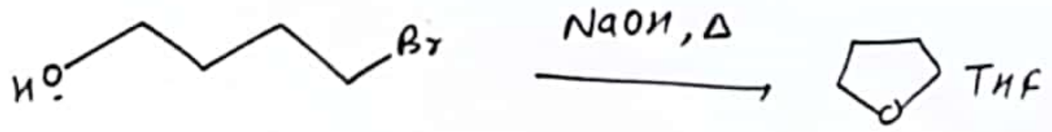


Intramolecular reactions are more favourable than intermolecular reactions

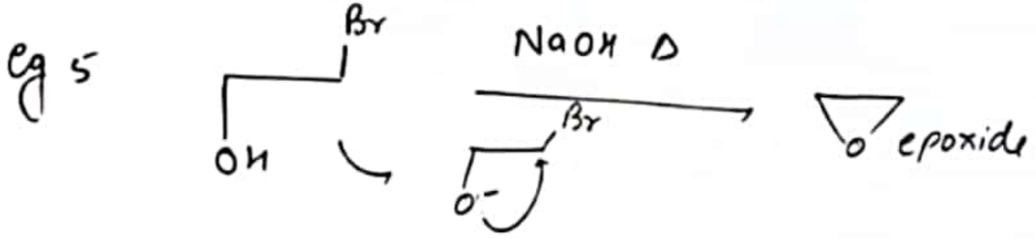
ORGANIC REACTION MECHANISM - III (ORM-III)

(IX) Intramolecular SN^2 Reaction:-

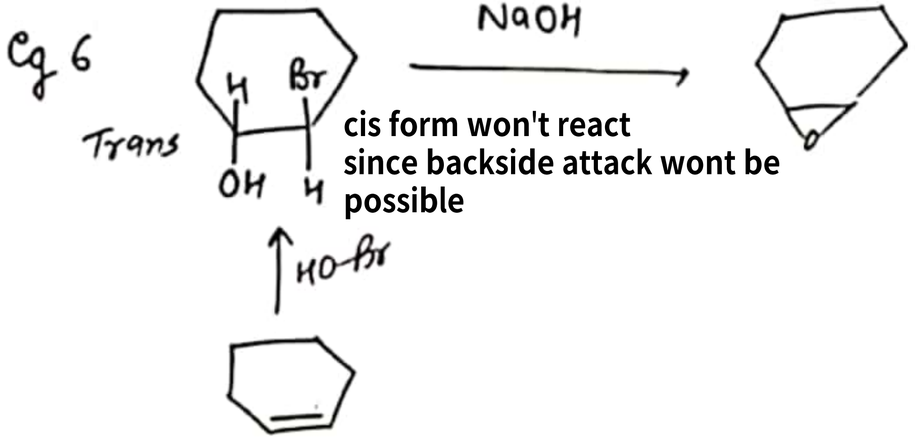
eg 4



eg 5

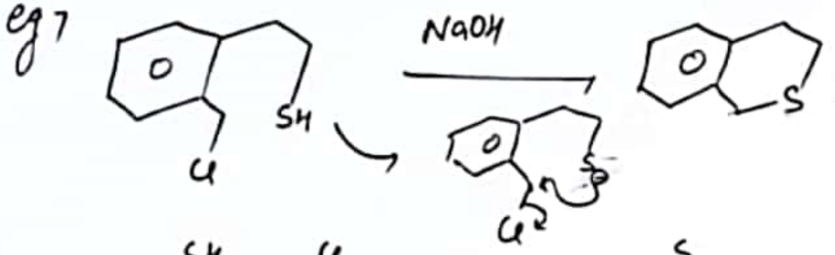


eg 6

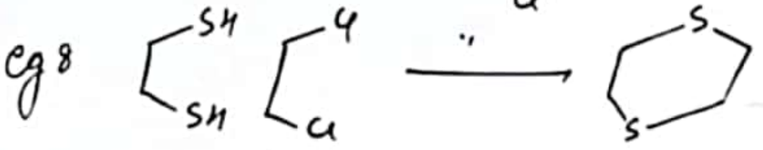


cis form won't react since backside attack wont be possible

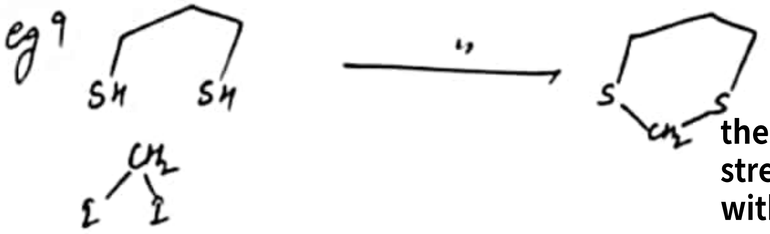
eg 7



eg 8

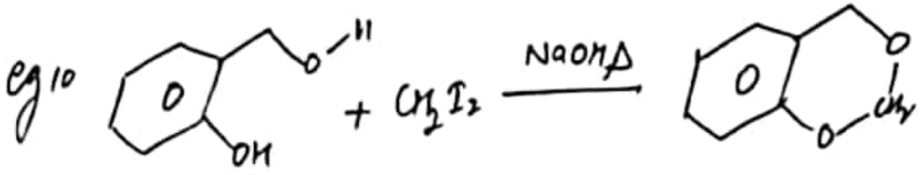


eg 9



the H also has some acidic strength (undergoes resonance with vacant d-orbitals of S)

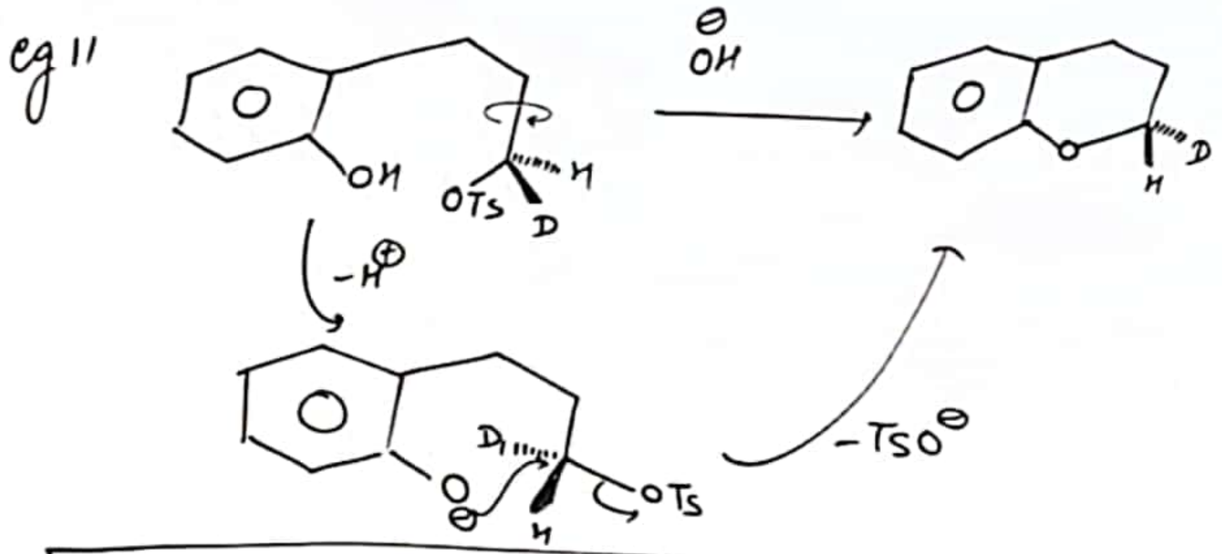
eg 10

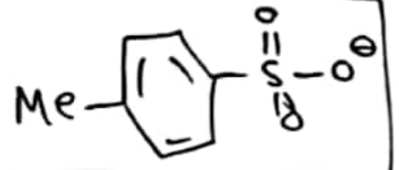


ORGANIC REACTION MECHANISM - III (ORM-III)

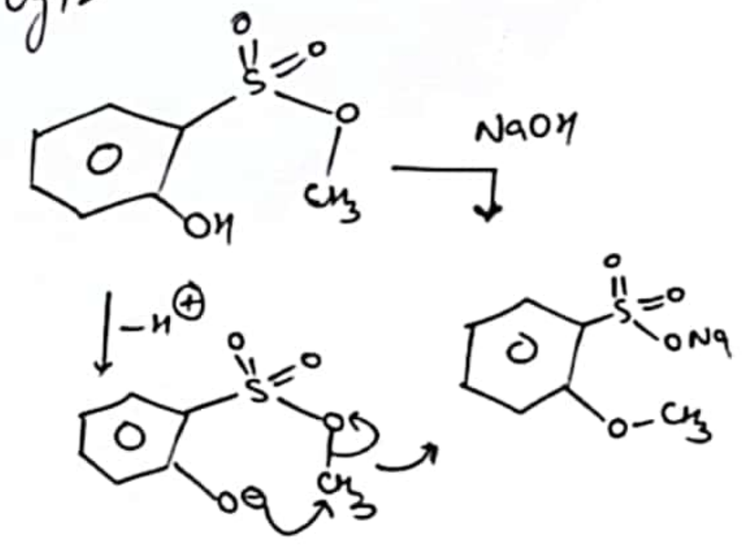
(IX) Intramolecular SN^2 Reaction:-

eg 11



$TsO^{\ominus}$ = 

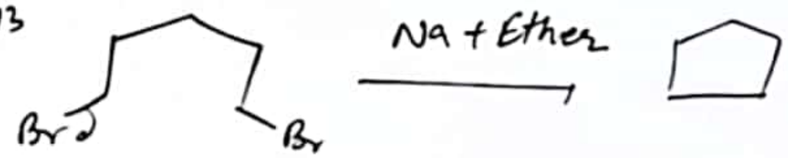
eg 12



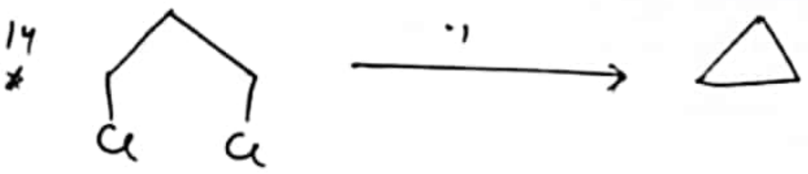
(IX) Intramolecular S_N2 Reaction:-

Examples of Intramolecular Wurtz Reaction

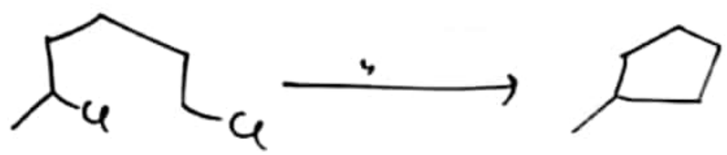
eg 13



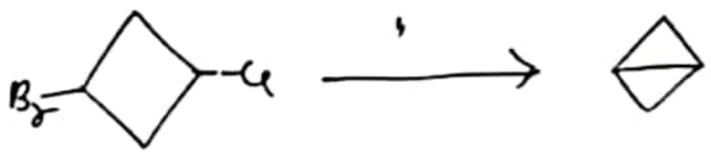
eg 14 *



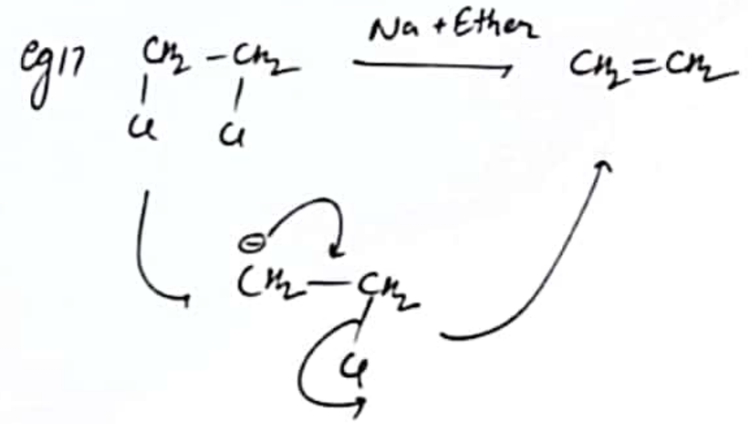
eg 15



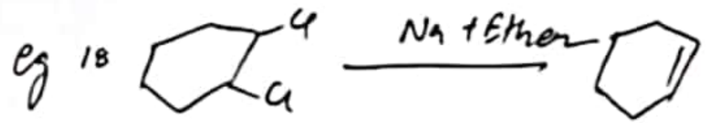
eg 16



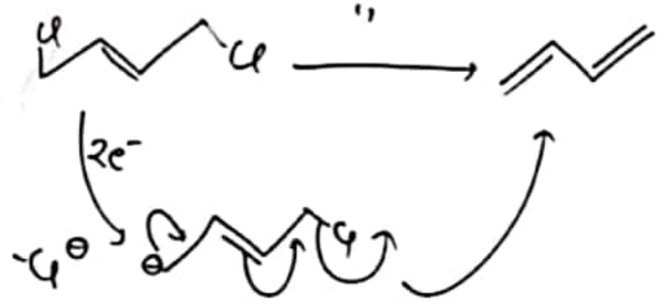
eg 17



eg 18



eg 19



ORGANIC REACTION MECHANISM - III

(X) S_N Reactions of Alcohols :-

- [A] → Reaction with HX
- [B] → " " PX_3 / PX_5
- [C] → " " $SOCl_2 / SOBr_2$

[X.A] Reaction of alcohols with HX :-

$$R\text{-}OH \xrightarrow{\text{Conc. HX}} R\text{-}X + H_2O$$

alkyl halide

lg (good)

$$R\text{-}OH \xrightarrow{\oplus\ \ominus} \text{(X)}$$

$\oplus$ Na $X^{\ominus}$

$$R\text{-}X + OH^{\ominus}$$

Poor l.g

⇒ Reactivity of HX ⇒ HI > HBr > HCl > HF

HF → No R_x^n

⇒ Lucas Reagent = anhydrous $ZnCl_2$ + conc. HCl

⇒ 3° R-OH / 2° R-OH / B-crowded alcohol / Benzylic / Allylic

↓ Conc. HX

S_N1

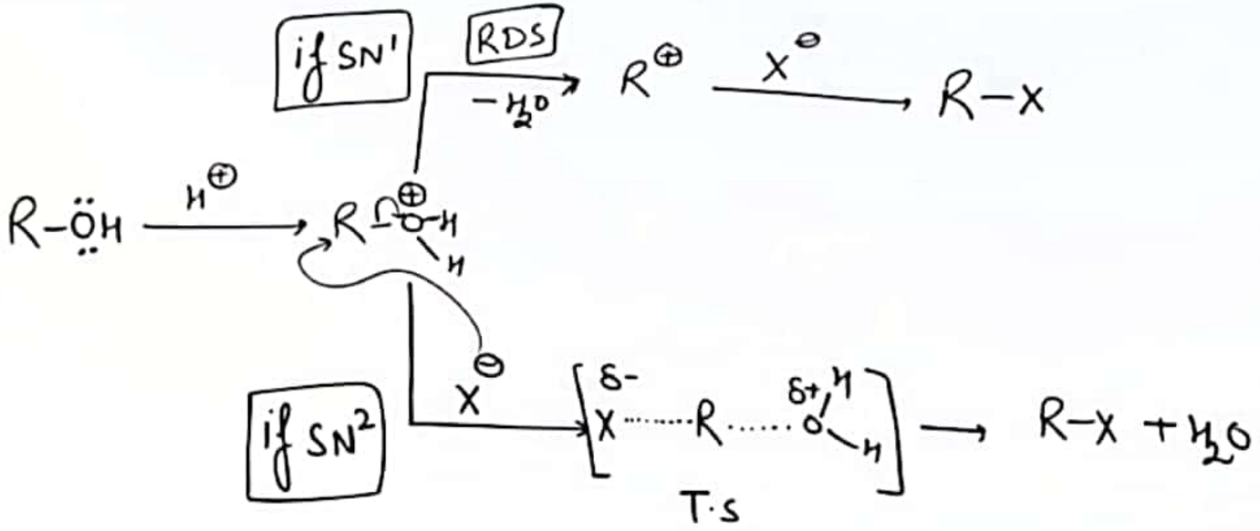
⇒ 1° R-OH $\xrightarrow{\text{Conc. HX}}$ S_N2

the rex. of primary alcohols and some secondary alcohols with HCl is v slow

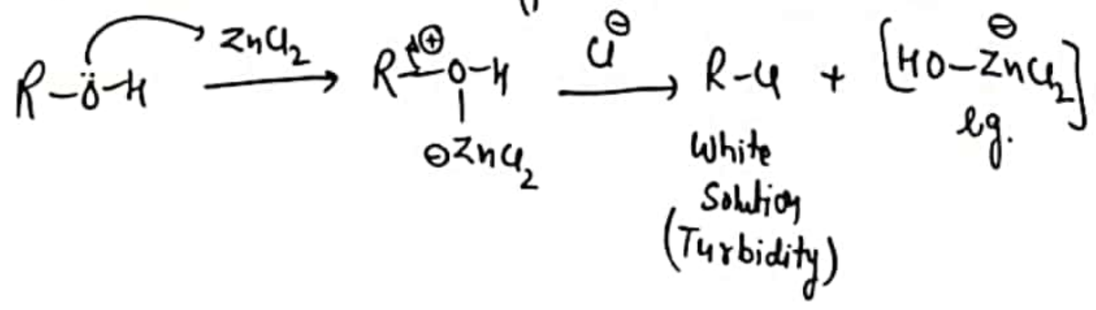
ORGANIC REACTION MECHANISM -III

(X) S_N Reactions of Alcohols :-

Mechanism



Mechanism with Lucas Reagent :-



⇒ Reactivity of HX ⇒ HI > HBr > HCl > HF

(No R_x^n)

⇒ Lucas Reagent = anhydrous $ZnCl_2$ + conc. HCl

⇒ 3° R-OH / 2° R-OH / B-Crowded alcohol / Benzylic / Allylic

↓ conc. HX

S_N1

⇒ 1° R-OH $\xrightarrow{\text{Conc. HX}}$ S_N2

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ORGANIC REACTION MECHANISM - III

(X) S_N Reactions of Alcohols :-

⇒ Lucas Test

3° R-OH / Benzylic / Allylic alcohol $\xrightarrow{ZnCl_2 + HCl}$ R-Cl

instant White turbidity

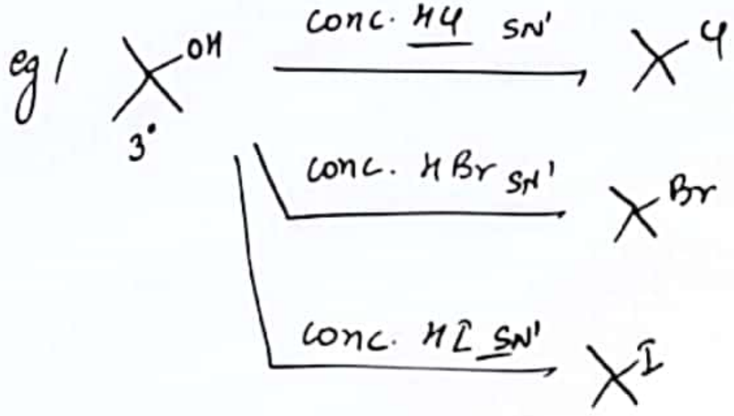
2° alcohol $\xrightarrow{(ZnCl_2 + HCl)}$ R-Cl

white turbidity after 5 to 10 min.

1° Alcohol $\xrightarrow{HCl + ZnCl}$ R-Cl

White turbidity (after 30 min / on heat)

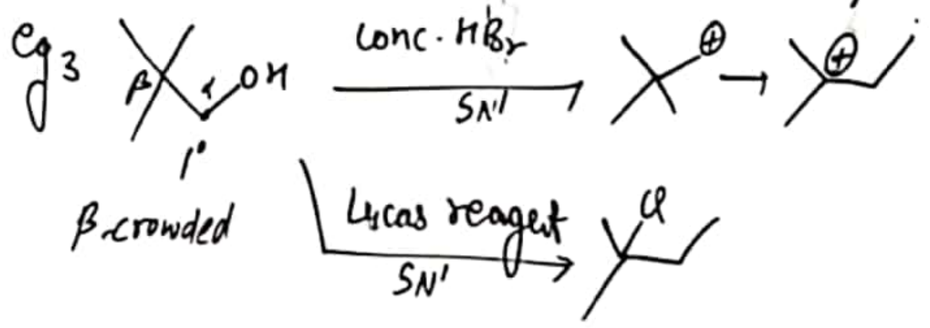
eg 1



eg 2



eg 3

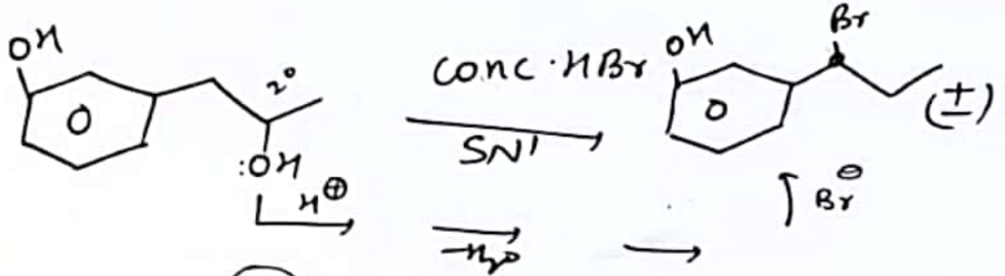


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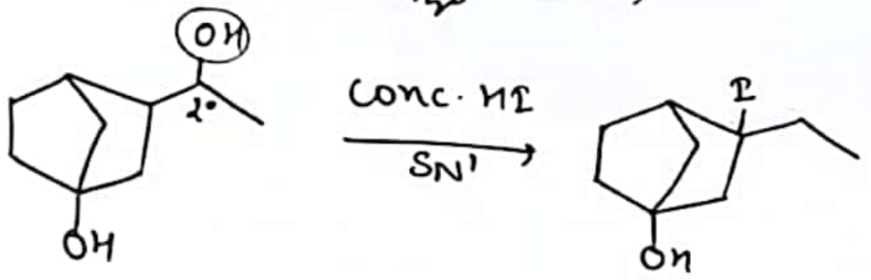
ORGANIC REACTION MECHANISM -III

(X) S_N Reactions of Alcohols :-

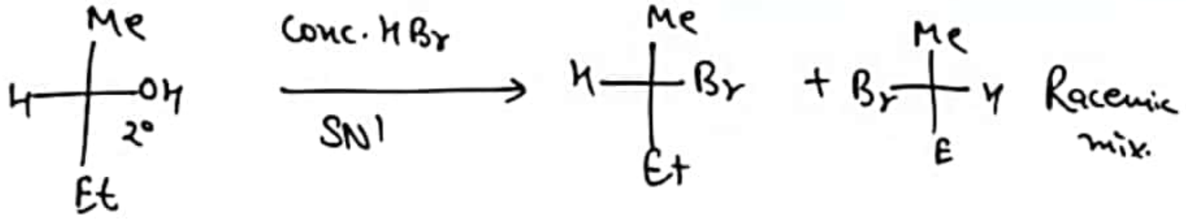
Eg 4



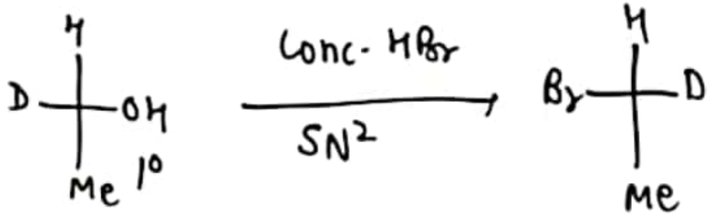
Eg 5



Eg 6



Eg 7



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(X) S_N Reactions of Alcohols :-

[X.B] S_N Reactions of alcohols with PX_3, PX_5

$$\left[\begin{array}{l} PX_3 = PCl_3,\ PBr_3,\ PI_3 \\ PX_5 = PCl_5,\ PBr_5 \\ P + I_2 \longrightarrow PI_3 \end{array}\right]$$

PI5 nahi bnta due to the very large size of I

$$3R\text{-}OH + PX_3 \xrightarrow{\text{mainly } SN^2} 3R\text{-}X + \underset{\text{Phousphorous acid (l.g)}}{H_3PO_3}$$

$$R\text{-}OH + PX_5 \xrightarrow{\text{" " } SN^2} R\text{-}X + HX + \underset{\text{l.g.}}{POX_3}$$

$1^\circ R\text{-}OH,\ 2^\circ R\text{-}OH \xrightarrow{PX_3/PX_5} SN^2$

$3^\circ R\text{-}OH \xrightarrow{\text{" "}} SN^1$

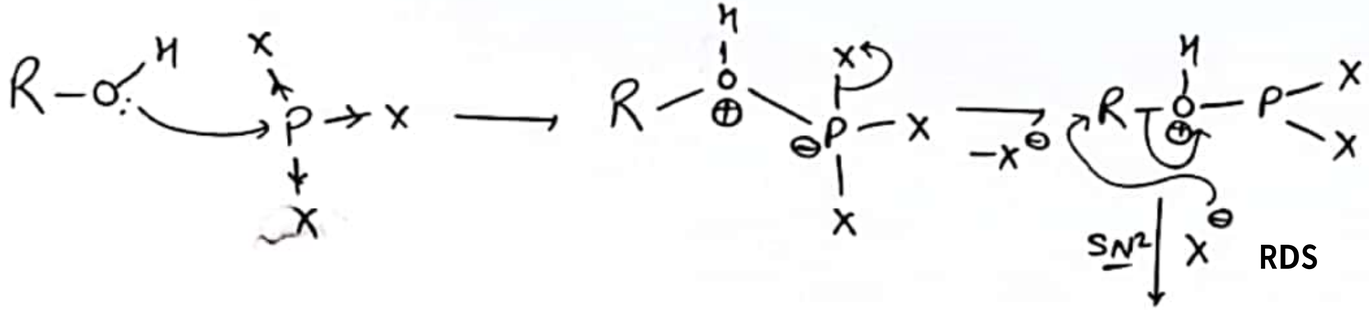
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ORGANIC REACTION MECHANISM -III

(X) S_N Reactions of Alcohols :-

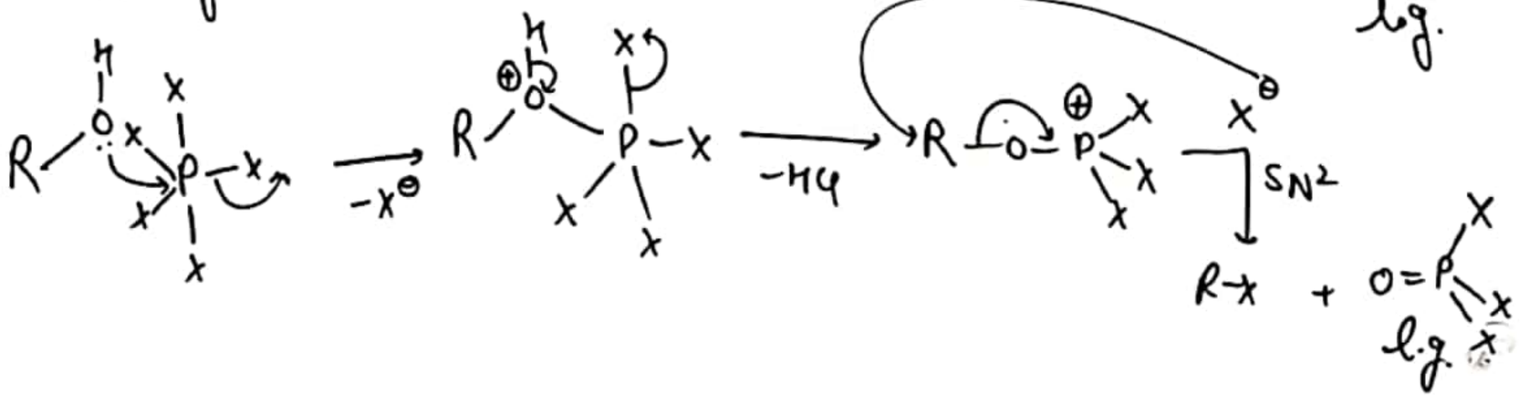
[X.B] S_N Reactions of alcohols with PX_3 , PX_5-

mechanism with PX_3



now, this will go and attack two more molecules of alcohol and finally form H3PO3 at the last step

mech. for PX_5



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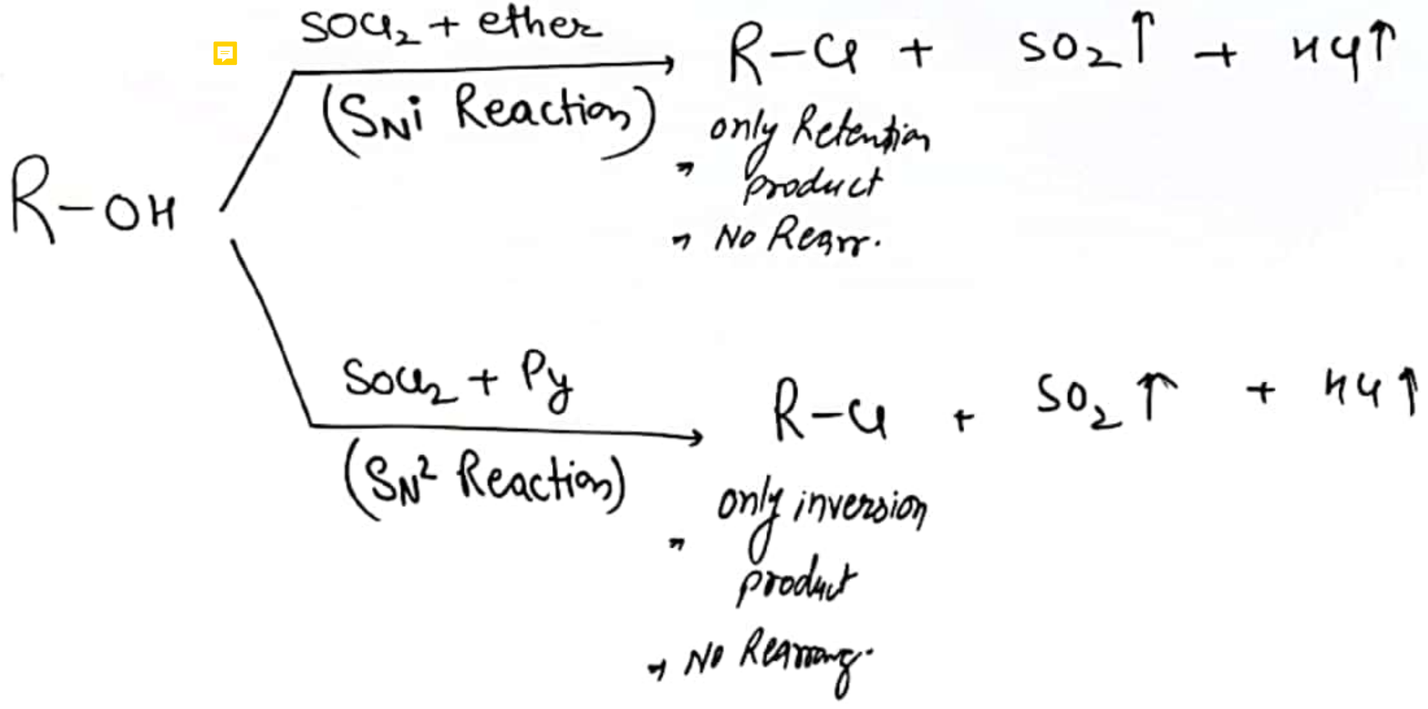
ORGANIC REACTION MECHANISM - III

(X) S_N Reactions of Alcohols :-

(X.c) Reaction with $SOCl_2$ / $SOBr_2$

$SOCl_2$ ↑ Thionyl chloride

⇒ Darzen's process. = industrial method for formation of alkyl chloride due to formation of gaseous product

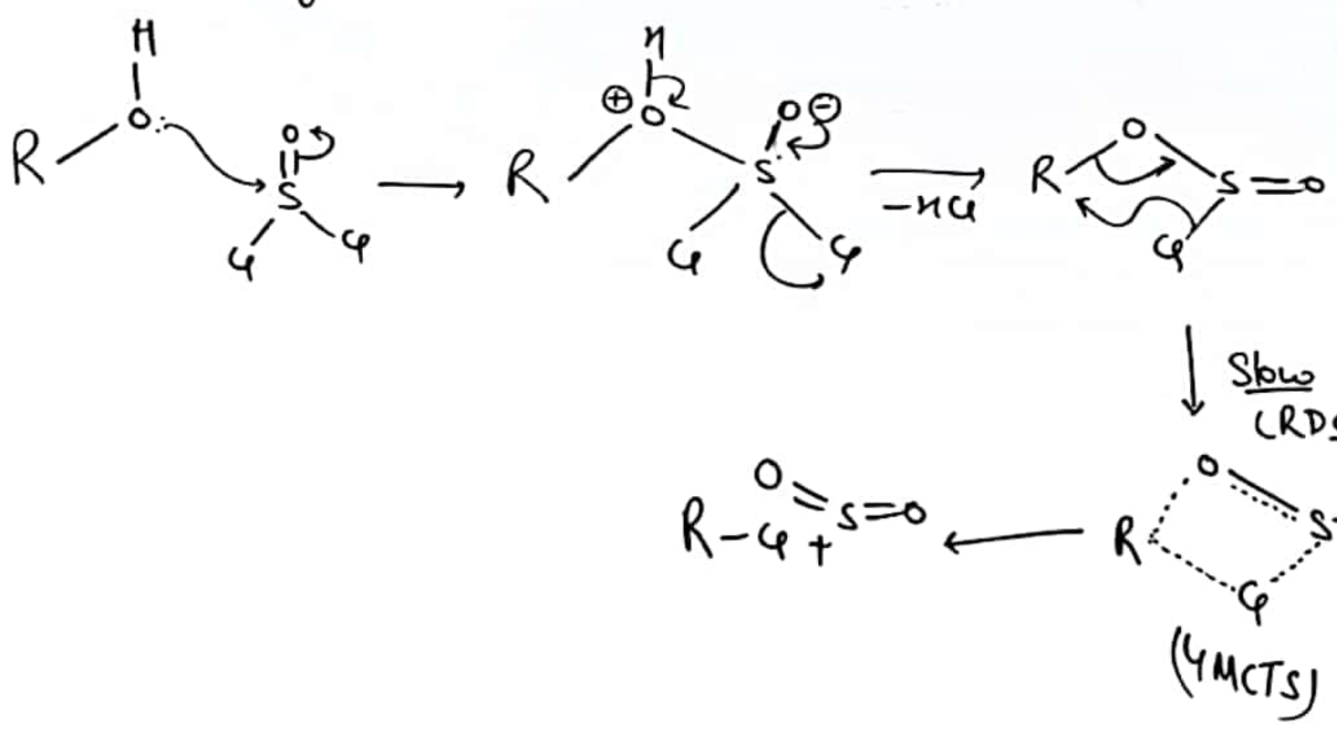


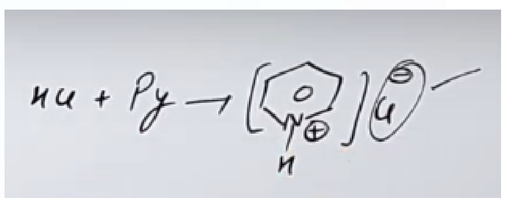
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(X) S_N Reactions of Alcohols :-

(X.c) Reaction with $SOCl_2$ / $SOBr_2$

mechanism of $S_N i$ (i = internal)





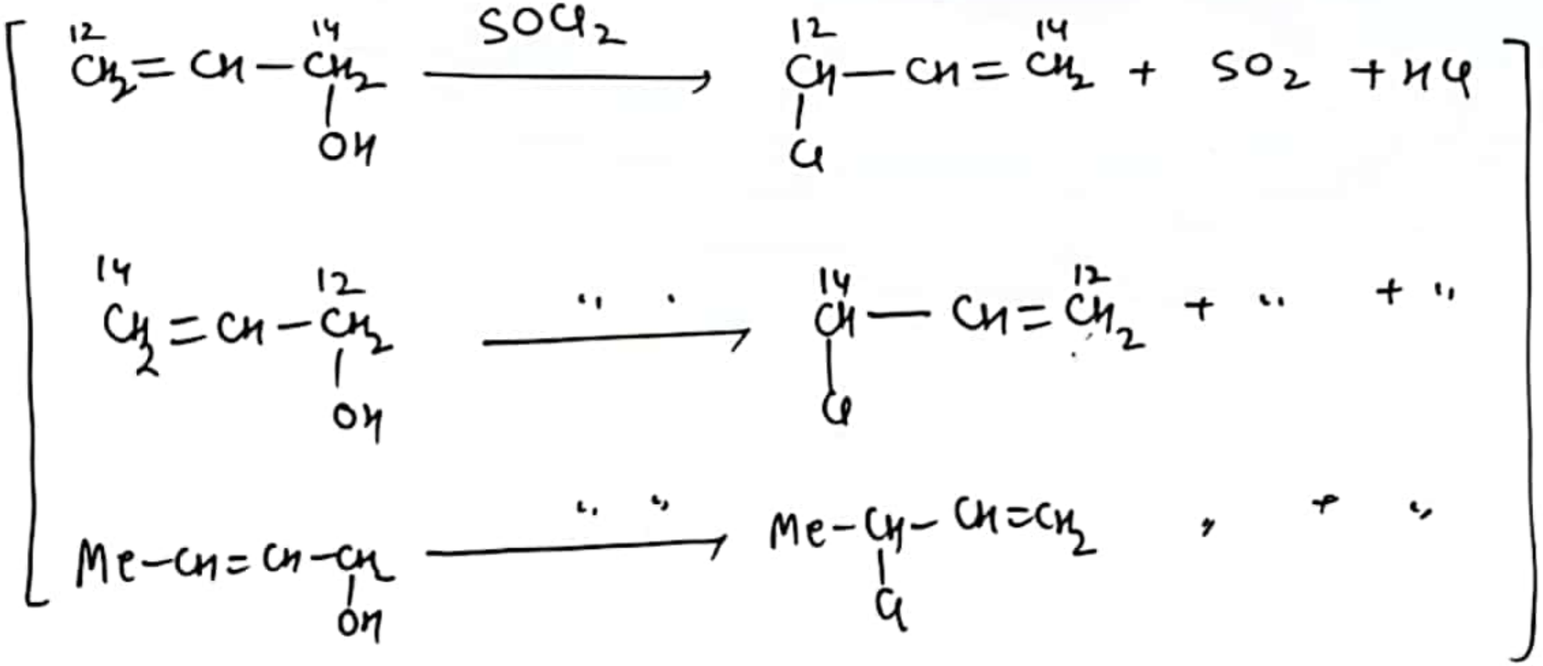
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ORGANIC REACTION MECHANISM - III

(X) S_N Reactions of Alcohols :-

(X.c) Reaction with $SOCl_2$ / $SOBr_2$

⇒ Reaction of $SOCl_2$ on allylic alcohols.



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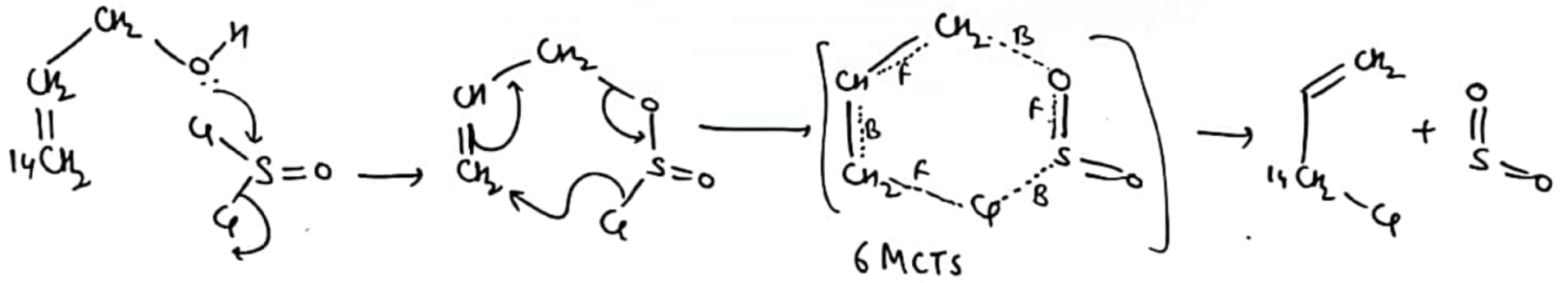
ORGANIC REACTION MECHANISM -III

(X) S_N Reactions of Alcohols :-

(X.c) Reaction with $SOCl_2$ / $SOBr_2$

⇒ Reaction of $SOCl_2$ on allylic alcohols.

mechanism



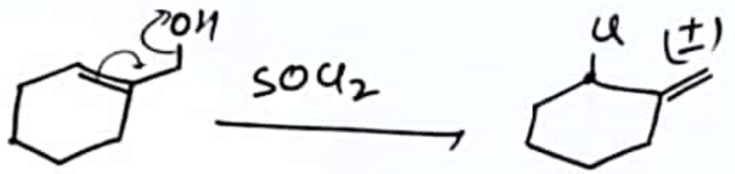
6 MCTS is more stable than 4 MCTS

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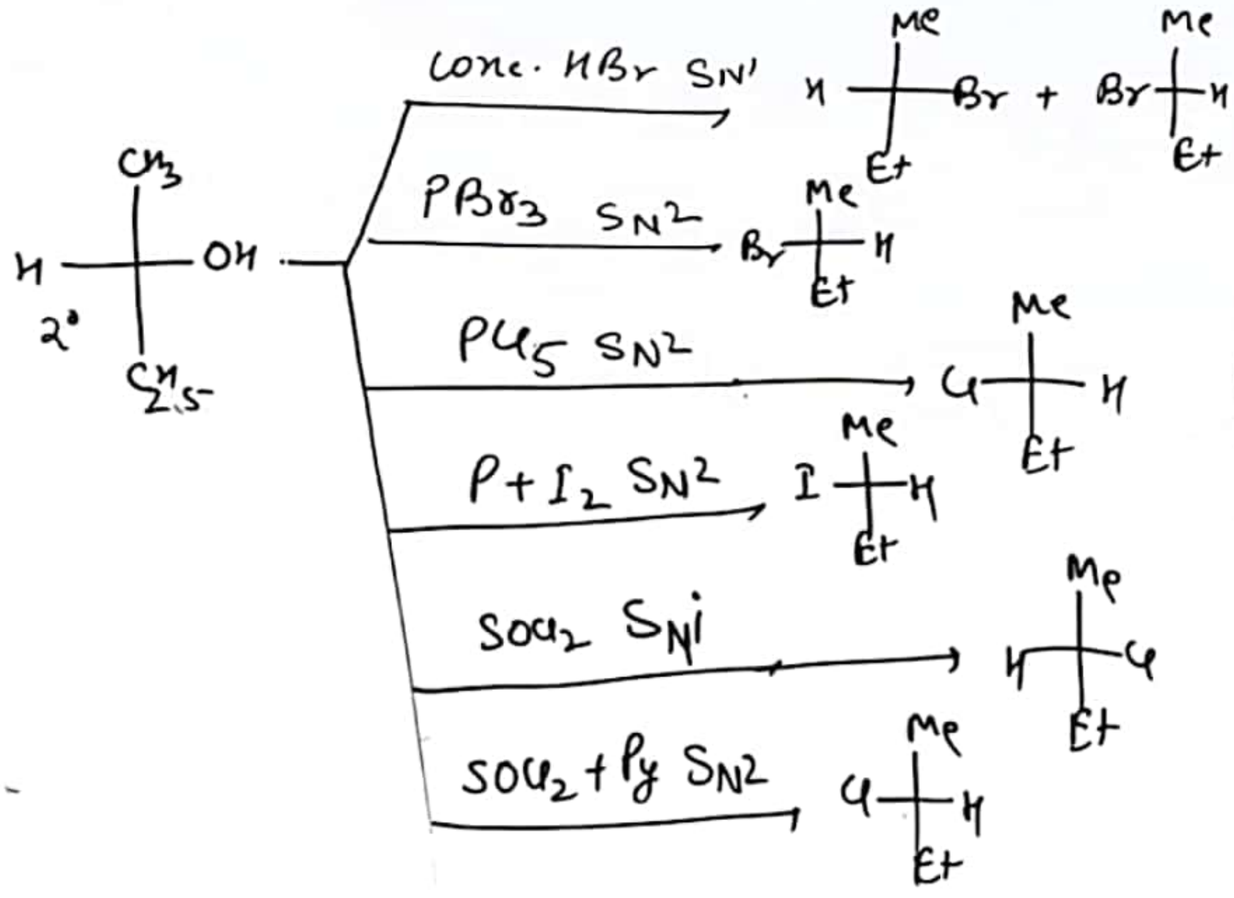
ORGANIC REACTION MECHANISM -III

X) S_N Reactions of Alcohols :-

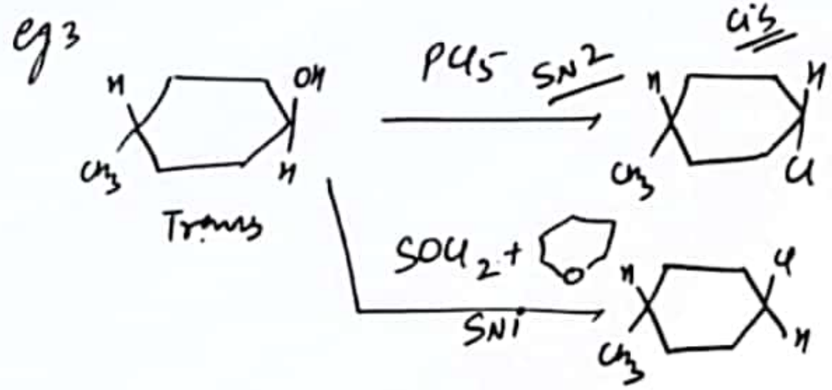
eg 1



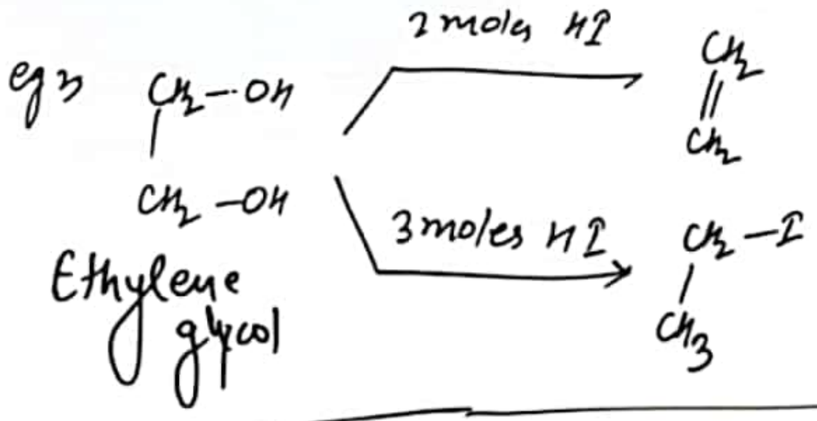
eg 2



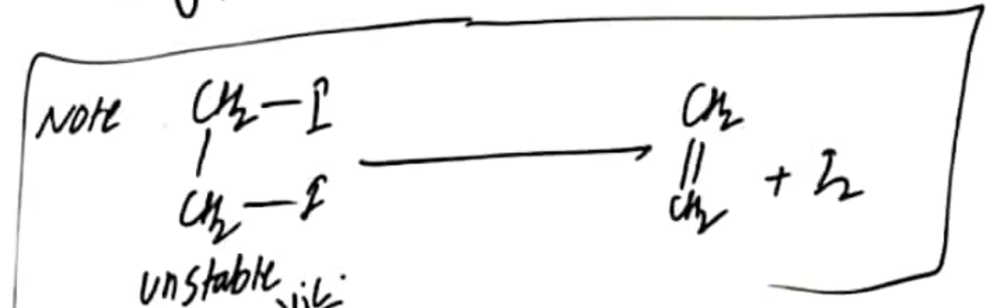
eg 3



eg 3



Note

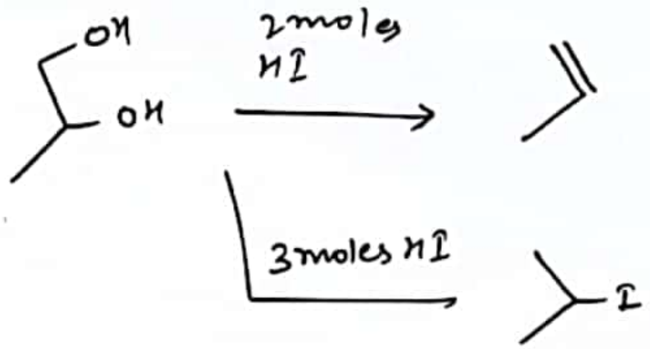


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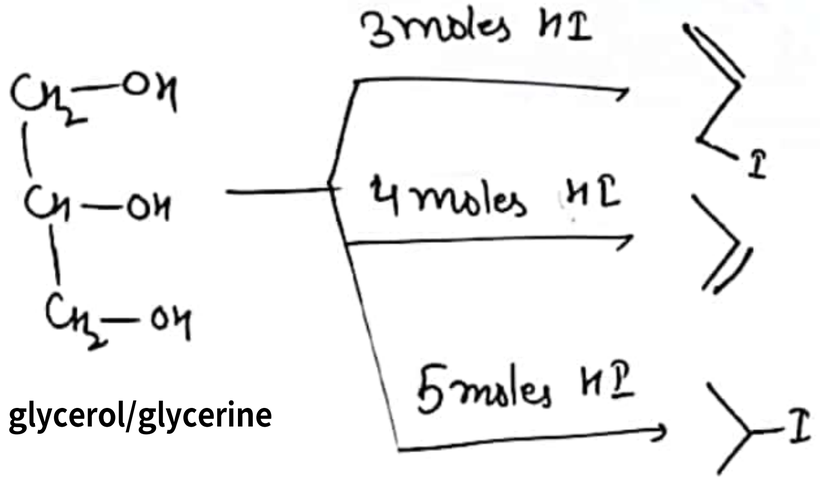
ORGANIC REACTION MECHANISM -III

(X) S_N Reactions of Alcohols :-

eg 5

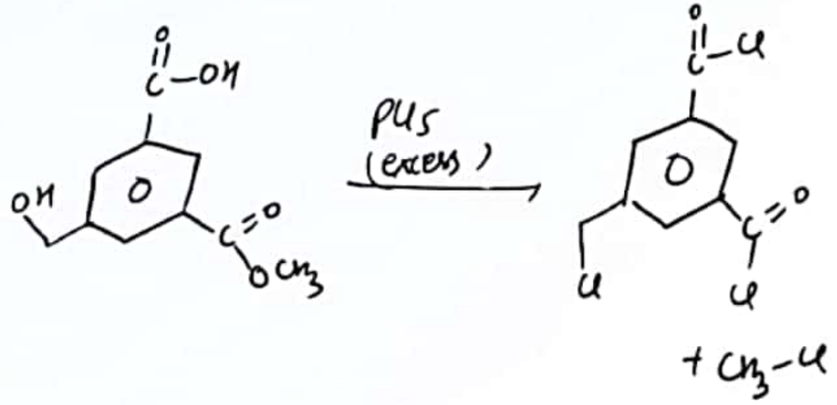


eg 6

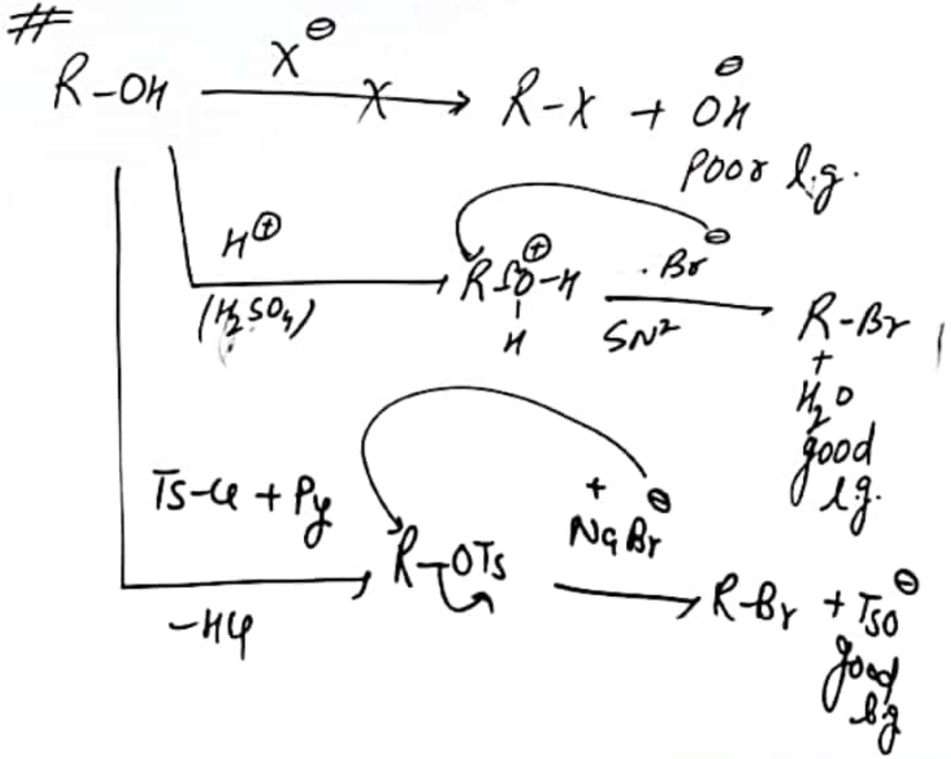


glycerol/glycerine

eg 1



#

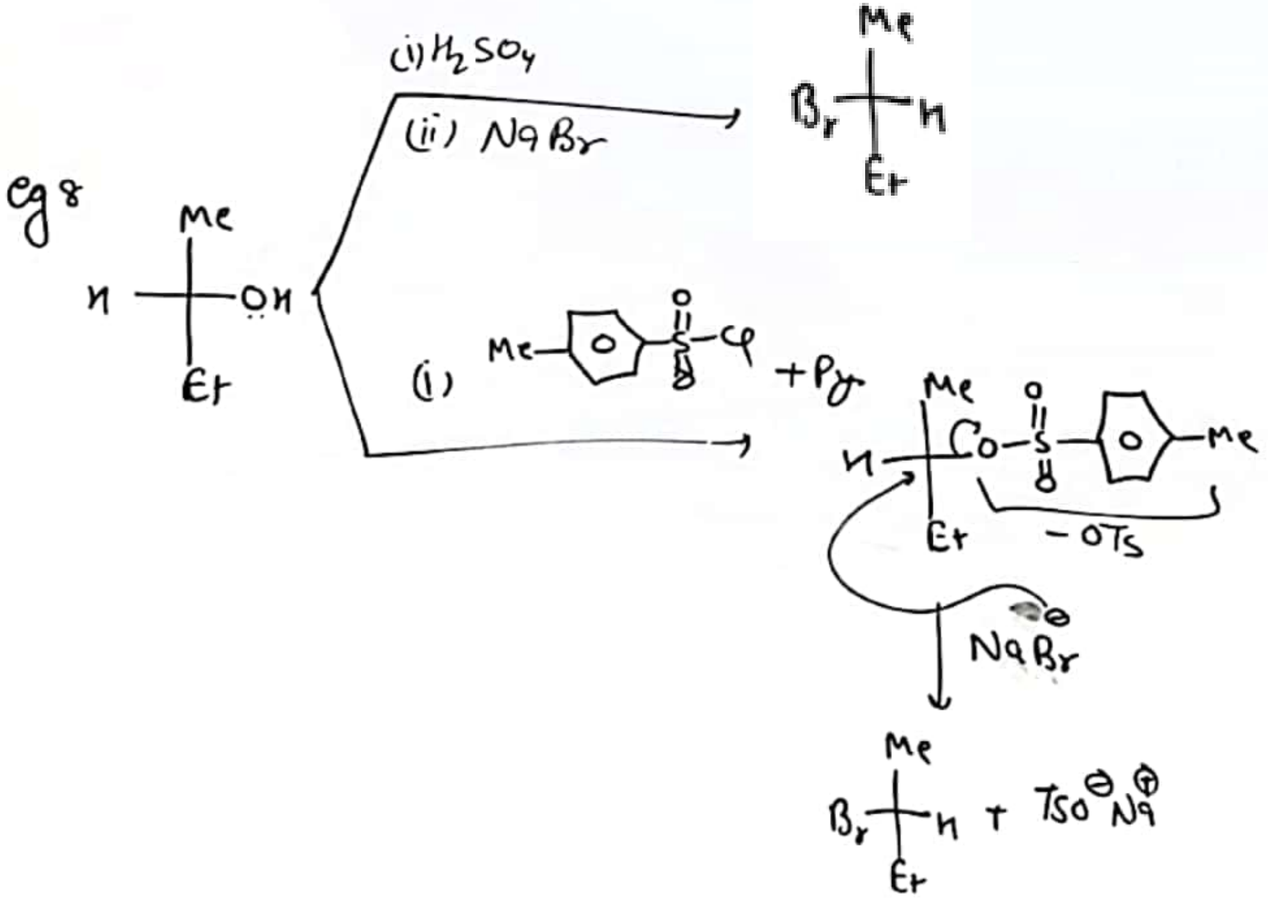


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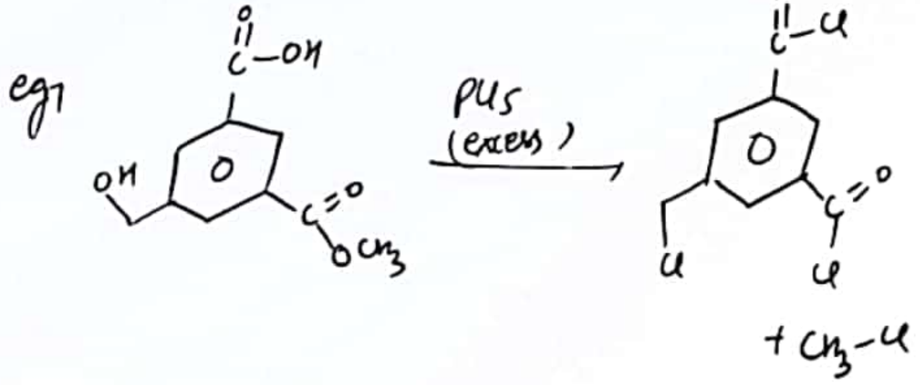
ORGANIC REACTION MECHANISM -III

(X) S_N Reactions of Alcohols :-

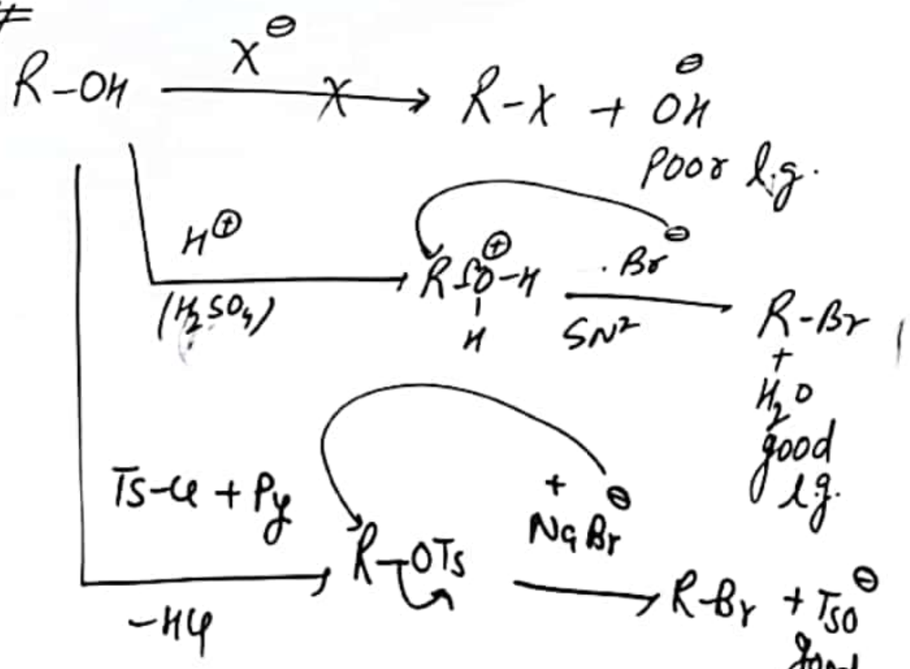
eg 8



eg 7



#



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ORGANIC REACTION MECHANISM - III

(XI) Nucleophilic Sub. Rxn of Ethers:-

- [A] → Reaction with HX
- (B) → Hydrolysis of ethers

[XI.A.] Reaction of ethers with HX

$$R-\ddot{O}-R \xrightarrow{\text{Conc. HX (1 mol)}} \underset{\text{l.g}}{R-OH} + R-X$$

$$\xrightarrow{\text{Conc. HX (excess)}} 2\,R-X + H_2O$$

$$\xrightarrow{NaX} \text{(X)} \quad R-X + \underset{\text{poor l.g.}}{R-O^{\ominus}}$$

⇒ Reactivity of HX

$$HI > HBr > HCl > \underset{\text{No Reaction}}{HF}$$

⇒ When any alkyl group of ether is 3° alkyl or benzylic (or resonance stabilised) then S_N1 mech. take place. otherwise S_N2 take place.

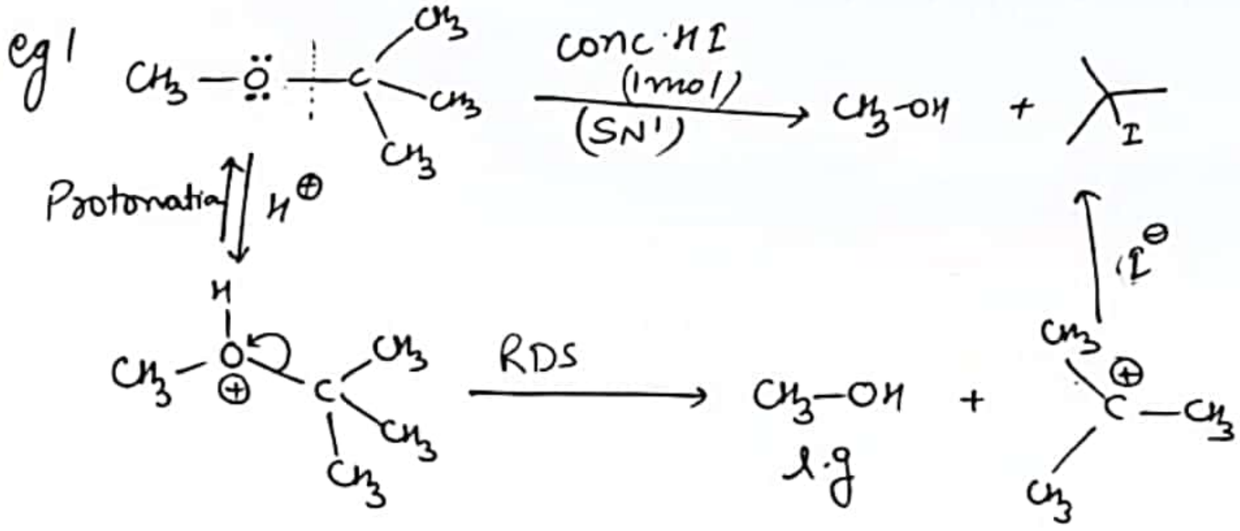
⇒ anhydrous HI give S_N2 reaction with ethers. since there is no water for solvation of ions

ORGANIC REACTION MECHANISM – III

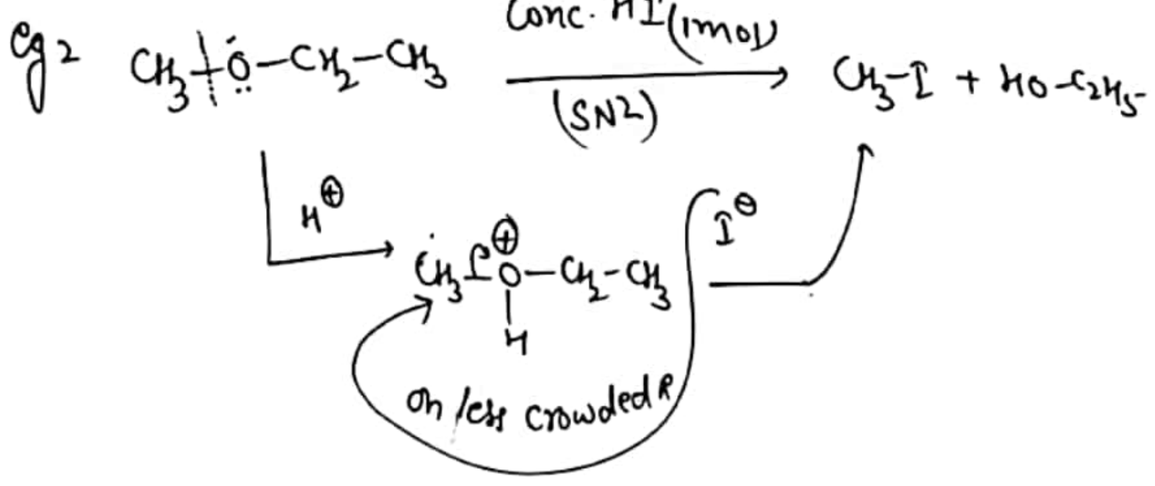
(XI) Nucleophilic Sub. Rxn of Ethers:–

[XI.A] Reaction of ether with HX:–

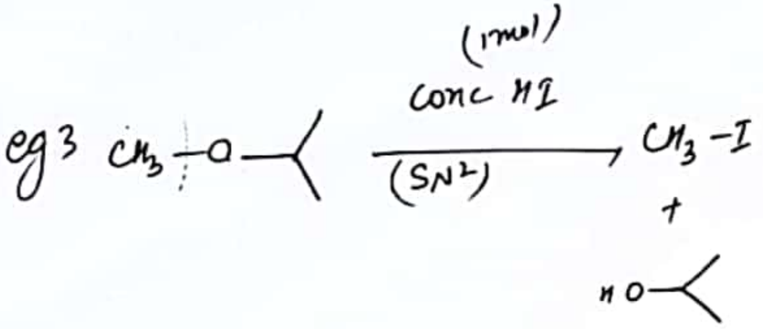
eg 1



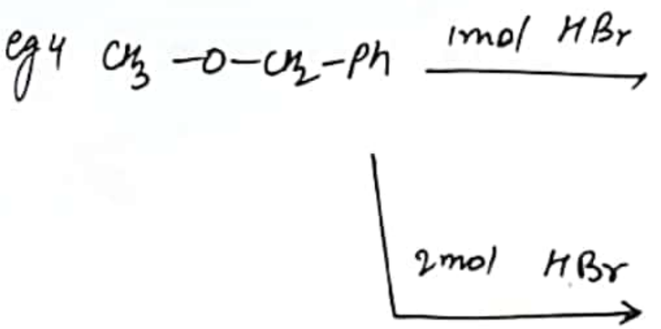
eg 2



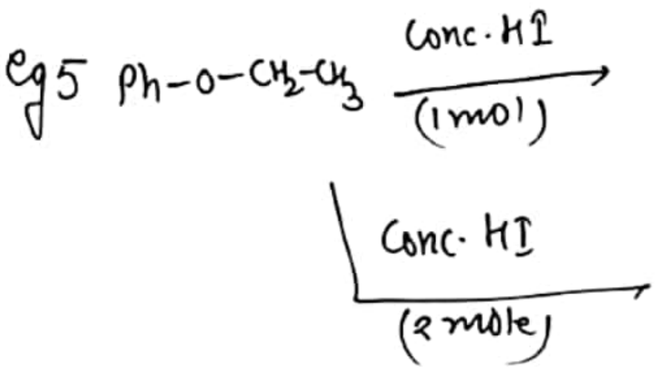
eg 3



eg 4



eg 5

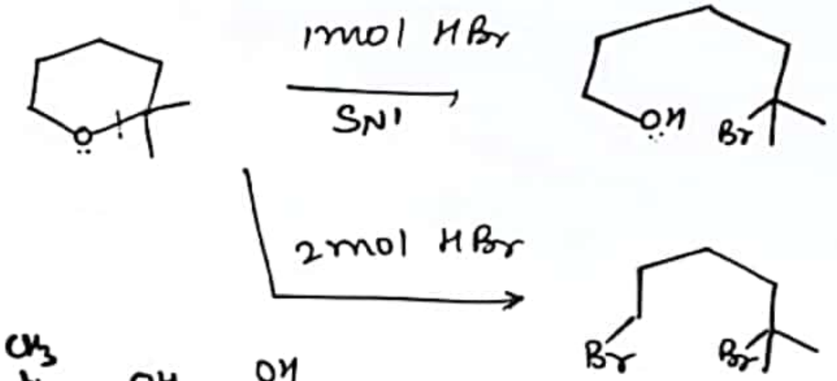


ORGANIC REACTION MECHANISM – III

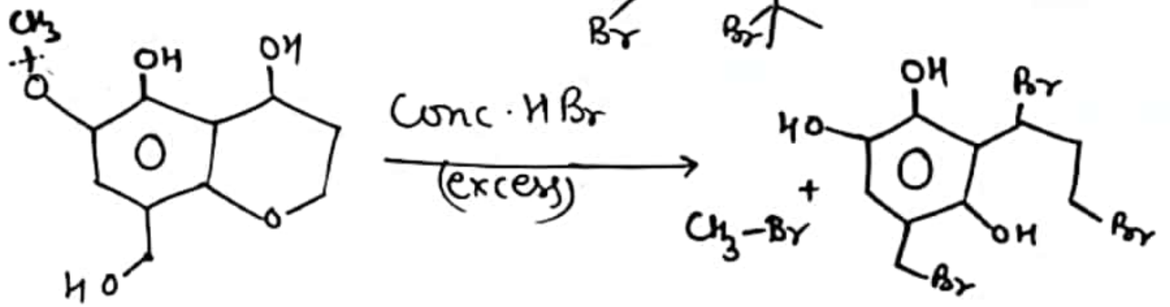
(XI) Nucleophilic Sub. Rxn of Ethers:–

[XI.A] Reaction of ether with HX:–

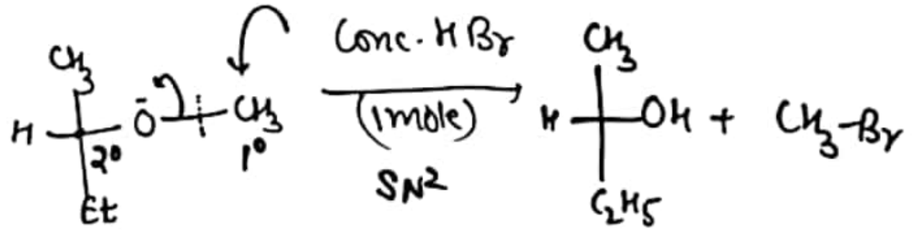
eg 6



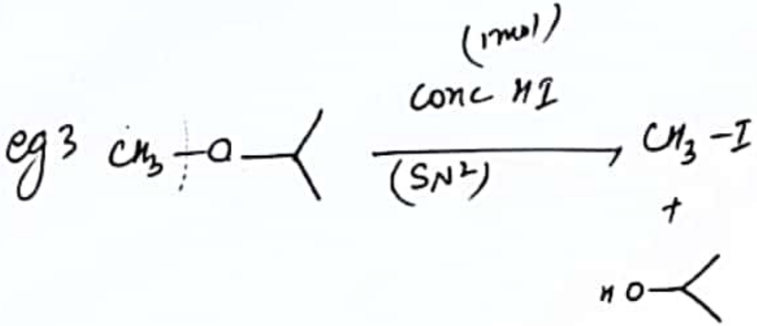
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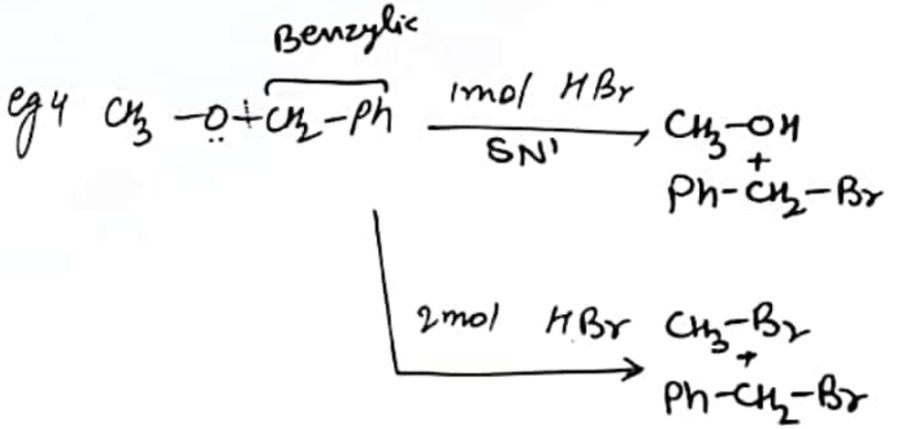
eg 8



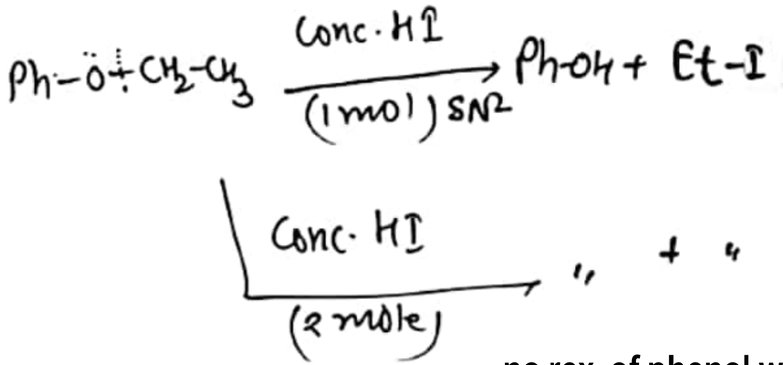
eg 3



eg 4



eg 5



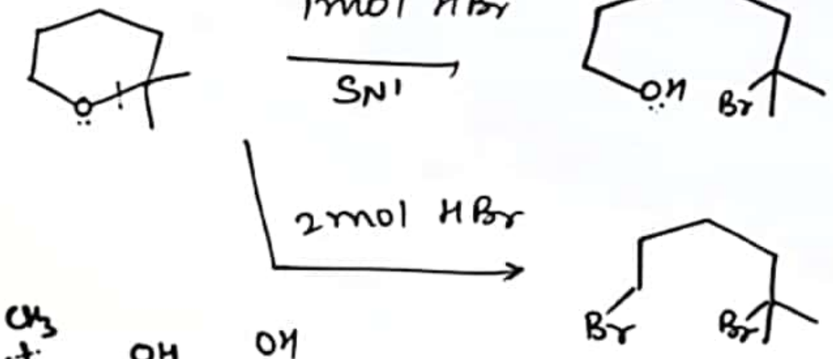
no rex. of phenol with HI

ORGANIC REACTION MECHANISM – III

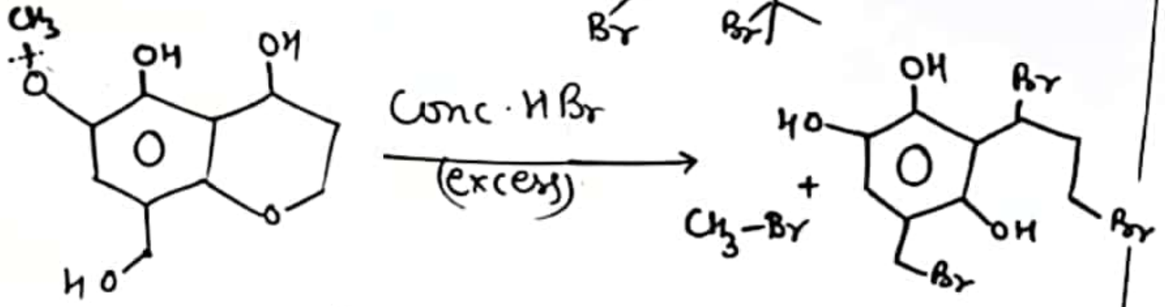
(XI) Nucleophilic Sub. Rxn of Ethers:–

[XI A] Reaction of ether with HX:–

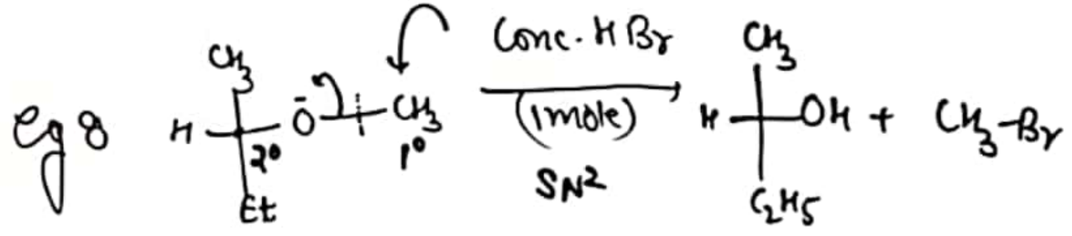
eg 6



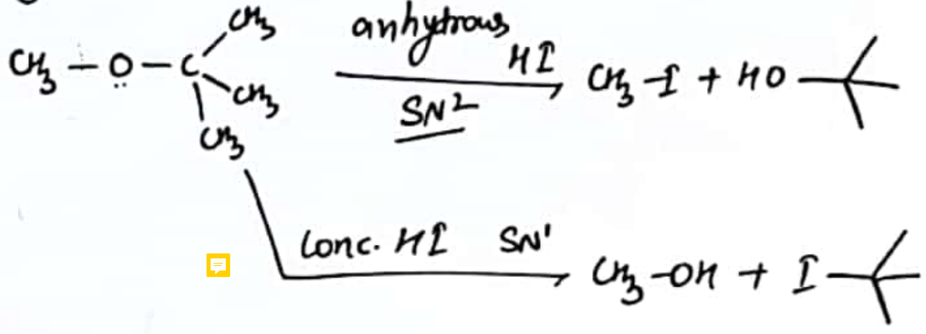
eg 7



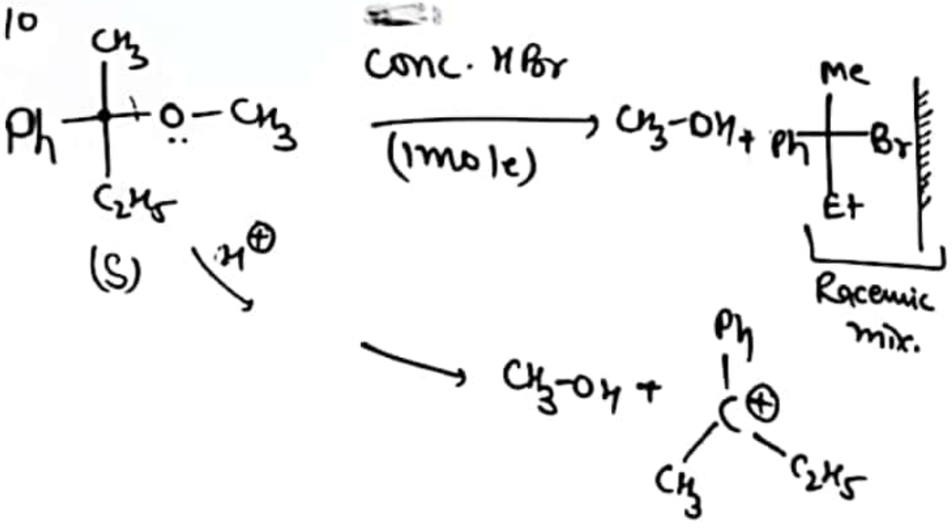
eg 8



eg 9



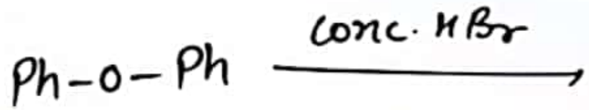
eg 10



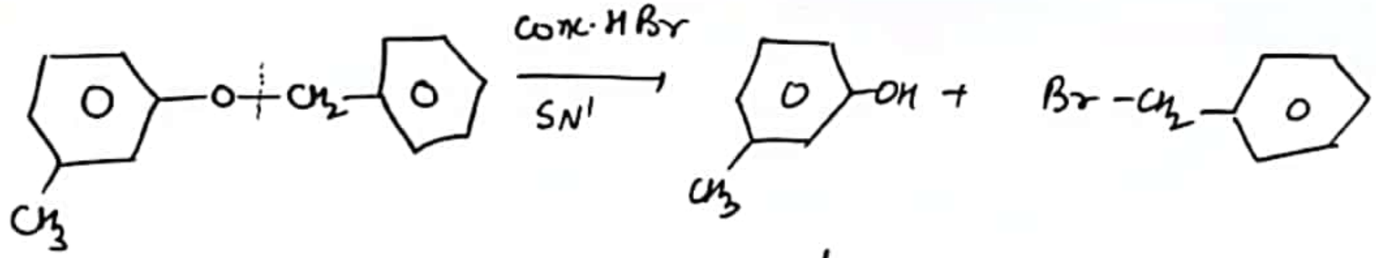
(XI) Nucleophilic Sub. Rxn of Ethers:-

[XI.A] Reaction of ether with HX:-

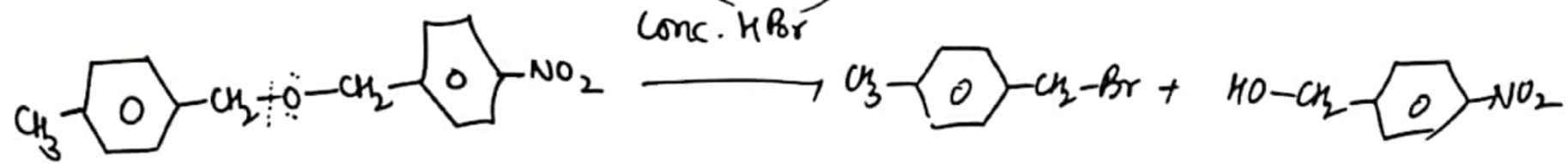
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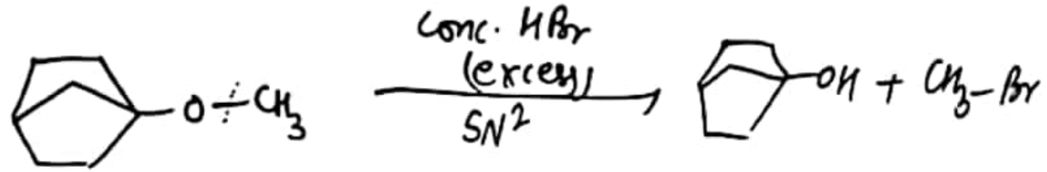
eg 12



eg 13



eg 14

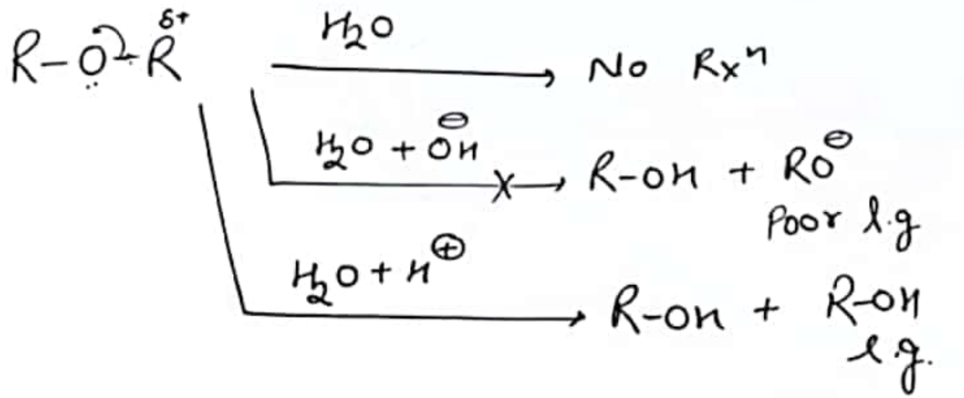


Bredt's rule

ORGANIC REACTION MECHANISM - III (ORM-III)

(XI) Nucleophilic Substitution Reactions of Ethers :-

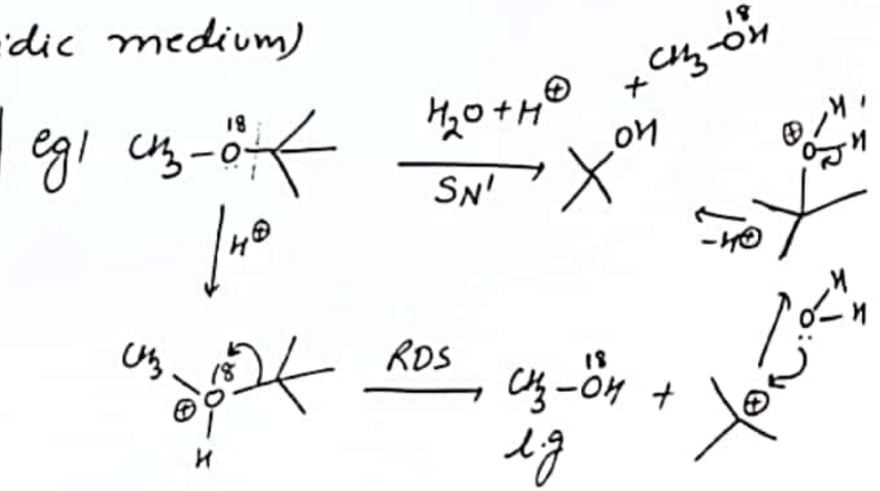
(XI.B) Hydrolysis of ether :- (possible only in acidic medium)



⇒ any alkyl group of ether is 3°, 2°, Benzylic } $\xrightarrow[\text{Hydrolysis}]{H_3O^{\oplus}}$ S_N1

⇒ When both alkyl groups of ether is 1° } $\xrightarrow{\text{Hydrolysis}}$ S_N2

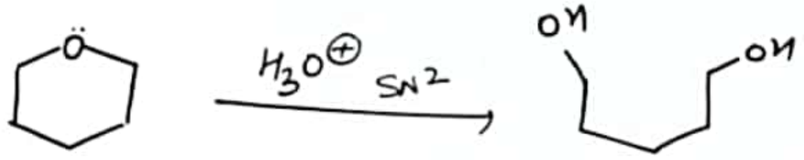
eg!

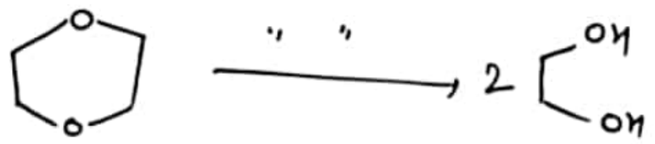


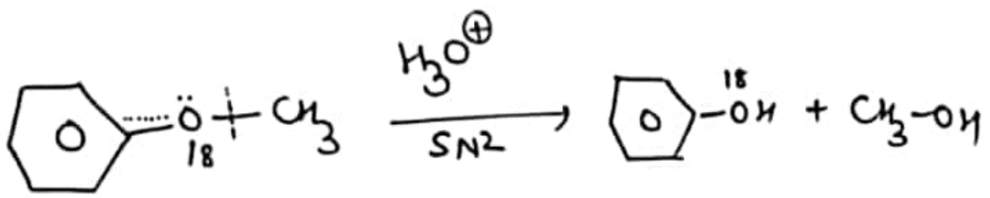
ORGANIC REACTION MECHANISM - III (ORM-III)

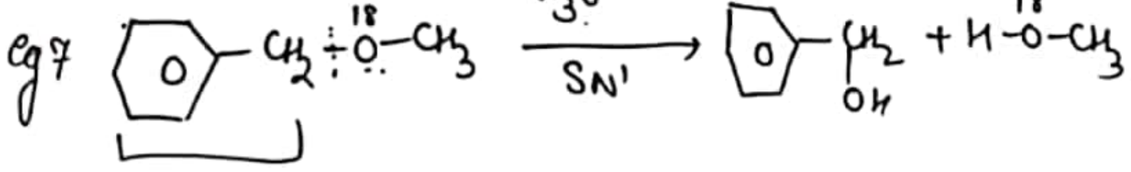
(XI) Nucleophilic Substitution Reactions of Ethers :-

(XI.B) Hydrolysis of ether :- (possible only in acidic medium)

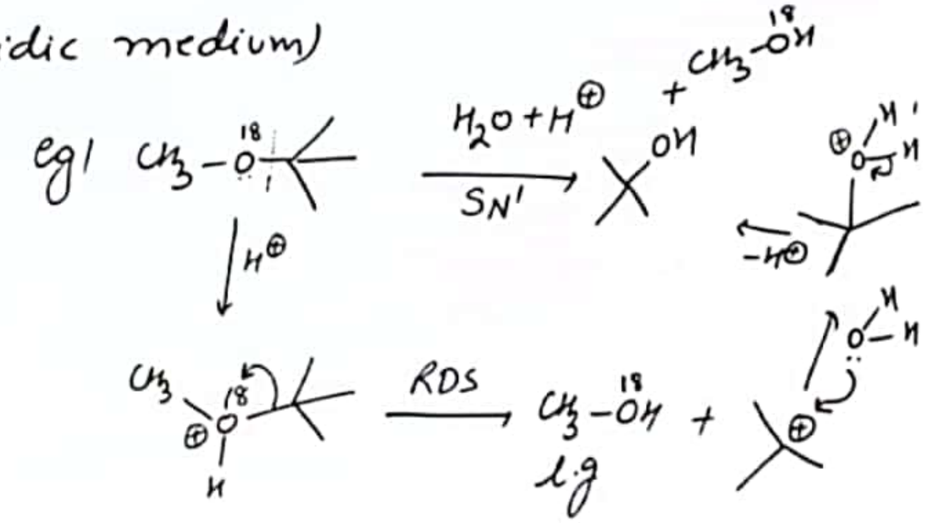
eg 4 

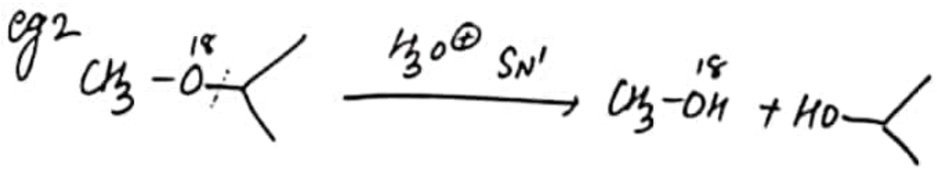
eg 5 

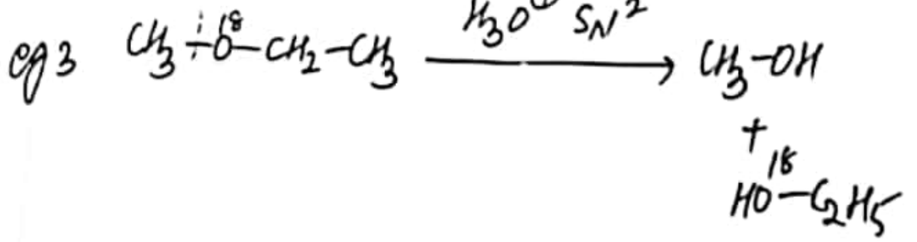
eg 6 

eg 7 

Benzylic

eg 1 

eg 2 

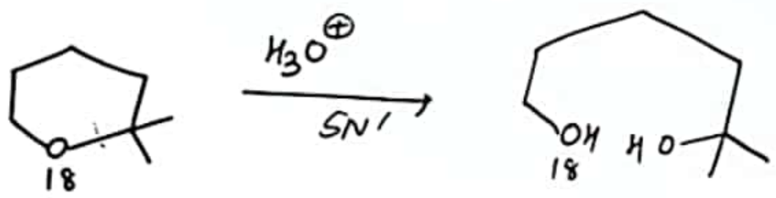
eg 3 

ORGANIC REACTION MECHANISM – III (ORM-III)

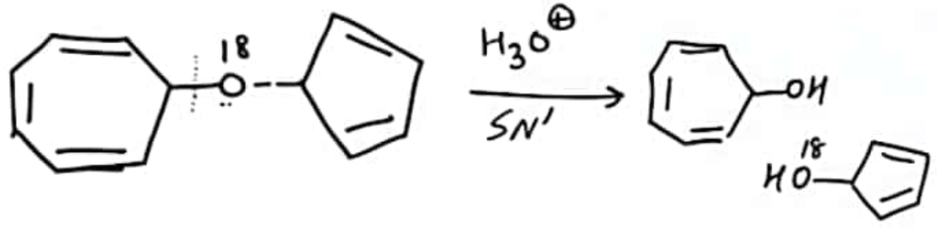
(XI) Nucleophilic Substitution Reactions of Ethers :-

(XI-B) Hydrolysis of ether :- (possible only in acidic medium)

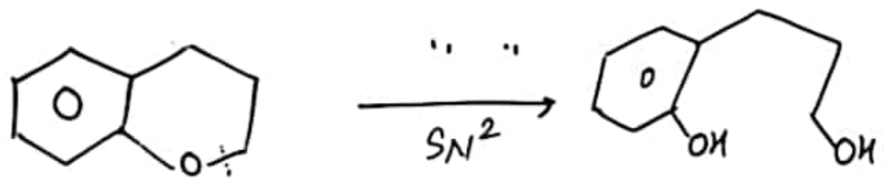
Eg 8



Eg 9

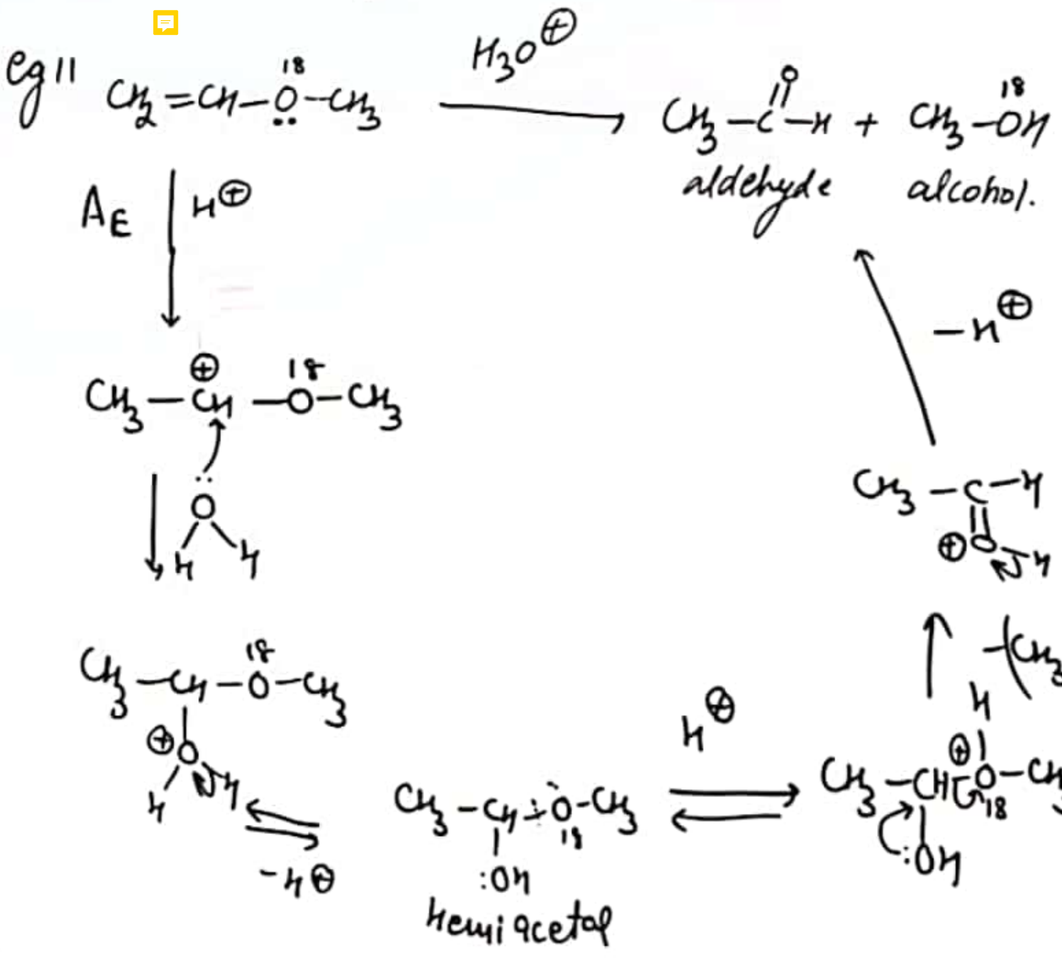


Eg 10



→ Hydrolysis vinylic esters.

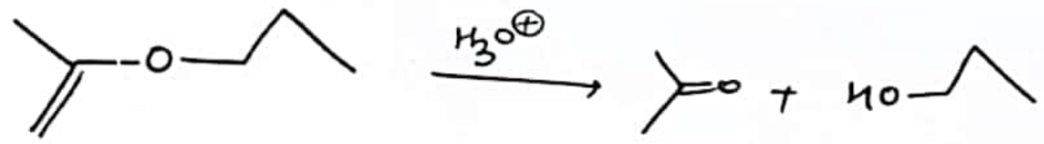
Eg 11

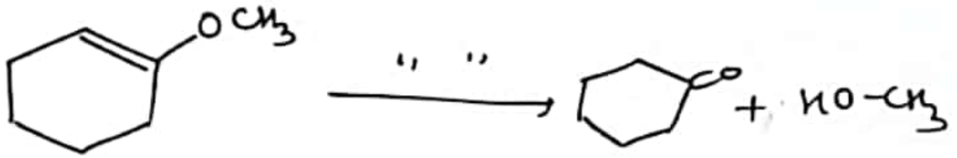


ORGANIC REACTION MECHANISM - III (ORM-III)

(XI) Nucleophilic Substitution Reactions of Ethers :-

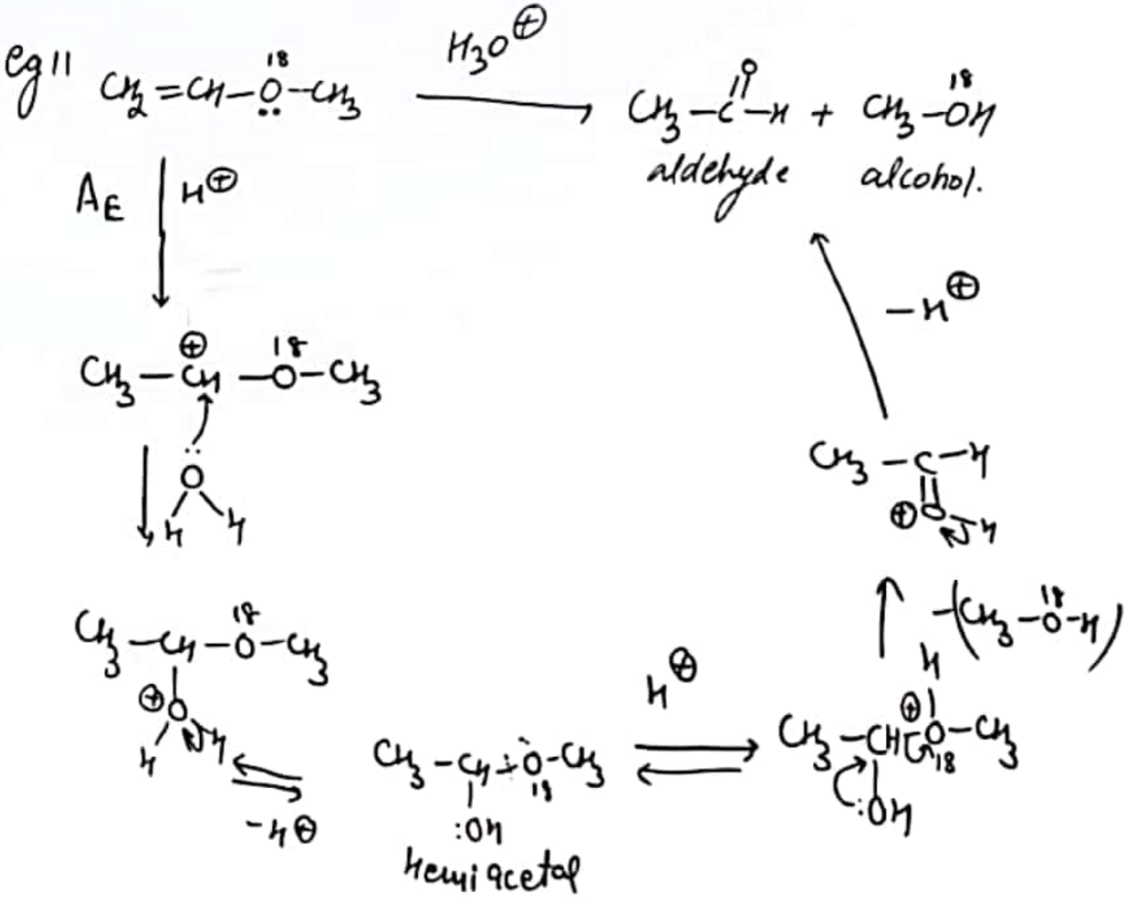
(XI.B) Hydrolysis of ether :- (possible only in acidic medium)

eg 12 

eg 13 

eg 14 $Ph-O-Ph \xrightarrow{\text{..}} \text{No Rx}^{n}$.

→ Hydrolysis vinylic esters.

eg 11 

ORGANIC REACTION MECHANISM – III (ORM-III)

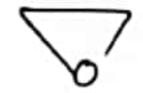
(XI) Nucleophilic Substitution Reactions of Ethers :-

XI.c Reaction of ether with PCl_5 :-

$$R-O-R \xrightarrow[S_N2]{PCl_5} 2\,R-Cl + POCl_3$$

ORGANIC REACTION MECHANISM - III (ORM-III)

(XII) Nucleophilic Substitution Reactions of Epoxides

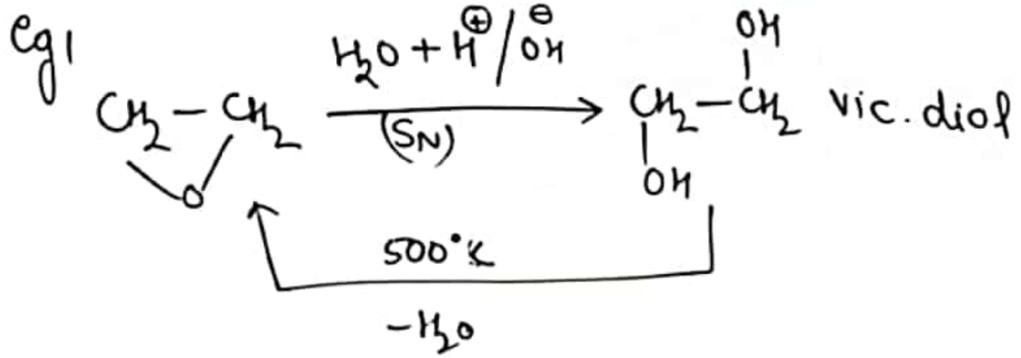


epoxide

⇒ High Ring Strain

⇒ higly reactive

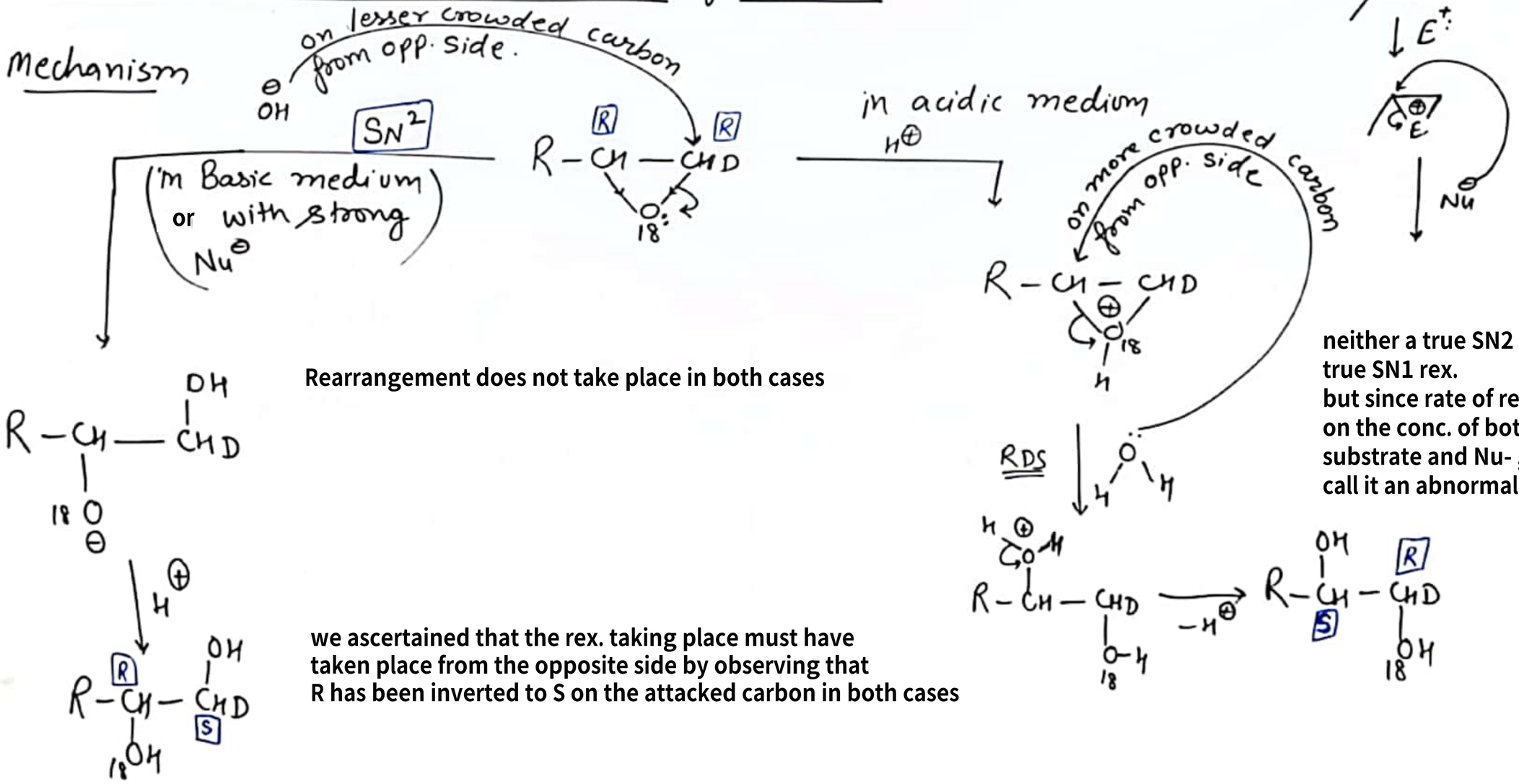
⇒ Can give S_N Reactions in acidic medium as well as basic medium.



ORGANIC REACTION MECHANISM – III (ORM-III)

(XII) Nucleophilic Substitution Reactions of Epoxides

Mechanism



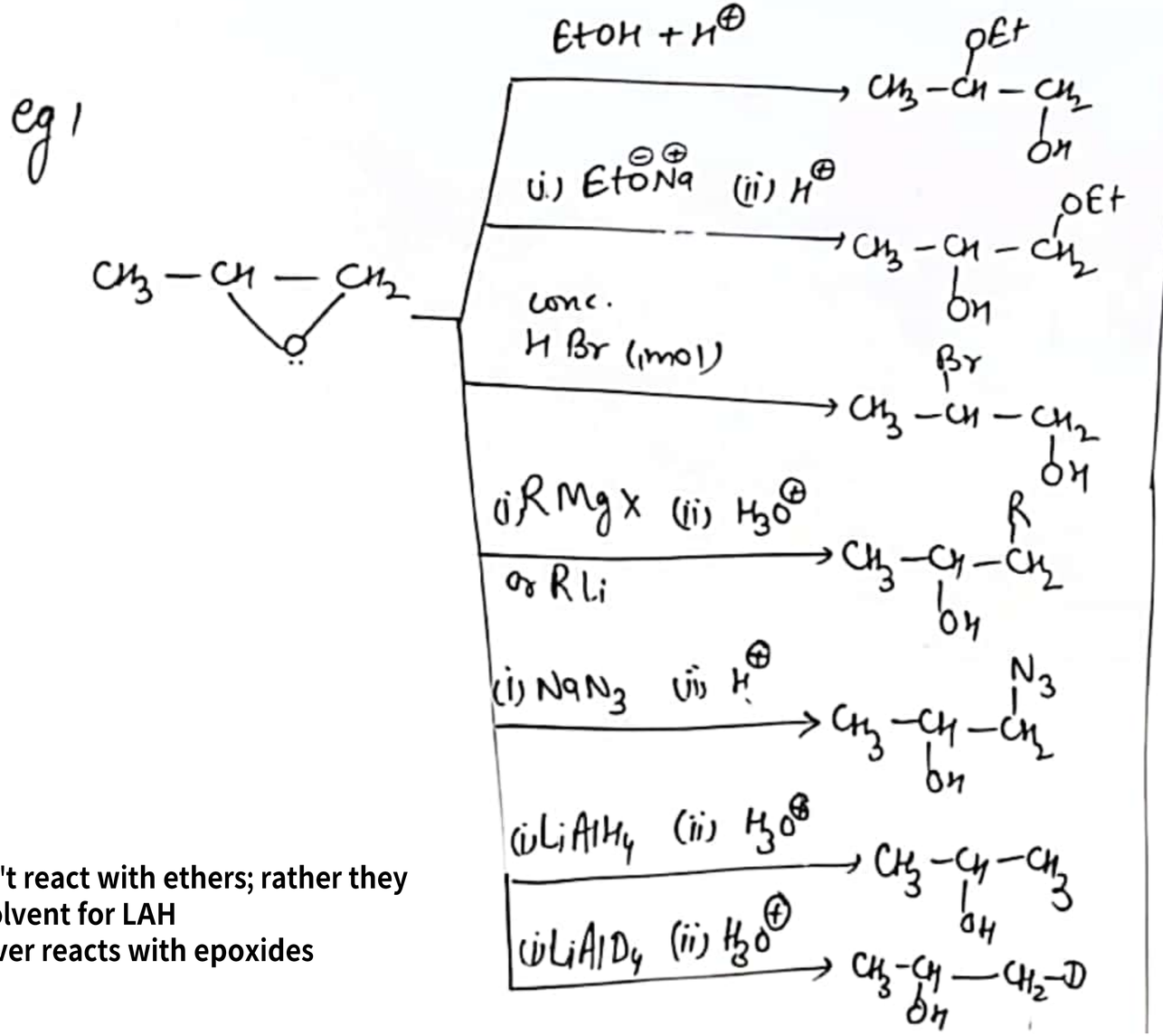
Rearrangement does not take place in both cases

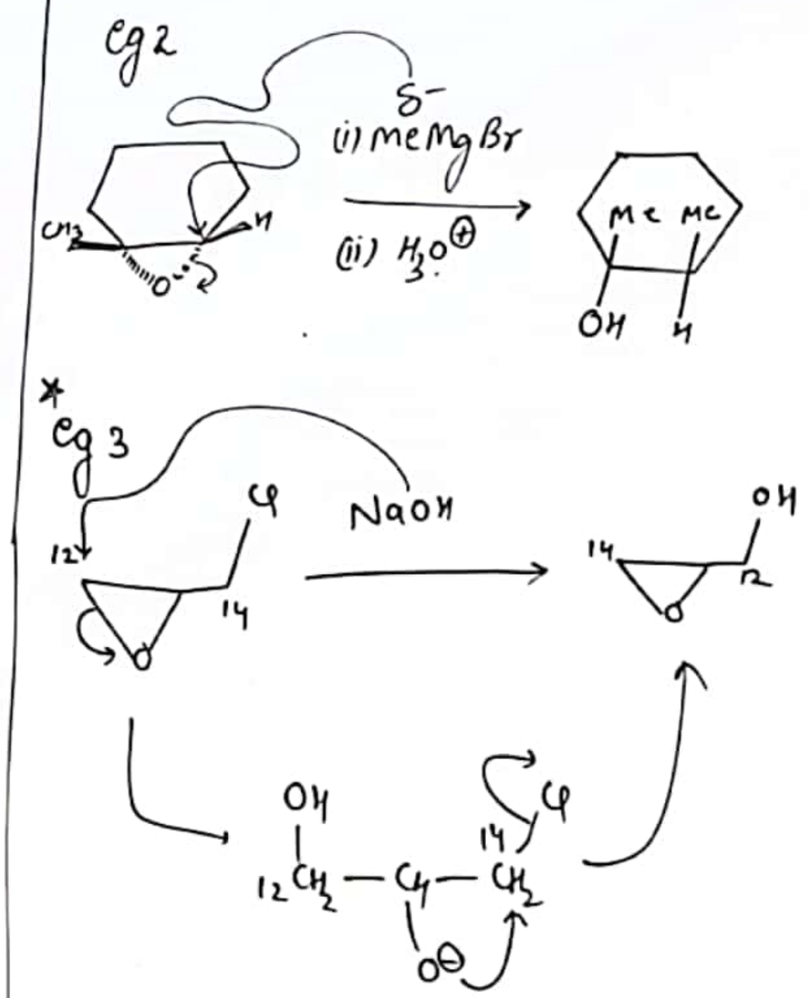
we ascertained that the rex. taking place must have taken place from the opposite side by observing that R has been inverted to S on the attacked carbon in both cases

neither a true SN2 rex. nor a true SN1 rex.
but since rate of rex. depends on the conc. of both substrate and Nu- , we can call it an abnormal SN2

ORGANIC REACTION MECHANISM - III (ORM-III)

(XII) Nucleophilic Substitution Reactions of Epoxides





LAH doesn't react with ethers; rather they acts as a solvent for LAH
LAH however reacts with epoxides

ORGANIC REACTION MECHANISM – III (ORM-III)

(XIII) $Ar S_N$ Reaction (Nucleophilic sub. Rxn on aromatic ring)

(A) ArS_N1

(only in diazonium salt)

in aromatic compound

(B) $Ar S_N2$

(When strong EWG is/are present on ortho / Para position of leaving group.)

(c) Benzyne mechanism

(In difficult condition with strong base/$Nu^{\ominus}$)

ORGANIC REACTION MECHANISM – III (ORM-III)

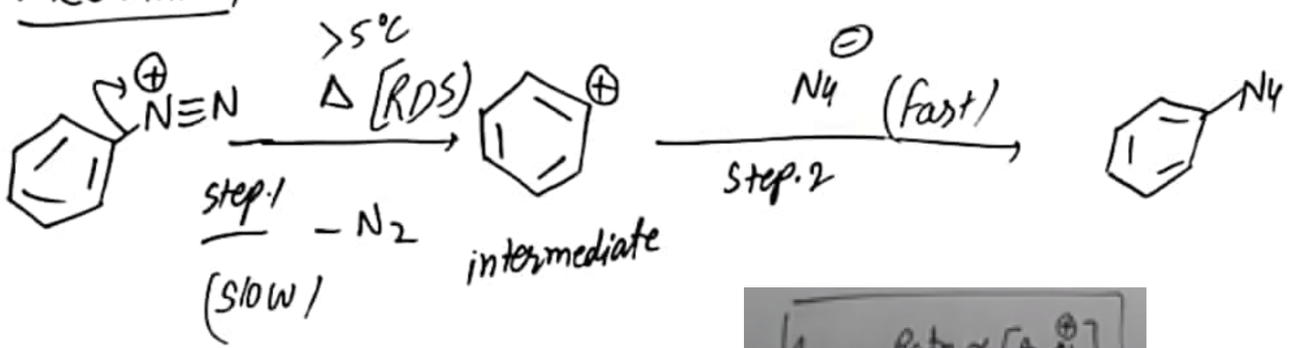
(XIII) $Ar S_N$ Reaction (Nucleophilic sub. Rxn on aromatic ring)

(XIII.A) $Ar S_{N}1$ Reaction (only introduction)



Benzene diazonium Chloride.
(Stable upto 5°C)

mechanism



$Ar S_{N}1$ Rate $\propto [Ar\overset{\oplus}{N_2}]$

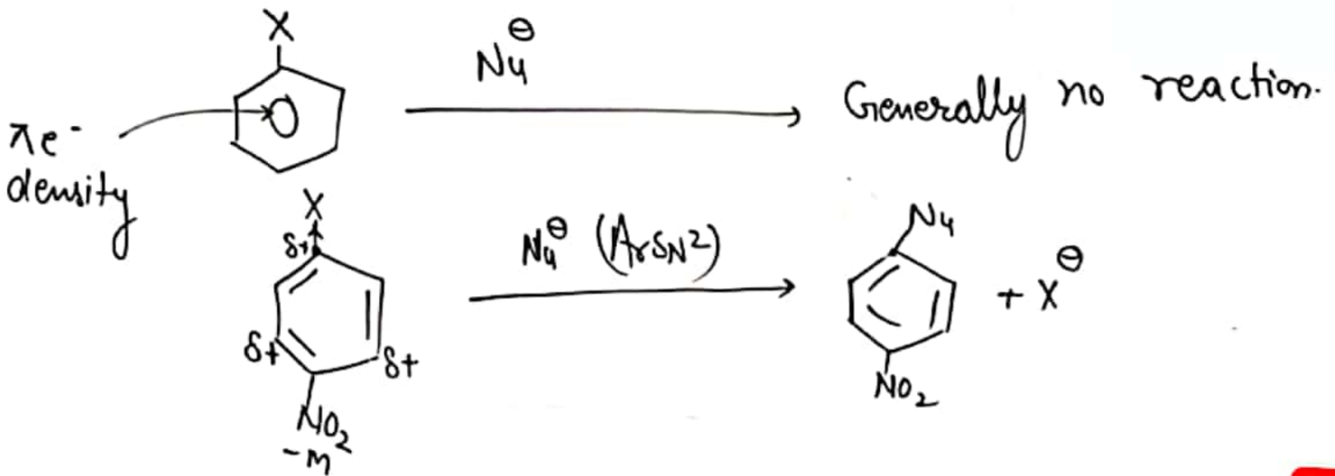
ORGANIC REACTION MECHANISM – III (ORM-III)

(XIII) $Ar S_N$ Reaction (Nucleophilic sub. Rxn on aromatic ring)

(XIII.B) $Ar S_N 2$

Generally Nucleophilic attack on aromatic ring do not possible due to high e^- density on aromatic ring.

But in presence of strong e^- withdrowing group on ortho or para position. of leaving group Nucleophilic attack possible.



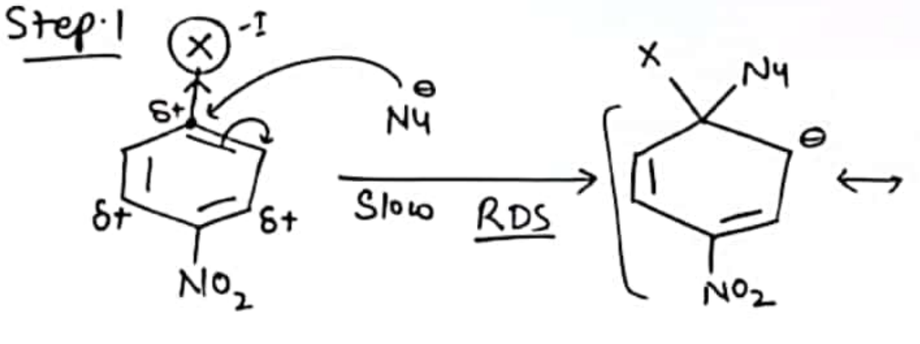
ORGANIC REACTION MECHANISM - III (ORM-III)

(XIII) $Ar S_N$ Reaction (Nucleophilic sub. Rxn on aromatic ring)

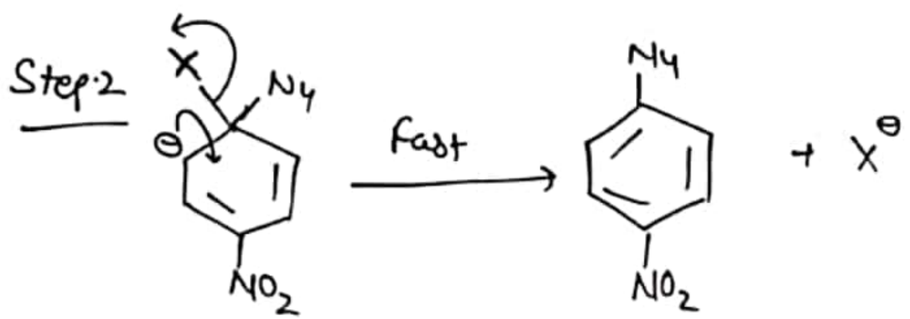
(XIII.B) $ArSN^2$

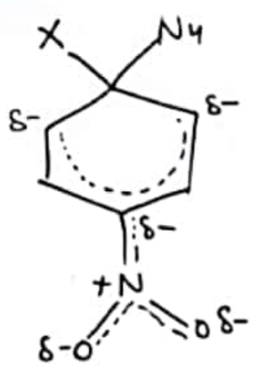
Mechanism

Step·1



Step·2





⇒ Rate of $ArSN^2 \propto [Ar\text{-}X]\,[N\bar{u}]$

⇒ 2nd order

⇒ Reactivity of $ArSN^2$ ↑ in presence of EWG.

⇒ ∴ ∴ ∴ ∝ Stability of intermediate

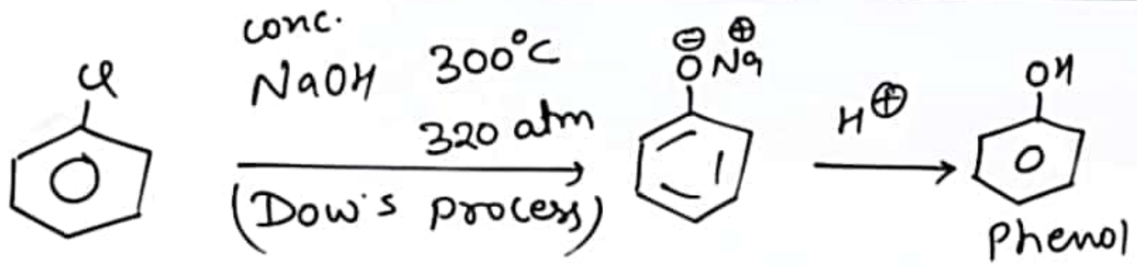
⇒ $X^{\ominus} > N_4^{\ominus}$ stability

Imp. ⇒ Reactivity w.r.t. Halogen l.g ⇒ Ar-F > Ar-Cl > Ar-Br > Ar-I

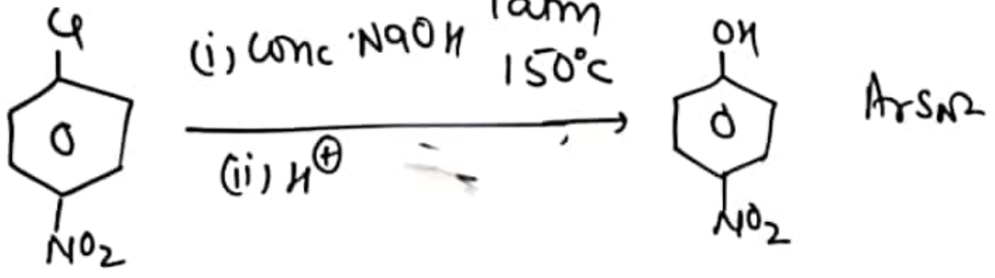
due to greater EWG of F>Cl>Br>I

(XIII) $Ar S_N$ Reaction (Nucleophilic sub. Rxn on aromatic ring)

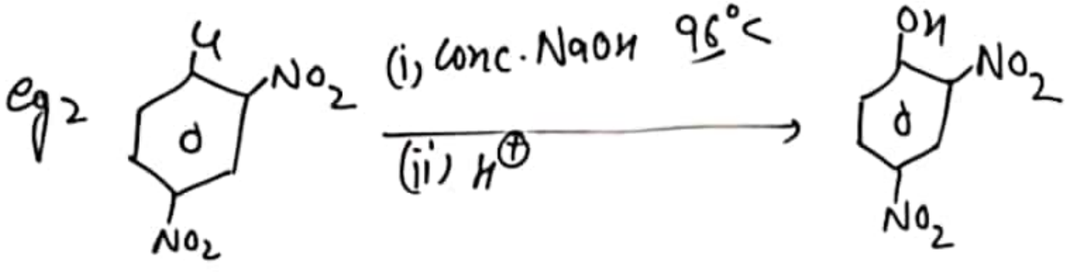
(XIII.B) $ArSN^2$

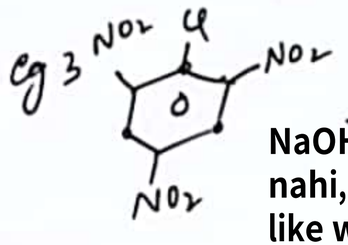


Benzyne mechanism (not $ArSN^2$)

eg 1 

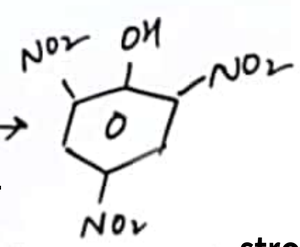
$ArSN^2$

eg 2 

eg 3 

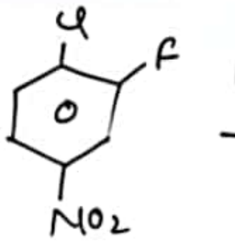
H_2O, Δ

NaOH ki bhi jarurat nahi, even weak Nu- like water can also give

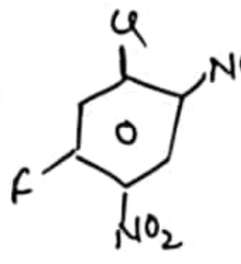


Picric acid

stronger acid than acetic acid

eg 4 

NaOH Δ

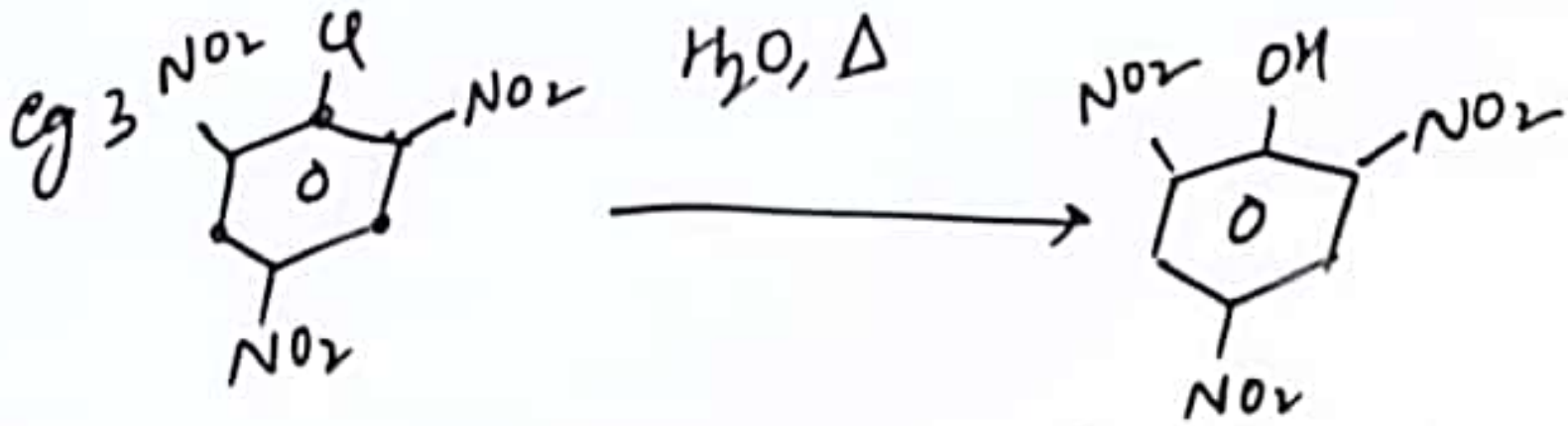
eg 5 

1 mol MeS^-Na^+

2 mole MeS^-Na^+

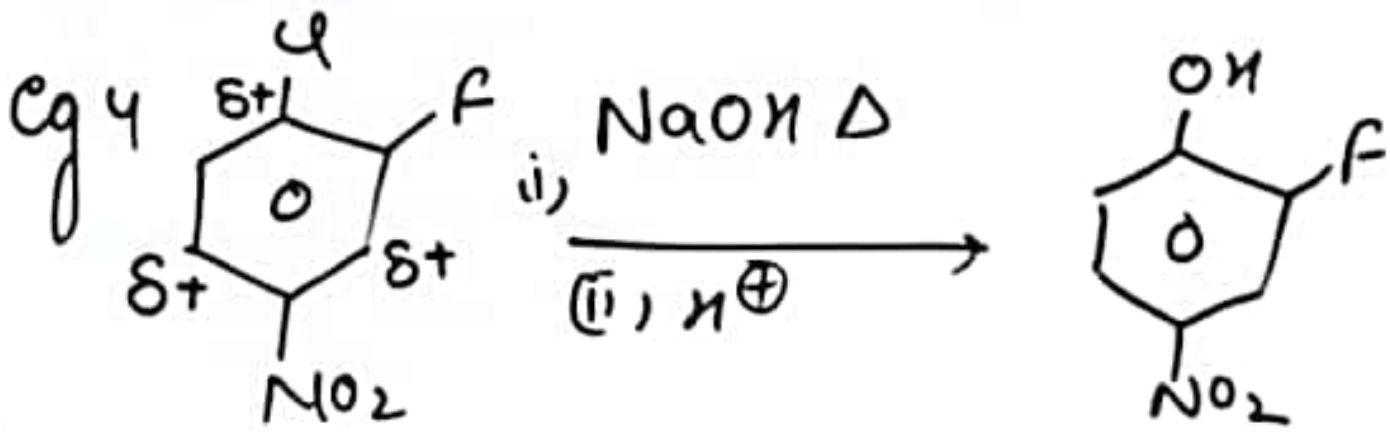
aromatic ring)

eg 3

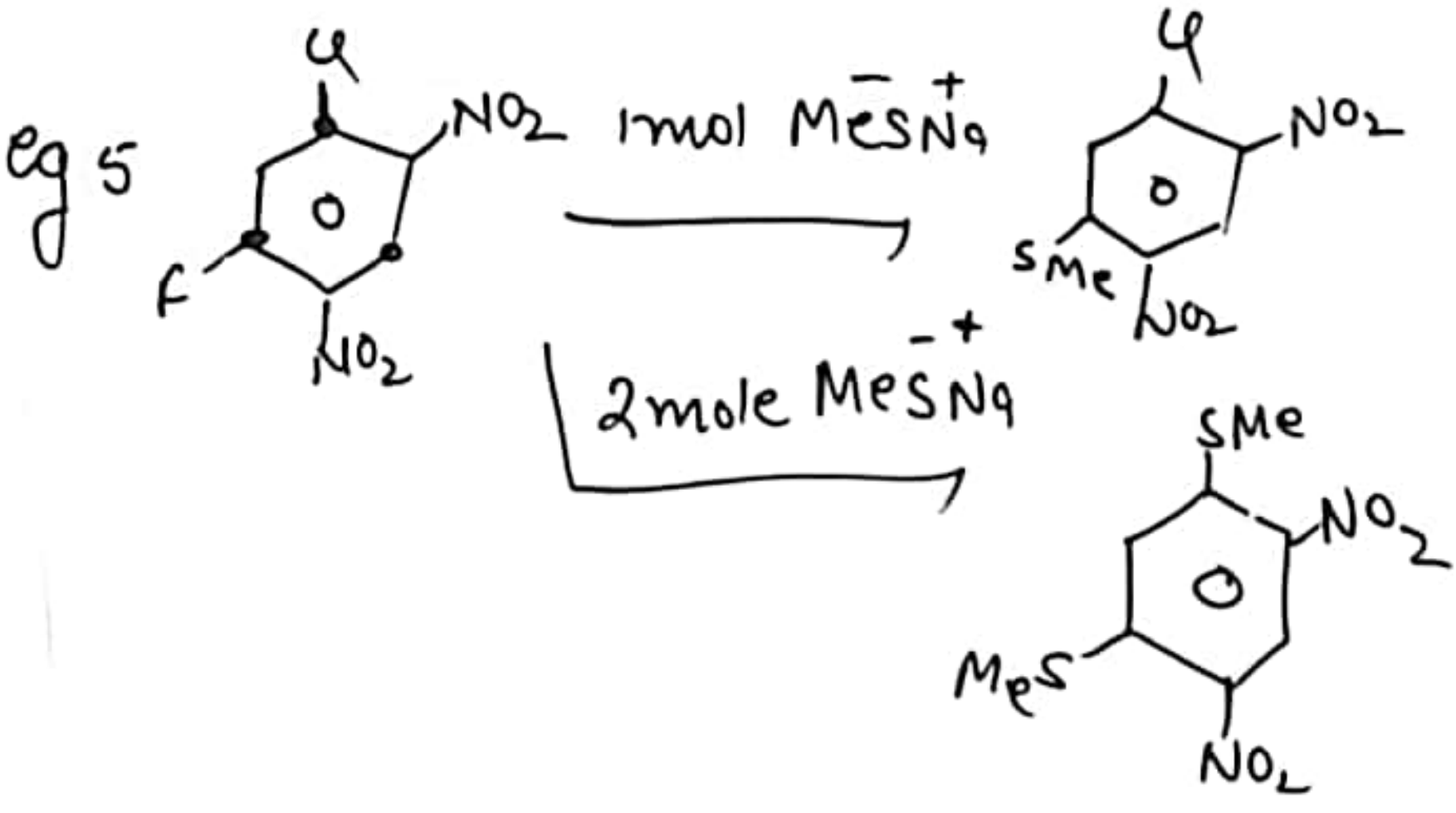


Picric acid

eg 4



eg 5



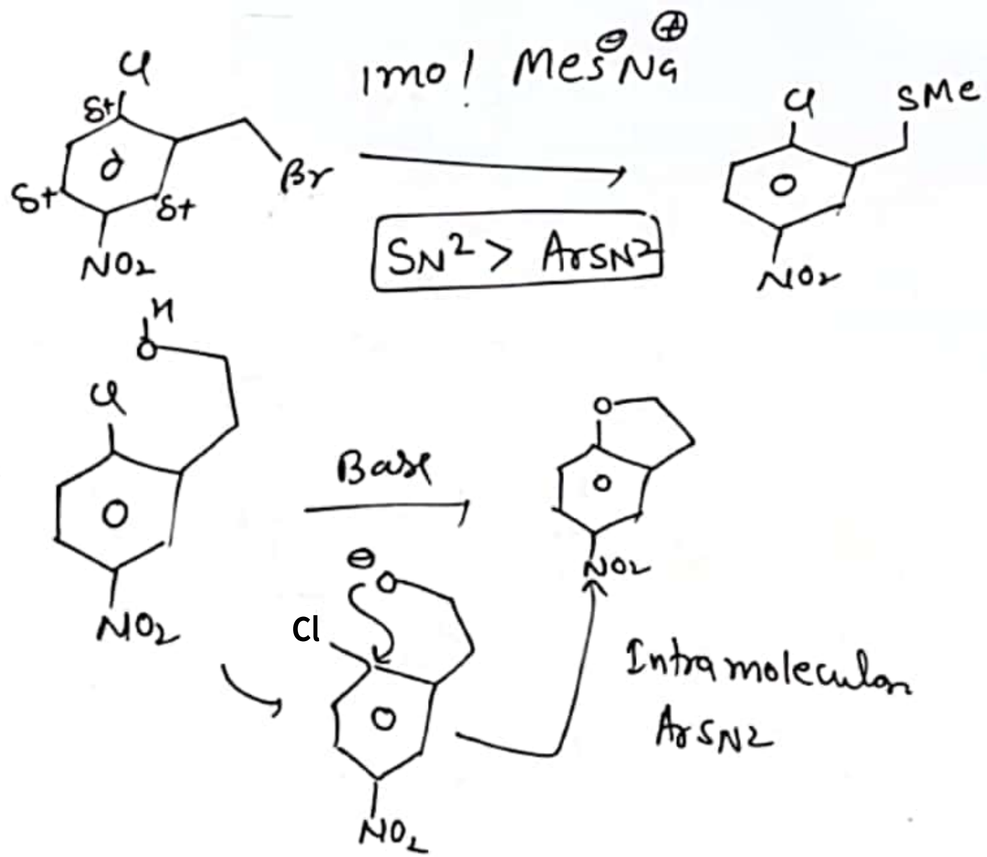
due to greater reactivity of F

ORGANIC REACTION MECHANISM – III (ORM-III)

(XIII) $Ar S_N$ Reaction (Nucleophilic sub. Rx^n on aromatic ring)

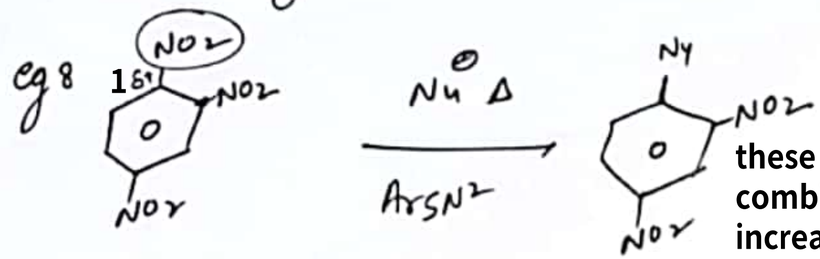
(XIII.B) $ArSN^2$

eg 6



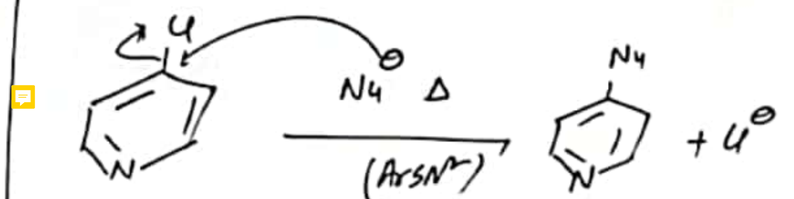
eg 7

eg 8



these two NO2 combined and increased the partial +ve charge at position 1

Chichibabin Rx^n



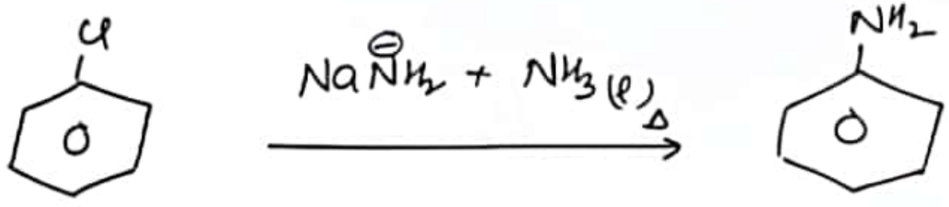
** pyridine has less e- density and can also give such reactions

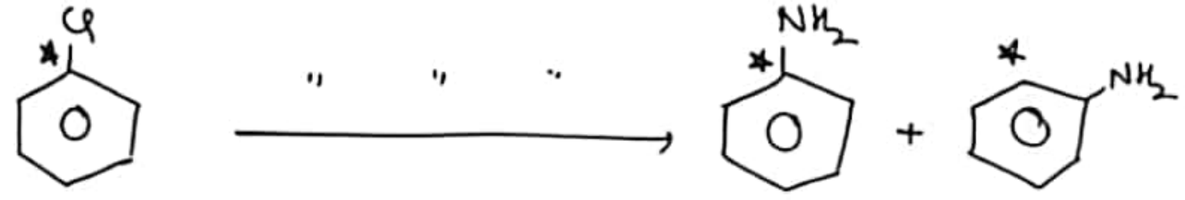
ORGANIC REACTION MECHANISM - III (ORM-III)

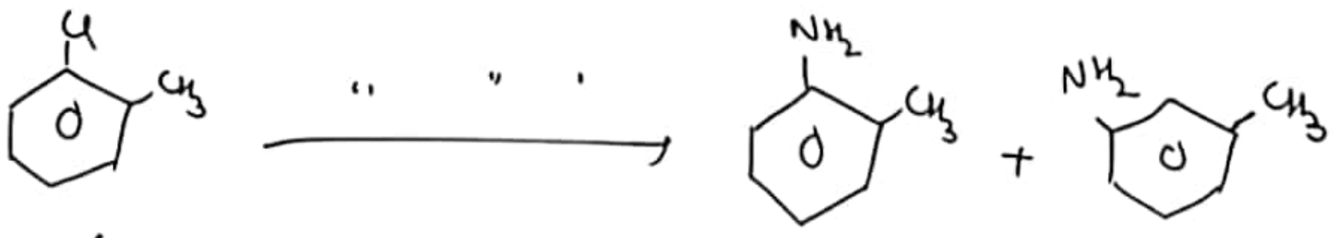
(XIII) Nucleophilic Substitution Reactions of Aromatic compounds.

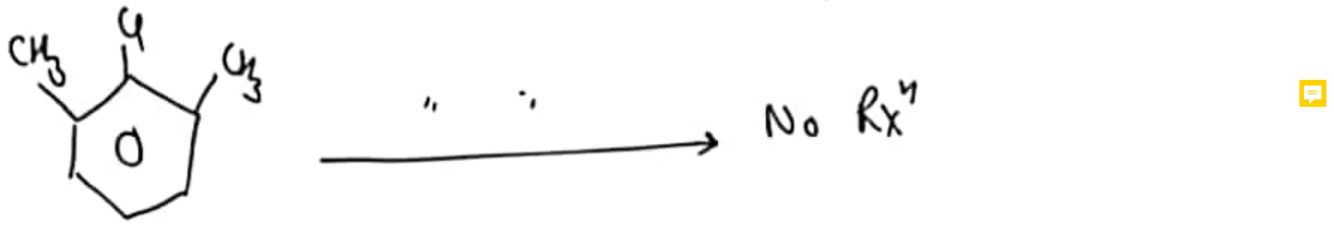
(XIII.C) Benzyne mechanism:-

⇒ It is possible in difficult conditions (Strong base + high temp)

eg1 

eg2 

eg3 

eg4 

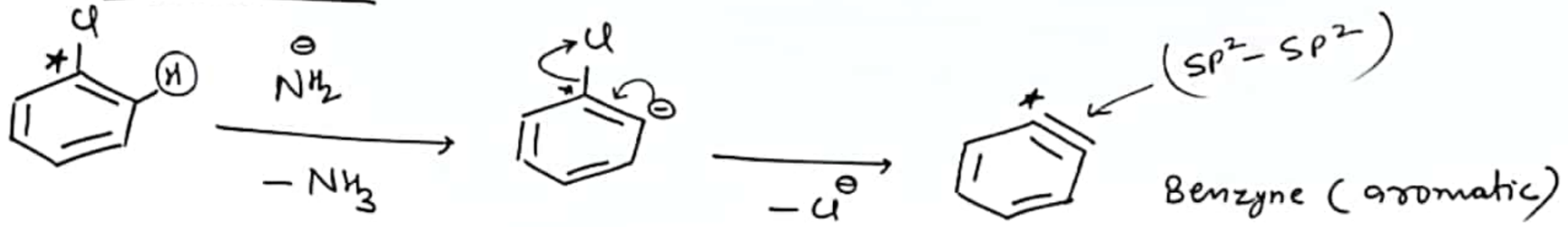
ORGANIC REACTION MECHANISM - III (ORM-III)

(XIII) Nucleophilic Substitution Reactions of Aromatic compounds.

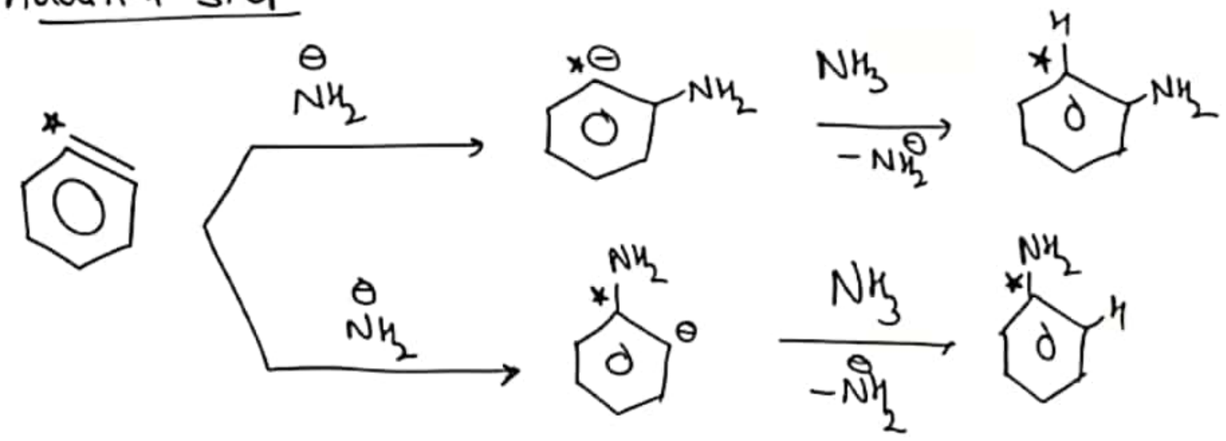
(XIII.C) Benzyne mechanism:- (Elimination - Addition mechanism)



⇒ Elimination step.



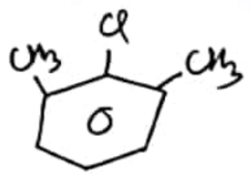
⇒ Addition step



ORGANIC REACTION MECHANISM - III (ORM-III)

(XIII) Nucleophilic Substitution Reactions of Aromatic compounds.

(XIII.C) Benzyne mechanism:- (Elimination - Addition mechanism)

eg 4 

$\xrightarrow{NaNH_2 + NH_3(l)}$ No R_x^n

since no possibility of benzyne formation

eg 5 